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THE ASSAYER



In which
With a delicate and precise scale
will be weighed the things contained in
The Astronomical and Philosophical Balance
of Lothario Sarsi of Siguenza

Written in the form of a Letter
to the Illustrious and Reverend Monsignore

DON VIRGINIO CESARINI

Lincean Academician
Lord Chamberlain to His Holiness

BY

SIGNOR

GALILEO GALILEI

Lincean Academician
Gentleman of Florence
Chief Philosopher and Mathematician
to the
Serene Grand Duke of Tuscany



R O M E

MDCXXIII

IMPRIMATUR
SI VIDEBITUR REVERENDIS. P. MAG. SAC. PAL. APOST.
A. Episc. Hieracen. Vicesg.



I have read, by order of the Most Reverend Father, Master of the Sacred Palace, this work *The Assayer*; and besides having found here nothing offensive to morality, nor anything which departs from the supernatural truth of our faith, I have remarked in it so many fine considerations pertaining to natural philosophy that I believe our age is to be glorified by future ages not only as the heir of works of past philosophers but as the discoverer of many secrets of nature which they were unable to reveal, thanks to the deep and sound reflections of this author in whose time I count myself fortunate to be born—when the gold of truth is no longer weighed in bulk and with the steelyard, but is assayed with so delicate a balance.

In the College of St. Thomas on the Minerva at
Rome, 2 February 1623.

F. NICOLÒ RICCARDI

Imprimatur

*F. Dominicus Paulaccius Mag. & socius Reverendis.
P. Fr. Nicolai Rodulfi Sac. Apost. Palatii Mag.*

TO HIS HOLINESS
POPE URBAN VIII



In this universal jubilee of literature and of virtue itself, when the whole City and especially the Holy See is more resplendent than ever from the placing there of Your Holiness by celestial and divine disposition,¹ every heart in veneration is inflamed to praiseworthy studies and worthy actions. Imitating so eminent an example, we come to appear before you, mindful of infinite obligation for the benefits continually received from your benign hand and full of joy and contentment to see on so sublime a throne such an exalted Patron. As evidence of our devotion and as a tribute from our true fealty, we bring you *The Assayer* of our Galileo, the Florentine discoverer not of new lands but of hitherto unseen portions of the heavens, containing investigations of those celestial splendors that usually attend great wonders. This we dedicate and present to Your Holiness as to one who has filled his soul with true ornaments and splendors and has turned his heroic mind to the highest undertakings, and we hope that this discourse upon unusual torches in the sky will be to you a sign of our live and ardent passion to serve Your Holiness and to merit your grace. Meanwhile, humbly inclining ourselves at your feet, we supplicate you to continue favoring our studies with the gracious rays and vigorous warmth of your most benign protection.

At Rome, the 20th of October, 1623
From Your Holiness' most humble
most devoted servants,

THE LINCEAN ACADEMICIANS

JOHANN FABER
LYNCEAN OF BAMBERG, ROMAN PHYSICIAN
AND PONTIFICAL PHARMACIST
TO GALILEO GALILEI
LYNCEAN OF FLORENCE, PRINCE OF MATHEMATICIANS
OF OUR AGE, DISCOVERER OF MARVELOUS THINGS
IN THE SKY THROUGH THE TELESCOPE
THE NEW EYE OF NATURE

Porta possesses the first claim; you, German, may have the second. The third claim, Galileo, rests on your labor.^a But you, Galileo, surpass the others by as great a distance as the celestial stars are separated from the earth. With the telescope they may measure a few miles of the earth and venture at sea; you surpass them by an infinite distance in the art of observation with the telescope while you ascend the famous Olympus. Yield, Vespucci, and let Columbus yield; let each take his course through the unknown sea; yet the land of the antipodes was not unknown to the ancients; nor did each pole flee former astronomers. But you were the one who gave the succession of the stars and the new constellations of the heavens to mankind. Hail to your telescope, Galileo; as it approaches the stars so mortals of city and state love you. Did you perhaps with marvelous skill so fit spectacles to an aging world that with mind still sound but eyes dimmed and body weakened it might see through two glasses? Through them new progeny of stars descend from the celestial heights as well as the countless ones which are in the Milky Way. By them also cold Saturn, marvelous to see, is observed with twin ears; horned Venus shines at night, a new Diana; Jupiter is surrounded by four companions. By them, you also, the brightest lights of the great universe, Sun and your sister, change your appearances. You, Sun, who shine so very brightly as the pure orb, appear marred by strange spots. And that smooth, round Moon in which we have believed, behold, this globe is conspicuous for its swollen mountains. With this Eye, we see bright stars, once called nebulae, shining in

the sky. Nor yet is the limit reached; shining through the tail of the Comet, by this Eye, they settle in a better-known place. Under a learned appearance, Aristotle deceived minds when he approved an aerial home for those torches; and he believed that the comet, a wholly harmless brightness, when it kindled the air was to be dreaded. Which, if yet you may believe, speaking with fateful countenance, it announces the destruction of its spectators; I fear it portends a sterile year alone for him for whom the greedy fire devours the sceptre of mathematics. But Galileo smooths all; the leveller of the telescope teaches what the comet is; observing with the eye of the lynx he has exposed the phenomena and the meaningless nonsense believed by princes of learning. O bold deed, to have penetrated the adamantine sky and walls of crystal with such fragile aid. It is granted to your mind that by this tube of yours, Galileo, you are able to illuminate the happy citadel of the gods.

TO THE ABOVE-MENTIONED
SIGNOR GALILEI
FROM SIGNOR
FRANCESCO STELLUTI
LINCEAN ACADEMICIAN

Unlike other things alive—
Birds that wing their way through air
Fish that in chill waters dive
Beasts that tread the earth and leave their footprints there—
Man, by reason made aware,
Solely judges and discerns;
Man approaches angels by the things he learns.

Nature models tasks for man;
Things by her to him denied
He by artifice and plan
Creates, as wonders numberless have testified.
Man has mind to serve as guide;
Dextrously his work begun
Master artists beautify what he has done.

If he would some beast portray
Or himself, in image true
Man can make this so it may
Move its eyes and hands and feet as creatures do.
Once Archytas made and flew
Wooden doves that soared in air
Seeming then to have both life and senses there.

Then a sage is famous who
 (Called 'the Great' by all men still)
Did by wondrous art endue
Lifeless lips with speech and storytelling skill.
Cleverly he forms at will
Other artificial birds
Giving them sweet songs to carry and make heard.

Other men have made of glass
 Models of the universe
And around these cause to pass
All our many girdling heavens so diverse;
Tiny planets thus rehearse
All the motions of the world
Rolling starlike just as those on high are whirled.

Overhead, to our alarm
 In the spaces of the sky
Sometimes lightnings threaten harm,
Loud and fearful thunders crash and terrify.
Man makes also flames that fly;
Thunder never louder roared
Than when man shoots lead through tubes in iron bored.

Some men dare to turn their backs
 On the Gates of Hercules
Sailing where there are no tracks
Hollow wood their home, impelled by ocean's breeze
Seeking roads through high-piled seas
Storming palisades of cold
Sailors add a new world to the world of old.

Braver yet in leaving land
Bold beyond all human kind
Daedalus with skillful hand
Wings of feathers made, upon his back to bind;
Boldly hoping he would find
Means of flight, he did indeed
Cleave the moist and salty air with birdlike speed.

You, when I consider well,
Vigorous and proud must be,
So to feast your sight, and tell
Things transcending human thought that you can see;
Lynxes blind as moles are we,
GALILEO, without you;
Argus eyeless is beside you, Linceans too.

But your piercing rays of sight
Do not owe their strength alone
To that great resplendence bright
Lent by glassy lenses you have made your own,
For a finer light has shone
Gathered there within your mind;
This light conquers vision, leaves it far behind.

Thus what Nature has concealed
In her breast from others' view
By your gaze serene revealed
Now is opened wide for other men by you.
Some believe they know this, too
From the sage old Stagirite,
Who but saw it dimly, as a shadowed sight.

That which you now teach of it
Did not come from pages old
Nor from new books filled with wit—
No, and you have not by friendly stars been told;
Labors long and vigils cold
Proofs and searches, studies deep
These are what aroused such powers from their sleep.

Scarce a man can understand
Though he use his senses well
How our touch is in our hand
How our eyes see sights, ears hear, and noses smell
This we ask you now to tell,
Though the tongue can scarce relate
How it knows the varied flavors on a plate.

Any man would gladly know
How both cold and heat are felt
How the sky for us below
Makes the wonders that are seen in heaven's belt—
Riddles that your soul has spelt;
Nothing near or far from you
Ever keeps its secrets from your piercing view.

Once an ancient sought the cause,
Near to Chalcis so men tell,
Why Euripus' wave no pause
Made in ebb and flow as tides now rose and fell;
Had you told him what so well
You have proved of your belief
He would not have plunged into those waves from grief.

Of those persons long since dead—
Wise Hipparchus and his kind—
It respectfully is said
They share high and starry thrones with your great mind;
Yet they never thought to find
That celestial lights on high
Numbered many thousand stars within the sky.

They would not have turned the back
With a show of mannered grace
Nor have strayed from truth's straight track
Nor established *seven* in some special place
Nor would they have tried to trace
Sites and bounds improperly
For those errant stars with golden hair we see.

Further planets you did place
High among the starry ways
Where in far celestial space
Mighty Jupiter benignly sheds his rays;
Where the slowest planet stays
Through new forms you see him run,
Likewise her whose rapid course outstrips the sun.

Of the fixed stars with their rays
You have seen so many turn
Through those vast and distant ways
That to number these the eye can scarcely learn;
Stars so countless we discern
It would seem that they must be
Numbered as the sands on shores beside the sea.

Glory must be yielded by
Daring modern Tiphyses
Celebrated to the sky
Having charted unknown shores and new-found seas;
You, however, unlike these
Add not just some new lands here
But new stars in skies sublime you make appear.

It is news to us when these
Find new lands in any place;
Not to aborigines—
Paths they found across the ocean's briny face
And arrived before our race.
You discover, far more sage,
Orbs to all men new and to our very age.

For these objects, new indeed,
Men are much in debt to you;
Your frail glasses lend us speed
Lightly soaring up to heaven with our view,
And the sky's your debtor too
For this brighter beauty there
And for more and larger lights in regions bare.

Had your eyes not soared on high
Spying out wide heaven's ways,
All that is within the sky
Would have been just as it was before our days;
You these globes with golden rays
Veiled from us and hid to fame
Dressed, and gave their motion, glory, place, and name.

Therefore if from present view
Of these lights by you perceived
Heaven gains much glory new
She will not by death's oblivion be bereaved;
You from whom she much received
Shall, so long as heaven turns,
Shine as bright as any glorious star that burns.

THE ASSAYER

BY

GALILEO GALILEI

Lincean Academician
Chief Philosopher and Mathematician
to the Most Serene Grand Duke of Tuscany

Written in the form of

A LETTER

TO THE MOST ILLUSTRIOUS AND REVEREND
SIG. DON VIRGINIO CESARINI
Lincean Academician
Lord Chamberlain to His Holiness



I have never been able to understand, your Excellency, how it comes about that every one of my studies which, in order to please or to be of service to others, I have seen fit to place before the public has occasioned in many a certain animus to detract, steal, or deprecate that modicum of esteem to which I had thought I was entitled, if not for the work, at least for my intention. In my *Starry Messenger* were revealed many new and marvelous discoveries in the sky that ought to have pleased all lovers of true science. Yet it had scarcely been printed when men arose on all sides who envied the praises due to the discoveries thus made, and those were not lacking who merely to contradict what I said did not scruple (*p.* 2) to cast doubt upon things they had seen at will again and again with their own eyes. My Lord the Most Serene Grand Duke Cosimo II, of glorious memory, once ordered me to write my opinions on the causes of things floating or sinking in water, and in order to comply with this command I set down upon paper everything beyond the teachings of Archimedes

that occurred to me, which perhaps is as much as may be truly said about the facts of this matter. And behold! immediately the whole press was filled with attacks upon my *Discourse*. My opinions were contradicted without the least regard for the fact that what I had set forth was supported and proved by geometrical demonstrations, and such is the strength of men's passions that they failed to notice that the contradiction of geometry is the bald denial of truth. How many men, and under what disguises, combatted my *Letters on the Solar Spots*! The material contained therein, which should have opened to the mind's eye so much room for admirable speculation, was completely scorned and derided by many who either disbelieved it or little appreciated it; others, not wanting to agree with my conceptions, advanced ridiculous and impossible opinions against me; and some, conquered and convinced by my reasons, attempted to rob me of that glory which was mine by pretending not to have seen my writings and subsequently trying to make themselves the original discoverers of such impressive marvels. I say nothing of some of my private discussions, proofs, and propositions (many of them not published by me) having been seriously impugned or deprecated as worthless; yet even these have sometimes been chanced upon by others who have then exerted themselves with admirable adroitness to appropriate these honors as inventions of their own ingenuity.

Of such usurpers I might name not a few, but I shall pass them over now in silence, as it is customary for first offenses to receive less severe punishment than subsequent ones. But I shall not remain silent any longer about a second offender who has too audaciously tried to do to me the very same thing which he did many years ago by appropriating the invention of my geometric compass despite the fact that (*p.* 3) I had many years previously shown it and discussed it before a large number of gentlemen and had eventually publicized it in print. May I be pardoned this time if, against my nature, my habit, and my present intentions, I show resentment and cry out, perhaps with too much bitterness, about a thing which I have kept to myself these many years. I speak of Simon Mayr of Guntzenhausen;³ he it was in Padua, where I resided at the time, who set forth in Latin the use of the said compass of mine and, appropriating

it to himself, had one of his pupils print this under his name. Forthwith, perhaps to escape punishment, he departed immediately for his native land, leaving his pupil in the lurch as the saying goes; and against the latter, in the absence of Simon Mayr, I was obliged to proceed in the manner which is set forth in the *Defense* which I then wrote and published.⁴ Four years after the publication of my *Starry Messenger*, this same fellow, desiring as usual to ornament himself with the labors of others, did not blush to make himself author of the things I had discovered and printed in that work. Publishing under the title of *The World of Jupiter*, he had the temerity to claim that he had observed the Medicean planets which revolve about Jupiter before I had done so. But because it rarely happens that truth allows herself to be suppressed by falsehood, you may see how he himself, through his carelessness and lack of understanding, gives me in that very work of his the means of convicting him by irrefutable testimony and revealing unmistakably his error, showing not only that he did not observe the said stars before me but even that he did not certainly see them until two years afterward; and I say moreover that it may be affirmed very probably that he never observed them at all.

Although from many places in his book I could draw the most evident proofs of what I say, I shall adduce but a single one, reserving the rest for some other occasion in order not to become excessively diffuse and be distracted from my main purpose. In the second part of his *World of Jupiter*, under the sixth phenomenon that he considers, Simon Mayr writes of having diligently observed that the four satellites of Jupiter are never found in a straight line parallel to the ecliptic except at their greatest elongations from Jupiter, and (p. 4) that when they are outside of these they always tilt with a noticeable deviation from the said line. He says they always tilt northward from this when they are nearest us in their orbits, and conversely slope toward the south when farthest away. To preserve this appearance, their orbits would be inclined to the plane of the ecliptic toward the south in the upper regions and toward the north in the lower. Now this doctrine of his is full of fallacies which openly show and testify to his fraud.

In the first place, it is not true that the four orbits of the satellites incline from the plane of the ecliptic; rather, they are always parallel to it.⁵ Second, it is not true that they are never exactly mutually aligned except when they are located at their maximum elongations from Jupiter; indeed, it sometimes happens that any distances—maximum, mean, or minimum—they are seen in a perfectly straight line and fall in with one another; also, when two of them have opposite movements and are very close to Jupiter, they meet so exactly that the two appear one. And finally, it is false that when they [apparently] slope from the plane of the ecliptic they always tilt toward the south when in the superior half of their orbits and toward the north when in the inferior. Rather, at some times only do they have their tilts in this fashion, while at other times they tilt the other way; that is, toward the north when they are in the upper semicircle and toward the south when in the lower. But Simon Mayr, from not having understood or observed this affair, has inadvertently disclosed his error. Now, the facts are as follows.

The four orbits of the Medicean planets are always parallel to the plane of the ecliptic, and, since we are located in that same plane, it happens that whenever Jupiter has zero latitude (lying also in the ecliptic), the movements of these stars appear to us to be made along the same straight line, and their conjunctions at any place will always be made bodily and without deviation. But when Jupiter lies outside the plane of the ecliptic and its latitude is (*p.* 5) north from that plane, the four orbits of the satellites remaining parallel to the ecliptic, those parts of them which are farthest from us (we being always in the plane of the ecliptic) show themselves as tilted to the south with respect to the closer parts of these orbits, which to us appear more northerly. Conversely, when Jupiter's latitude is southerly, the farther parts of the same little orbits seem to us more northerly than the closer parts, and thus the tilting of the stars is seen in the opposite manner when Jupiter's latitude is northerly from what happens when Jupiter is southerly. That is, in the former case, they will be seen to tilt toward the south when they are in the upper parts of their orbits, and toward the north in the lower parts, but in the latter case they will tilt the contrary way; to wit, toward the north

in the upper half and toward the south in the lower. And this tilt will be greater or less accordingly as the latitude of Jupiter is greater or less.

Now Simon Mayr writes of having observed that the said four stars always slope toward the south when in the upper half of their orbits; hence his observations were made at a time when Jupiter had a northerly latitude. But when I made my first observations, Jupiter was southerly, and it remained so for a long time, nor did it become northerly until after more than two years. Thus the latitudes of the four stars could not have appeared as Simon writes of them until more than two years later, so if he ever saw and observed them it was not until two years after me.

Behold him, then, already convicted by his own depositions of lying about having made such observations before me. But I shall go further and say that it may be much more probably believed that he never made them at all, inasmuch as he affirms that he has not observed or seen them mutually arranged in a precise straight line except when they were at their greatest elongations from Jupiter. The truth is that for four months together, from the middle of February to the middle of June, 1611, in which time Jupiter had little or no latitude, the disposition of these four stars was always along a straight line in all their positions.

Next, notice the craft with which he tries to show himself prior (*p.* 6) to me. I wrote in my *Starry Messenger* of having made my first observation on the seventh of January, 1610, continuing then with others on the succeeding nights. Along comes Simon Mayr, and, appropriating my very observations, he prints in the title of his book and again in the opening part of the work that he had already made his observations in the year 1609, trying to give people the idea that he was first. Now the earliest observation that he produces as made by him is the second one made by me; yet he announces it as made in the year 1609. What he neglects to mention to the reader is that since he is outside our church and has not accepted the Gregorian calendar, the seventh day of January of 1610 for us Catholics is the same as the twenty-eighth day of December of 1609 for those heretics. So much for the priority of his pretended

observations. He also attributes to himself falsely the discovery of their periodic movements, found by me after long vigils and wearying labors, and revealed in my *Solar Letters* and again in the treatise which I published on things placed in water. That my book was seen by the said Simon may be clearly deduced from his book, and from mine he undoubtedly extracted those movements.

But I find that I have allowed myself to run on in too long a digression beyond what was required by the present occasion. Therefore, getting back to the discussion we have begun, I shall go on to say that after such very clear proofs there remained to me no longer any room for doubt of the ill feeling and stubborn opposition against my works. I had considered remaining perfectly silent about it, in order to save myself further occasion for the unpleasant sensation of being a target for such frequent sharpshooting, as well as to remove from others any matter capable of exciting such reprehensible talents. Certainly I have not lacked opportunities to put out other works of mine, perhaps no less surprising in the philosophical schools and no less consequential in science than those hitherto published. But the reasons recited above were so effective that I was content merely with the opinions and judgment of some gentlemen, my real and sincere friends, to whom I communicated my thoughts and in discussion with whom (*p.* 7) I have enjoyed that pleasure which accompanies the privilege of imparting what the mind gradually brings forth, at the same time avoiding any renewal of those stings which I had previously felt with so much vexation. These gentlemen, my friends, demonstrating in no small way their approval of my ideas, have managed with various reasons to change my mind about the resolutions thus made. In the first place, they tried to persuade me not to be concerned about these obstinate attacks, saying that in the end these would in fact rebound upon their authors and would render my arguments more vivid and attractive, becoming a clear proof of the uncommon nature of my compositions. They pointed out to me the common maxim that vulgarity and mediocrity receive little or no consideration and are left behind; the mind of man turns only on that which reveals marvels and transcendent things, and it is this which in turn gives rise in ill-tempered minds to envy and defamation.

These and similar arguments, coming to me upon the authority of those gentlemen, nearly destroyed my resolve to write no more. Yet my desire to live tranquilly and without such disputes prevailed, and thus fixed in my purpose I believed I had silenced in this way all the tongues which previously had shown such eagerness to contradict me.

But it was in vain that I had reached this frame of mind, nor have I been able by remaining silent to avoid my persistent fate of having always to concern myself with those who write against me and quarrel with me. It was useless for me to forbear from speaking, for those people who are so anxious to make trouble for me have had recourse to making mine the writings of other men. In thus stirring up a fierce fight against me, they have been willing to do something which in my opinion never occurs without clearly indicating a spirit impassioned beyond all reason. Surely Sig. Mario Guiducci, in carrying out the requirements of his office, should have been allowed to lecture in his Academy and then to publish his *Discourse on Comets*, without Lothario Sarsi, a completely unknown outsider, having jumped upon me for this, and, without the slightest regard for that fine man, (*p.* 8) having made me the author of this *Discourse*, in which I had no part beyond the regard and honor done me by Sig. Guiducci in his concurrence with my opinions—an agreement he had expressed in the said discussions held with those gentlemen, my friends, with whom he was pleased to meet frequently. And even if the entire *Discourse on Comets* had been the work of my hand (an idea which could never occur to anyone wherever Sig. Mario is known), what kind of behavior would this be, for Sarsi to reveal my face and unmask me so zealously after I had shown my wish to remain incognito? Upon this action, finding myself constrained by such unexpected treatment of so unusual a kind, I come to break my previously established resolve not to publish my writings any more. I shall do what I can to see that the impropriety of his action shall not escape unnoticed, in the hope of discouraging anyone who will not let sleeping dogs lie, as the saying goes, and who stirs up trouble with men who would hold their peace.

Now I am aware that this name Lothario Sarsi, which has never

before been heard in the world, serves as a mask for someone who wants to remain unknown. It is not my place to make trouble for another man by tearing off his mask in the fashion of Sarsi himself, as this does not seem to me an action deserving emulation or one which can in any way bestow aid or favor upon my writing. Indeed, I think that to deal with him as an unknown person will be to gain a wider field in which to make my arguments plainer and explain my ideas more freely. I am taking into account the fact that many times those who go masked are either low persons who try in this guise to gain esteem among gentlemen and scholars and to utilize for some purpose of their own the dignity which attends nobility. Again, they may be gentlemen who, thus unrecognized, lay aside the respectful decorum accorded to their rank and make free use (the custom in many cities of Italy) of speaking openly about anything with all and sundry, getting as much pleasure as anyone can in this raillery and contention devoid of all respect. And I believe that it must be one of these latter (*p. 9*) whom this mask of Lothario Sarsi hides, for if he were one of the former it would have been in poor taste for him thus to have tried to impose upon the public. Consequently, I believe in addition that just as he, thus unknown, has allowed himself to say some things against me which to my face he would perhaps not say, so it ought not to be taken amiss if I, availing myself of the privilege accorded against masqueraders, deal with him quite freely. Nor should he or anyone else suppose me to be weighing my every word when perhaps I may speak rather more frankly than will please him.

I wanted your Excellency above all to be the first witness of this reply of mine, for, as a most understanding man of the noblest quality and devoid of any partiality of spirit, you will rightly understand my cause and will not fail to reprove the audacity of those who, not ignorantly but in passion (for the ignorant matter but little), try evilly to distort my arguments before the common people who do not comprehend. And though upon first reading Sarsi's essay it was my intention to include my answer in a simple letter to your Excellency, when I came to the task I found that the matters contained in his essay which required some attention multiplied under my hands, and I have been compelled to pass far beyond the bounds of a letter.

I have nevertheless adhered to my original decision to address myself to your Excellency, however extended my reply may be. It has been my thought to call it *The Assayer*, restricting myself to the metaphor employed by Sarsi; but since it seemed to me that he used too crude a steelyard in his weighing of Sig. Guiducci's propositions, I have elected to employ an assayer's balance precise enough to detect less than the sixtieth part of one grain. Using this with all possible care and neglecting not a single proposition set forth by him, I shall put every one to the test. I am going to distinguish these weighings, denoting each by a number, and thus if they are ever seen by Sarsi and he wishes to respond he may the more easily do so (*p.* 10) without omitting anything.

Coming at last to particular considerations, perhaps it will be well for me to say something about the title page of the work, in order that nothing may escape weighing. Sig. Lothario Sarsi calls this *The Astronomical and Philosophical Balance*. After the title, in the subjoined epigram, he gives the motive which induced him so to name it, which is that this comet mysteriously hinted to him by originating and appearing in the sign of Libra that he should balance and weigh on accurate scales the things contained in the treatise on comets published by Sig. Mario Guiducci. I note that Sarsi confidently begins altering things to suit his purposes at the very first opportunity, a style maintained thereafter throughout his essay. He chanced to think up this pun on the correspondence between his balance and the celestial Balance, and, since it seemed to him that his metaphor would be considerably enhanced if the comet had first appeared in Libra, he freely asserted that it arose there. He felt no concern about contradicting the truth, nor even in a certain way himself, for he contradicts his own master, who on page 10 of his *Disputation* draws this conclusion: 'But finally, whatever of these dates saw the first light of the comet, it is Scorpio which was its true native land.' And a dozen lines farther on, '... it was born in Scorpio, that is, in the principal house of Mars'; and a little later, 'I, in so far as I am able, have sought its native land which, in agreement with all, I declare to have been Scorpio.' Hence, much more appropriately (and more truthfully, when we consider what Sarsi has actually written), he

might have entitled his work, 'The Astronomical and Philosophical Scorpion'—that constellation which our sovereign poet Dante called the

. . . figure of that chilly animal

Which pricks and stings the people with its tail;⁶

and truly there is no lack of stings here for me. Much worse ones too than those of scorpions; for the latter, as friends of mankind, do not injure unless first offended and provoked, whereas this fellow would bite (*p.* 11) me, who never so much as thought of offending him. But as luck will have it, I know the antidote and speedy remedy for such stings, and I shall crush the scorpion and rub him on the wounds where the venom will be reabsorbed by its proper body and leave me free and sound.

I

Now we come to the treatise, and the first weighing shall be performed upon some words of the proem; namely, the sentence beginning, 'One person, however . . .' But let us set forth this proem in full, precisely as in the printed text of which we do not wish to leave out one iota:

FIRST WEIGHING

OF THOSE THINGS WHICH WERE OPPOSED

BY GALILEO TO OUR *Disputation*

'Last year when . . . and rather sharply.' (*p.* 69, l. 4 to l. 23).

In these final words, he affirms that we have sharply criticized his master's *Disputation*. To this, I do not see at the moment that anything must be replied, inasmuch as his statement is absolutely false. Diligently searching in Sig. Mario's treatise, I have been unable to find any such place, and Sarsi does not cite one. But we shall have more to say about this later.

II

Next follows—and this is to be the second weighing—'At first, we were grieved . . . (*p.* 12) before just magistrates.' (*p.* 69, l. 24 to l. 31).

Here he says that from the first he has felt pain that the *Disputation* displeased me, but he adds that he has since been consoled by seeing even Aristotle, Tycho, and others criticized with the same severity, so there was no need of additional defenses for those accused when they took the same side as the most eminent minds; their very cause would speak for itself should they remain silent, and would defend itself in the minds of the just. From these words, I gather that to Sarsi's mind those who impugn authors of the most eminent intelligence should be held in little esteem, for he takes no account of the fact that some people do spring to the defense of those who are attacked, even when the bare authority of the latter suffices to maintain their credit in the eyes of the learned. Here, I should like your Excellency to note how Sarsi, whether by choice or through inadvertence, muddies the reputation of his teacher, Father Grassi, whose chief aim in his *Problem* was to attack Aristotle's opinion about comets. This is clearly seen in his book, and Sarsi himself here repeats and confirms it on page 73. So if the assailants of great men should be ignored, then Father Grassi should be also. Now we have not only not ignored him but we have given him the same consideration as the most eminent minds, placing him in a class with them, so in this regard he is as much exalted by us as he is debased by his pupil. I do not see what Sarsi can offer in his own excuse unless it be that his meaning was that one ought to ignore the vulgar among the opponents of eminent geniuses, while on the other hand one should esteem those who happen also to be very eminent, and that among the latter he meant to include his master while the rest of us fell among the common herd. But then in this (p. 13) regard, we are reproached for doing that which it was suitable for his master to do.

III

Next follows, and this is to be the third weighing: 'But since it appeared . . . brief consideration to both of them . . .' (p. 69, l. 31 to p. 70, l. 2).

The sense of these words, continuing that of the above, seems to me to imply that while one need not take account of people who contradict the most eminent geniuses but may rather pass them over in

silence, still if they must be answered then this task should be given to inferior persons rather than otherwise. Thus in our case the wisest men have deemed it proper that not Father Grassi or someone of equal importance should reply to Galileo, but *saltem aliquis*—just anyone. Here I have nothing to say or reply except that knowing and confessing my lowly estate, I bow my head to the sentence of such men. Indeed, I marvel not a little that Sarsi should of his own volition choose to be that mere anybody who would tuck up his sleeves and welcome a task which, in the judgment of the wisest men (and himself), should not be given to any but the meanest servant. Nor can I well understand, since it is the natural instinct of every man to attribute to himself much merit rather than little, why this Sarsi should now lower his position so greatly as to be induced to pass himself off as a mere anybody. The improbability of this has aroused in me some suspicion, and in the end has made me think that in these words there may be a slight misprint, and that where it says that, ‘somebody should consider the argument of Galileo more carefully,’ it should read, ‘somehow the argument of Galileo should be considered more carefully.’⁷ This reading I consider to be the correct one, inasmuch as it fits in with all the rest of the treatise while the other reading accords but ill with the regard which I still like to think Sarsi entertains for himself. Your Excellency will see, as you go along examining his essay with me, how true it is—as I say here—that of the things (*p.* 14) written by Sig. Mario he has merely made an examination ‘somehow,’ or even just ‘anyhow,’ attending to some minutiae which have little bearing upon the main topic and passing over the chief conclusions and reasons. He has done this because in his heart he knew that he could not fail to approve the latter and confess them to be true, and this would have run counter to his main purpose of only condemning and refuting them—as he himself writes on page 105 in these words: ‘These things which have been said regarding the opinion of Galileo are among those which immediately concern the comet. For he, who in a very long disputation has presented his beliefs in many and confused words, objects to our saying more and prevents us from presenting many refutations against his position. For how can we divine and

refute what he does not offer?' In these words, I note two things in addition to his exclusive intention to refute. One is that he pretends not to have understood many things because of their being (he says) obscurely written, though these turn out to be the things for which no counterattack was discovered. The other is that he says he has been unable to confute things which I have not set forth and which he has not been able to divine. Yet your Excellency will see that as a matter of fact most of the things he undertakes to refute are not set forth by me but are divined (or better, let us say imagined) by him.

IV

'... hoping that thereby ... (*p.* 15) ... than with the consul' (*p.* 70, l. 2 to l. 23).

In all this remainder of the proem, I note first that Sarsi claims to have done with his impugnations something welcome to many people; and perhaps he may have done so for people who happen not to have read Sig. Mario's essay. But if they have taken Sarsi's word for it, given privately and confidentially, this will have departed a good deal from what is written, since in his own printed and published work he does not refrain from setting forth as written by Sig. Mario a great many things that were never in his essay nor even in our imaginations. He then adds that he wants to abstain from words that would indicate an exasperated and angry spirit rather than a scientific one. We shall see as we go along how well he has adhered to this; for the present I merely take note of his confession of being inwardly exasperated and angry, for, if he were not, his giving expression to this intention of abstaining would have been I shall not say erroneous, but superfluous; abstinence has no place where there is not an inclination and a disposition.

As to what he writes next, of wishing to serve as an intermediary in relating things concerning my latest discoveries which he has heard from Father Horatio Grassi, his teacher, I certainly do not believe any such thing. I am sure that the said Father would never have spoken or thought or willingly have seen Sarsi write such fantastic things, far removed in every respect from the doctrine taught

in Father Grassi's college, as I hope I shall make clearly recognized. Indeed, without proceeding a step further, let me ask who on earth, merely knowing the prudence of those Fathers, could be induced to believe that one of (p. 16) them would have publicly written that in private letters to friends at Rome I had openly declared myself the author of Sig. Mario's treatise? This is not true, and if it were its publication could not fail to reveal some pleasure in sowing seeds of dissension among close friends. And what sort of behavior is it to take the liberty of printing the private statements of others? It is best that your Excellency should be informed of the truth of this matter. During the whole time that the comet was being seen, I was confined by illness to my bed, where I was often visited by friends. Discussions frequently took place there, during which I had occasion to voice some thoughts of mine which raised considerable doubt about the doctrines previously held on this subject. Among my other friends, Sig. Mario was oftentimes present, and one day he told me that he had considered speaking on comets before the Academy, and that if it pleased me he would there cite what he had learned from me, along with things which he had gathered from other authors or had thought of himself, inasmuch as I was in no condition to write. The courtesy thus offered I regarded as my good fortune, and I not only accepted but I thanked him and expressed my indebtedness to him. Meanwhile from Rome and other places, insistent requests to know whether I had anything to say on this matter came from other friends and patrons who perhaps did not know of my indisposition. To these I replied that I merely had several questions which I was unable to set down on paper because of my infirmity, but that I hoped these ideas and doubts of mine might shortly be seen inserted in a discourse by a gentleman who was a friend of mine and who in order to do me honor had taken the trouble to collect these and insert them in a treatise of his. That is all that was given out by me, as has also been written in several places by Sig. Mario himself. Hence there was no call for Sarsi to embellish the truth by introducing my letters, nor should he assign to Sig. Mario such a small part in the *Discourse* (with which he had much more to do than I) as to pass him off as a copyist.

But since he is pleased to have it this way, let it be so; and meanwhile let Sig. Mario (*p.* 17) accept my defense of his treatise in return for the honor he has done me.

V

Returning to the treatise, your Excellency may reread the following words: 'First of all, . . . by false arguments.' (*p.* 70, l. 24 to *p.* 71, l. 13).

Concerning these particular writings, I say in the first place that I have never lamented over being maltreated in the *Disputation* of Father Grassi, in which I am sure that his Reverence never thought to offend me. Yet if, for the sake of argument, I had held the view that Father Grassi intended to include me in criticizing those who held a low opinion of the argument taken from the slight enlargement of the comet, Sarsi should not think that on that account there was cause for me to take offense and to lament. Perhaps there might have been, had my position been false, and had it been publicly revealed to be so. But since my statement was quite correct, (*p.* 18) and the other one was false, the multitude of my opponents (and especially of those as worthy as Father Grassi) could only cause me pleasure and not pain, inasmuch as there is more delight in victory over a valiant and numerous host than over few and feeble opponents.

News arrived from many parts of Europe, going (as Sarsi writes) to his master, but some people in passing this way enabled us also to hear how generally all the most celebrated astronomers were taking a great interest in that argument. Nor were there lacking even in our region and in our very city men who held the above opinion. As soon as I learned of this, I let it be very plainly understood that I deemed the reasoning quite vain. Many people made a joke of this, the more so when there appeared on their side authoritative support and confirmation from the mathematician of the Collegio Romano, nor shall I deny that this caused me a bit of trouble. Finding myself placed under the necessity of defending my statement against so many opponents (who, strengthened by such assistance, rose up against me still more imperiously), I did not see any way to contradict them without including Father Grassi too. So it was through a necessary though fortuitous circumstance and not by choice that I pointed my

opposition in the direction which I least wished to. But I never pretended, as Sarsi says, that my opinion was certain to be swiftly carried by the winds to Rome as usually happens with the words of great and celebrated persons; this really far exceeds the bounds of my ambition. It is certainly true that reading the *Balance* has made me marvel somewhat that what I did say should not have penetrated to Sarsi's ears; is it not surprising that so many things have been reported to him which I never said nor even thought (of which a large number are set down in his discourse), while not a single syllable has reached him of other things that I have said over and over again? But perhaps the winds that blow the clouds and those chimeras and monsters which tumultuously take shape therein had not the strength to carry solid and weighty things.

From the words (*p.* 19) which come next, I seem to gather that Sarsi would attribute to me a great defect in not having equaled the courtesy of the Fathers at the Collegio by returning the compliment they had paid to me in public lectures upon my celestial discoveries and my ideas about bodies placed in water. And what was I supposed to do? To praise and approve the *Disputation* of Father Grassi, replies Sarsi. But since dealings between you and me are to balance, Sig. Sarsi, and must be businesslike, as they say, then I ask you whether those reverend Fathers consider my ideas to be true or take them to be false? If they recognize my ideas as true and praise them as such, then it is at exorbitant interest that you demand payment of what was lent when you expect me to extol with equal praise things I know to be false. But if they regard my ideas as foolish and still praise them, then I may indeed thank them for their urbanity, but I should have been much more grateful if they had lifted me out of my error and had shown me the truth, for I regard much more highly the utility of accurate correction than I do mere pomp and ceremony. And since I think that the same is true for all good philosophers, I feel no obligation in either case.

Perhaps you will tell me that I should have remained silent. To this I reply first that Sig. Mario and I were already too deeply obligated to let our ideas be known before Father Grassi's essay was published, so that to have remained silent would have been to invite

upon ourselves contempt and almost universal derision. I add that I should have asked and perhaps even entreated Sig. Guiducci not to publish his *Discourse* if there had been in it anything prejudicial to the dignity of the famous Collegio or any of its professors. But if all the opinions opposed by us were held by others before they were by the mathematician and professor of the Collegio, I do not see how his Reverence's merely having lent his assent to them placed us under any obligation to ignore the truth and conceal it in favor of the perpetuation of an error. Hence, any criticism of being a poor logician falls upon Tycho and the others who have committed the equivocation in the argument. This equivocation was revealed by us not to single anyone out or to blame them, but merely to remove others from error and to make evident (*p.* 20) the truth, and I do not know that actions of this sort can ever be reasonably blamed.

Sarsi has, therefore, no reason for saying that I am guilty of contempt for the dignity of the Collegio Romano. Quite the contrary; for if Sarsi's voice does emanate from the Collegio, I have reason to suspect that my doctrine and my reputation have been in bad odor there not merely at present but all along, inasmuch as in this *Balance* none of my thoughts are approved. Nothing is to be read there except opposition, full of accusation and blame, and, if one may believe the rumors, there is in addition to what is written an open boast of power to annihilate everything of mine.⁸ But as I do not believe this, nor that any of these ideas exist at the Collegio, I must suppose that Sarsi's philosophy is such as to give him the right equally to praise or to blame, confirm or refute the same doctrines accordingly as benevolence or wrath may move him. In this he reminds me of a teacher of philosophy in my time at the University of Padua, who, being angry at one of his colleagues (as frequently happened), said that if the latter did not mend his ways he would send someone to spy secretly on the opinions he voiced in his lectures, and in revenge would always maintain the contrary.

VI

Now, your Excellency, read: 'But lest we waste time . . . (*p.* 21) which he considers unfortunate.' (*p.* 71, l. 14 to l. 36).

From what Sarsi writes here, I may guess that he has not read carefully enough Sig. Mario's *Discourse* or even that of Father Grassi, since from both the one and the other he adduces propositions which are not to be found in either. It is indeed true that he would have needed such things to have been written in order to get an opening through which he could somehow brand me a Copernican, so in their absence he has attempted to supply them himself.

First, in Sig. Mario's essay it is not thrown in Father Grassi's teeth and attributed to a deficiency on his part that he has sworn fealty to Tycho and followed his foolish fabrications in every detail. Here are the places referred to by Sarsi. On page 36, 'Later I shall turn my attention to the Professor of Mathematics at the Collegio Romano, who in a recently published tract appears to have subscribed to Tycho's every statement, and even added some further reasoning in confirmation of that opinion.' And the other place, on page 52: 'The Mathematician of the Collegio Romano has accepted the same hypothesis for this last comet as well; beyond the little which that author has written in support of Tycho's opinion, I am led to affirm this by seeing how much he concurs with Tycho's other fantasies throughout the remainder of the work.' Now you may see, your Excellency, whether anything is here attributed to deficiency and omission. Moreover, it is perfectly clear that nothing is under consideration in the whole work except phenomena relating to comets, about which Tycho wrote a large volume. To say that the Mathematician of the Collegio agrees with other fantasies of Tycho does not extend to other views than those which pertain (*p.* 22) to comets. So I do not see that this is an appropriate place to call into comparison with Tycho such men as Ptolemy and Copernicus, who never treat of hypotheses concerning comets.

Sarsi then says that in the writing of his master nothing is to be found in which he has followed Tycho except the demonstrations for finding the location of the comet, but this is not true, with all due respect to Sarsi. Rather, nothing is less to be found there than such demonstrations. God forbid that Father Grassi should have imitated Tycho in these without noticing that Tycho, in his manner of investigating the distance of a comet by observations made from

two different places on the earth, shows himself needful of attention to the very first elements of mathematics. And so that your Excellency may see that I do not say this without grounds, reconsider the demonstration which begins on page 123 of Tycho's treatise on the comet of 1577, which is in the last part of his *Progymnasmata*.⁹ In this, he wants to prove that the comet was not below the moon by means of comparing the observations made by himself at Uraniborg with those made by Thaddeus Hagek at Prague. Drawing the chord AB of the arc of the terrestrial globe passing through the two places named, and sighting from the point A at a fixed star at D, he assumes

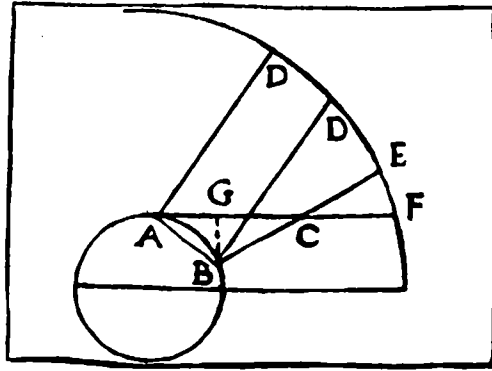


Fig. 16

the angle DAB to be a right angle, which is not even possible. The line AB being the chord of an arc of less than six degrees, as Tycho himself declares, it is necessary in order for the said angle to be a right angle that the fixed star D should be less than three degrees from the zenith of A. But this is false, since its minimum distance is more than forty-eight degrees, since according to Tycho himself the declination of the fixed star D, which is Aquila (or let us say Altair), is $7^{\circ}52'$ north, while the latitude of Uraniborg is $55^{\circ}54'$. He writes further that the same fixed star is seen from the two places A and B at the same place in the eighth sphere because the whole earth—let alone the small (*p.* 23) part AB—has no perceptible magnitude with respect to the immensity of the eighth sphere. Pardon me, Tycho; the largeness or smallness of the earth has nothing to do with the

case. Seeing the same star in the same place from all parts of the earth depends upon the star's truly being in the eighth sphere, and upon nothing else, in exactly the same way as the letters on this page will never change their apparent positions with respect to the page no matter how much you may change the position of the eye with which you look at it. But an object placed between the eye and the page will vary its apparent place with respect to the letters as the text is moved, so that the same letter is seen now to the right, now to the left, now higher, and now lower. It is in this way that the apparent places of the planets change in the stellar orb when seen from different parts of the earth, because that orb is very far beyond them. What the smallness of the earth does in this case is this: the planets more distant from us making a lesser change and the closer ones a greater, ultimately, for a very remote one, the size of the earth would not suffice to make its change perceptible.

Then when Tycho adds that what takes place is in obedience to the laws of arcs and chords, your Excellency may see how far he is from those laws and even from the first elements of geometry. He says that the two straight lines AD and BD are perpendicular to AB, which is impossible, since only the straight line that falls from the vertex is perpendicular to the tangent and its parallels, while these lines do not come from the vertex at all, nor is AB the tangent or parallel to it. Besides, he requires them parallel and then he says that they eventually meet at the center, whereas (in addition to the contradiction between meeting and being parallel) these lines if prolonged pass far from the center. And he finally concludes that coming from the center to the circumference at the points A and B, they are perpendiculars, which is impossible inasmuch as among the lines drawn from the center to all points of the chord AB, only that one which falls at its center is perpendicular to it, and those which fall at its extremities are the most inclined and oblique of all.

Thus your Excellency may see to what and (*p.* 24) to how many absurdities Sarsi would have lent his master's assent if what he had asserted in this statement were true; that is, that the latter had followed Tycho's reasoning and method of proof in seeking the location of the comet. Sarsi himself may see also how much better than

he, and, even without employing astrology or telescopes, I have penetrated I shall not say into the inner sense of his mind, for to spy that out I have neither eyes nor ears, but the sense of his writing, which is so clear and manifest that there is no need for the lynx eyes so gallantly brought in by Sarsi as a jest at our Academy. Since your Excellency and other princes and great gentlemen are involved with me in this jest, I shall profit from Sarsi's instruction and shall give little heed to his witticisms, taking shelter under your shadow; or, to put it better, I shall illuminate my shadow with your splendor.

But getting back to the point, you see how once more he will have it that I deemed it a great defect on the part of Father Grassi to have adhered to Tycho's doctrine, and he asks resentfully: 'Whom then should he have followed? Ptolemy, whose doctrine has been revealed to be false by the recent observations of Mars? Or perhaps Copernicus? But he must rather be rejected by everyone, in view of the ultimate condemnation of his hypothesis.' Here I note several things, and I reply first that it is quite false that I have ever criticized anyone for following Tycho, even though I might very reasonably have done so, as will indeed at last become clear to his adherents from the *Anti-Tycho* of the distinguished Chiaramonti.¹⁰ Hence, so far as this remark is concerned, Sarsi is very wide of the mark. Even more irrelevant is his introduction of Ptolemy and Copernicus, who are never to be found writing a word relative to the distances, magnitudes, movements, and theory of the comets, which (and which alone) are here under consideration. He might with as much reason have brought in Sophocles, Bartoli,¹¹ or Livy.

It seems to me that I discern in Sarsi a firm belief that in philosophizing it is essential to support oneself upon the opinion of some celebrated author, as if when our minds are not wedded to the reasoning of some other person they ought to remain completely barren and sterile. (p. 25) Possibly he thinks that philosophy is a book of fiction created by some man, like the *Iliad* or *Orlando Furioso*—books in which the least important thing is whether what is written in them is true. Well, Sig. Sarsi, that is not the way matters stand. Philosophy is written in this grand book—I mean the universe—which stands continually open to our gaze, but it cannot be understood

unless one first learns to comprehend the language and interpret the characters in which it is written. It is written in the language of mathematics, and its characters are triangles, circles, and other geometrical figures, without which it is humanly impossible to understand a single word of it; without these, one is wandering about in a dark labyrinth. Sarsi seems to think that our intellects should be enslaved to that of some other man (I shall disregard the fact that in thus making everyone, including himself, an imitator, he will praise in himself what he has blamed in Sig. Mario), and that in the contemplation of the celestial motions one should adhere to somebody else.

But even allowing this assumption, I do not see why he selects Tycho and sets him before Ptolemy and Nicholas Copernicus, for both of these have constructed and followed out to the end, with the greatest skill, complete systems of the universe. This I cannot see that Tycho has done, unless indeed Sarsi thinks it is enough for Tycho to have rejected the other two systems and promised us a new one, though failing afterward to carry out his promise. I wish that besides having convicted the other two of error, Sarsi had recognized something of Tycho's; for, as to the system of Ptolemy, neither Tycho nor other astronomers, nor even Copernicus, could clearly refute it, inasmuch as a most important argument taken from the movements of Mars and Venus stood always in their way. Since the disc of Venus shows itself in its two conjunctions and elongations from the sun to vary but little in magnitude, and that of Mars at perigee is scarcely three or four times larger than it is at apogee, one could never be persuaded that the former really shows itself forty and the latter sixty times as large in one position as in the other, as required if their revolutions were about the sun in accordance with the Copernicus system. Yet I have demonstrated this to be true and to be apparent to our senses, and (*p.* 26) by means of a fine telescope I have made it palpably evident to anyone who wished to look. Then as to the Copernican hypothesis, had not the most sovereign wisdom removed us Catholics from error for our own good and illuminated our blindness, I do not believe that this grace and benefit could have been derived from the reasons and experiences embraced by Tycho.

Thus the two systems being surely false and Tycho's being null, Sarsi should not reprehend me for desiring with Seneca the true constitution of the universe. The request is a large one, and though such knowledge is greatly desired by me, I do not (as Sarsi writes) deplore the misery and misfortune of our age amidst lamentations and tears, nor is there any trace of such a complaint to be found anywhere in Sig. Mario's essay. But Sarsi, under the necessity of camouflaging and supporting some ideas of his which he wished to explain, went about preparing and supplying attacks himself which have not been made against him by anyone else. And even if I did deplore this misfortune of ours, I fail to see how Sarsi could properly say that my complaints had been voiced in vain because I had no means of power to remove the distress. It seems to me that this is precisely what would give me cause for regret, and that on the other hand there would be no occasion for mourning if I were able to remove the misfortune.

VII

Now, at length, let your Excellency read this: 'Although . . . (p. 27) . . . treated in kind' (p. 72, l. 1 to l. 26).

Hurrying over in a few words what is written here, I say that Sig. Mario and I are not so austere as to be revolted by jokes or poetic graces. As evidence of this there are the various charming things inserted very airily by Father Grassi in his treatise of which Sig. Mario has not breathed a word in reproach. It is indeed delightful to read of the birth, the cradle, the dwellings, and the funeral of the comet, and of its having been lighted to illuminate the supper meeting of the sun with Mercury. It has given us no offense that these lamps were lighted twenty days after the supper; still less, that we know that where the sun is, candles are superfluous and useless, and that this was no supper but merely luncheon; that is, a feast by daytime and not by night, which time of day is unknown to the sun. All these things were passed by without question because, spoken as they were, they left nothing to be desired as to the truth of the concepts underlying these jests which was in itself so well known and obvious that it needed no other and more profound demonstration.

But now, in a fundamental and very difficult question, Sarsi tries to persuade me, all joking aside, that there exists in nature a particular celestial orb for comets, and whereas Tycho was unable to untangle himself in his own explanation of the irregularity of his comet's apparent motion, Sarsi expects my mind to be satisfied and set at rest by a little poetic flower which is not followed by any fruit at all. This is what Sig. Mario rejects when he quite correctly and truly says that nature takes no delight in poetry. This proposition is quite true, even though Sarsi shows that he does not (*p.* 28) believe it, and pretends not to recognize either nature or poetry and not to know that fables and fictions are in a sense essential to poetry, which could not exist without them, while any sort of falsehood is so abhorrent to nature that it is no more possible to find one in her than to find darkness in light.

But it is time for us to get on to more important matters; therefore let your Excellency read the following.

VIII

'Now I come . . . the same argument?' (*p.* 72, l. 27 to *p.* 73, l. 13).

In order to recognize just how important are the things written here, it will suffice to review briefly what it is that Sig. Mario says and what is raised against him here. Sig. Mario writes in general: 'Those who wish to determine the place of the comet by means of parallaxes need first to establish that it is a fixed and real thing and not a vague appearance, inasmuch as the reasoning from parallax is indeed conclusive for real objects but not for apparent ones.' This he exemplifies in many ways, and then he adds that absence of parallax renders incompatible the two (*p.* 29) propositions of Aristotle that a comet is a fire, which is a real thing, and that it is located in the air very near the earth. Here Sarsi rises up and says, 'All very well, but this is beside our purpose. For we are arguing against Aristotle, and it would be a waste of time to prove that the comet was not an appearance when we agree with him to the extent of its being a real thing. And taking it as a real thing, our argument from parallax is conclusive. And,' he adds, 'our adversary himself does

not employ any more valid an argument against Aristotle; now if he does not, then why may not we freely use it in the same cause?’

Well, here I do not know what Sarsi is asserting nor in what he thinks he is impugning Sig. Mario, seeing that both are saying the same thing; that is, that reasoning from parallax does not hold for mere appearances but holds very well for real objects, and consequently holds against Aristotle when he would have the comet be a real thing. If Sarsi will permit me to be frank, nothing more need be said here than that he has tried to dress up Sig. Mario’s remark so as to dazzle the reader’s vision and leave him with the idea that Sig. Mario had made a blunder. In order for Sarsi’s objection to have any force, it would be necessary for Sig. Mario, instead of saying generally to everyone: ‘Whoever wishes the argument from parallax to bear upon comets must first prove that comets are real things,’ to have said specifically: ‘If Father Grassi wants the argument from parallax to operate against Aristotle, who takes comets to be real things and not apparent only, then he must first prove that comets are real things and not apparent ones.’ In that way Sig. Mario would truly have been guilty of a bad error, as Sarsi wants to make it appear that he was. But Sig. Mario never wrote or thought any such foolishness.

IX

‘Furthermore the opinions . . . (*p.* 30) . . . drives and draws it.’ (*p.* 73, l. 14 to *p.* 74, l. 19). (*p.* 31)

Once more trying to show that the general question advanced by Sig. Mario was vain and superfluous, Sarsi says that no author worth considering, ancient or modern, ever supposed a comet capable of being a mere appearance; and for that reason that his master, who was disputing only with such men and who aspired to victory over them alone, never had any need to remove comets from the company of mere images. To this I respond that in the first place Sarsi might for this very reason ignore Sig. Mario and me also, since we are outside the number of those ancient and modern authors against whom his master was disputing. We intended to speak only with those men, whether ancient or modern, who try in all their studies to

investigate some truth in nature, completely avoiding those who merely engage ostentatiously in noisy contests in order to be popularly judged and pompously praised not as discoverers of truth but merely as victors over others. Nor need he have gone about so anxiously overthrowing something which is not prejudicial either to himself or to his master.

In the second place, he should have considered that a man may be much more readily excused who expresses a belief and happens not to think of some detail pertaining to it (especially one which had not occurred to a thousand others who have expressed the same belief) than may one who thinks up something foolish and useless on the subject. Thus he could, and should, confess that it had never passed through his master's mind or occurred to any of his predecessors that a comet might be an appearance, rather than call the consideration which occurred to us foolish. For the one course, without giving any offense to his master, would have indicated candor and frankness, while the other, necessarily damaging my reputation (if he had succeeded in it), betrays instead a spirit moved by some passion. Sig. Mario, in the hope of doing something profitable and welcome to men studious of truth, proposed with all humility that henceforth it would be good to consider the essence of the comet, and whether it might not be a mere appearance rather than a real thing. (*p.* 32) He did not criticize Father Grassi or anyone else who had not previously done this. Sarsi now rises up and passionately strives to prove the suggestion to be beside the point and obviously wrong. Still, in order to be prepared for anything (lest it should appear worthy of some consideration), he robs me of any possible credit by calling it an old notion of Cardan and Telesio, disparaged by his master as a fantasy of feeble philosophers who have no following. Under this pretense, and without any shame for his disrespect, he robs and strips them of their reputations in order to cover up a very small blemish in his master. Well, Sarsi, though you represent yourself as a pupil of these venerable Fathers in natural philosophy, make no such claim as to moral philosophy, for you will never be credited. I have not seen what Cardan and Telesio wrote, but from other indications which will presently appear I may readily guess that Sarsi did

not catch their meaning. At the moment, I cannot neglect, for his benefit and in their defense, to show the improbability of his deduction that their science was poor from the small number of followers they have.

Sarsi perhaps believes that all the hosts of good philosophers may be enclosed within walls of some sort. I believe, Sarsi, that they fly, and that they fly alone like eagles, and not like starlings. It is true that because eagles are scarce they are little seen and less heard, whereas birds that fly in flocks fill the sky with shrieks and cries wherever they settle, and befoul the earth beneath them. But if true philosophers are like eagles, and not like the phoenix instead, Sig. Sarsi, the crowd of fools who know nothing is infinite; many are those who know very little of philosophy, few, indeed, they who truly know some part of it, and only one knows all, for that is God. To say plainly what I am trying to hint, and dealing with science as a method of demonstration and human reasoning capable of pursuit by mankind, I hold that the more it shall partake of perfection, the smaller the number of conclusions (*p.* 33) it will promise to teach, and the fewer it will demonstrate; hence the less attractive it will be, and the smaller will be the number of its followers. On the other hand, magnificence of titles and grandeur and number of promises attract the natural curiosity of men and hold them perpetually involved in fallacies and chimeras, without ever offering them one single sample of the sharpness of demonstration, by which the taste might be awakened to the insipidity of its ordinary fare. Hence these will keep an infinite number of men occupied, and that man will be very fortunate who, led by some unusual inner light, shall be able to turn from the dark and confused labyrinths within which he might have gone forever wandering with the crowd and becoming ever more entangled. Therefore, in the matter of philosophy, I consider it not very sound to judge a man's opinion by the number of his followers. But though I think the number of followers of the best philosophy may be very small, I do not conclude conversely that therefore those opinions and doctrines are necessarily perfect which have few followers, for I know very well that some men may hold opinions so erroneous as to be rejected by everyone else. Now from which of

these two sources is derived the scarcity of followers of the two authors named by Sarsi as sterile and derelict, I do not know; I have not studied their works sufficiently to judge.

Returning to our subject, I say that it seems to me Sarsi is a trifle tardy in his attempt to persuade us that the reason his master made no mention of the idea that the comet might be a mere simulacrum to which the argument based on parallax does not apply was not that it had not occurred to him but that it had been dismissed by him as a foolish notion. This excuse comes late, I say, because his master had written in the *Problem*: 'Anything established between the firmament and the earth, if observed from different places, will correspond to different parts of the same firmament.'¹² This clearly shows that there had not occurred to him the matter of rainbows and haloes, mock suns, and other reflections that are not subject to that law, which he should have mentioned and excepted—especially since he himself, forsaking Aristotle, inclines toward Kepler's opinion that the comet (*p.* 34) might be a reflection.

But to continue, I see that Sarsi makes a great distinction between the head of the comet and its beard or curl. He holds that, as to the curl, this might truly be an optical illusion and an appearance (as was believed also by those Pythagoreans whom Aristotle mentions), but as to the head, he thinks that this must be a real thing and that nobody has ever thought otherwise. Now here, I should like a well-defined distinction between what Sarsi means by real and what he takes as apparent, and what his standard is for calling something real or apparent. If he calls the head real because it exists in a real substance and material, then I say that so is the curl real; for if anyone should take away those vapors in which our vision is reflected toward the sun, this would likewise remove the curl, just as removing the clouds takes away rainbows and haloes. And if he requires the curl to be unreal because it would not exist without the reflection of vision toward the sun, then I say that the same would follow for the head. The head no less than the curl is nothing but a reflection of rays in some material, whatever that material may be, and though as reflections they are merely appearances, as to the material they are real. If Sarsi admits that a change of position on the part of the

observer makes (or may make) a change in the place where the curl is generated in the material, I say that this may follow in the same way for the head, nor do I believe that those ancient philosophers thought otherwise. If, for example, they had thought that the head was inherently a consistently shining star while the tail alone was an appearance, then they would have said that when because of the obliquity of the sphere there was no reflection of our vision toward the sun, the curl would no longer be seen, while the star—that is, the head of the comet—would be seen as well as ever. This they did not say; they said that the whole comet was not seen, a clear indication that the generation of both head and curl is identical. And whether or not the ancients said this, it is now brought under consideration by Sig. Mario with very reasonable grounds for raising the question; this may be weighed, as is done by Sig. Sarsi, and in the proper place (*p.* 35) we shall consider what he has written about it.

X

Meanwhile let your Excellency proceed to read this: 'I must reply . . . (*p.* 36) . . . comets do not possess?' (*p.* 74, l. 20 to *p.* 75, l. 20).

Sig. Guiducci, with the very honorable intention of smoothing the road for seekers of truth, had brought into consideration the fallacy of those who argued that because the comet appeared to move in a straight line its movement was along a great circle. He averred that although it was true that motion along a great circle would always appear straight, it did not necessarily follow that motion which appeared straight was necessarily along a great circle, as assumed by those who had inferred from the comet's apparent straight motion that it moved along a great circle. Among these was Father Grassi, and, perhaps resting on the authority of Tycho (who had first equivocated thus), he neglected something which he might not have overlooked if his predecessor had not done so. To me this makes Father Grassi's small error quite excusable, though I believe from what Sig. Mario says that the Father made much of this and held it in high esteem. Now comes Sarsi, continuing in his predetermined course and contriving to make the remark appear as an inadvertence of trifling importance, believing he can rescue his master in this way;

but it seems to have the opposite effect (if the Father lends his assent to Sarsi's excuses and defenses), for in trying to save a single error he will run into many more.

First, still reputing it vain and superfluous to heed things which neither he nor others have noticed, Sarsi says that since his master was disputing against Aristotle and the Pythagoreans who never brought up a straight motion for the comet, it would have been beside the point to have attempted to dispose of this. But careful consideration shows that this excuse has not assisted the Father at all, for neither did the same adversaries ever mention a cometary motion along lesser circles, and hence it was equally superfluous to prove that comets move along great circles. Therefore Sarsi must either find that the ancients wrote that comets moved along lesser circles, or confess that it was as (*p.* 37) redundant for his master to consider the motion to be along a great circle as it would have been for him to have considered it straight.

The second point is that under Sarsi's own rule it was a still greater defect to omit consideration of straight motion, inasmuch as it was Kepler who had attributed this to comets, and he is named by Sarsi himself. Nor does the excuse he puts forth seem to me at all sufficient—that since this opinion was deduced by Kepler as a consequence of the earth's motion, a proposition that cannot be piously and religiously held, Sarsi has therefore deemed it worthless. Why, this must be so much the more motive for destroying it and demonstrating its impossibility; surely it is not a bad idea to prove also by physical reasons, if possible, the falsity of those propositions which have been declared repugnant to the Holy Scriptures.

Third, Sarsi's excuse is also defective because not only motion which is truly straight would appear to be made along a straight line, but any other motion, as long as it occurred in the same plane as that in which the observer's eye is located. This is even hinted by Sig. Mario; hence Sarsi will have to find some other way to persuade us that no movement except the circular has ever occurred to anyone as suitable to assign to comets. I do not see how he could entirely succeed in this, since, if nobody else had mentioned one, Sarsi himself did so in writing a few lines later, when, in defense of an

elongation from the sun of more than ninety degrees, he allowed a non-circular motion and admitted motion along an oval line, and even required the motion along this line to be irregular. Hence, it is necessary either for the movement in question to be now circular and now oval and now completely irregular, according to Sarsi's necessities, or for him to confess his defense of his master to be defective.

Fourth, what if I admit the motion of the comet to be circular, and not merely by popular opinion, but truly and necessarily? Will Sarsi perhaps suppose that it is conclusively proved to be along a great circle by its motion having appeared to his master and others as along a straight line? I know that Sarsi has believed this until now, but he has been mistaken; and (*p.* 38) if I thought it would not annoy him, I should remove him from error. In order to do this, I should ask him which circles in the sphere he calls great circles. I know that he would answer me, 'Those which passing through its center, which is also the center of the earth, divide it into two equal parts.' Then I should say, 'So the circles described by Venus, Mercury, and the Medicean planets are not great circles at all, but rather are quite small ones, the last named having Jupiter for their center, and the others having the sun for theirs. Yet if one observes how their movements appear, these will be found to be along a straight line.' This comes about because our eye is located in the same plane as that in which lie also the circles described by the stars mentioned. We conclude from this that nothing may be deduced from a motion appearing to be straight except that it is made in the plane which passes through the eye (that is, in the plane of a great circle); not that it is along the circumference of a great circle any more than a smaller one. In itself, this motion might be made in a circular line, or in some other line as irregular as you please, and still appear straight. Thus the two propositions already examined by us are not equivalent, and to take one for the other is an equivocation and an error in logic.

If I might rest assured that Sarsi would not take it ill, I should like to compare this fallacy with another and similar one which I see has been overlooked by some very great men, and upon which perhaps

Sarsi himself has not reflected. I should not wish to cause him displeasure by showing him that I have not also overlooked it, as have so many other people more perspicacious than myself, but, be that as it may, I shall impart it to your Excellency. It has been noted by very sharp observation that the tip of the tail, the head of the comet, and the center of the sun's disc are always perceived along the same straight line, and from this it has been very strongly conjectured that the tail is a refraction of the solar light extended diametrically opposite to the sun. Up to the present, as far as I know, no one has considered that the sun and the whole length of the comet appearing in a straight line does not necessarily prove that the straight line drawn from the tip of the tail to the head of the (*p.* 39) comet would, if prolonged, terminate in the sun. In order that three or more points may appear in a straight line, it is sufficient for them to be located in the same plane as the eye; thus, for instance, Mars or the moon is sometimes seen in line between two fixed stars, but the straight line joining the two stars would not for that reason pass through Mars or the moon. Hence nothing can be necessarily concluded from the tail of the comet appearing directly opposite to the sun, except that these are in the same plane as the eye.

Now, in the fifth place, let there be noted what I may call a certain inconstancy in the words toward the end of the passage just examined by your Excellency and me. Sarsi undertakes to show later how incorrectly I (that is, Sig. Mario) have attributed straight motion to the comet; then, three lines below, he says that there is no need to disprove straight motion, it being certain and evident that this has never been found in comets. But if the impossibility of this motion is certain and evident, to what purpose has he undertaken to disprove it? And in what way is this certain and evident if, according to Sarsi's statement, no one yet has so much as considered it, let alone refuted it? Such motion, he says, has been taken into consideration by Kepler alone. But Kepler does not refute it; rather, he introduces it as possible and true. It seems to me that Sarsi, feeling himself unable to do anything else, tries to get the reader tangled up. But I shall endeavor to undo the knots.

XI

'But, furthermore, in this place . . . (*p.* 40) . . . comets too close at hand.' (*p.* 75, l. 21 to l. 33).

Here, first of all, if I accept Sarsi's imputation of negligence because various motions that might have been attributed to the comet did not occur to me, then I do not see how he can exculpate his master from the same criticism for not having considered its possible straight motion. If he excuses his master by saying that such a consideration would have been superfluous on account of this movement's not having been introduced by any other author, I fail to see why I deserve to be criticized; I may be granted the same pardon, since no other author has introduced these alien motions which Sarsi now names. Besides, Sig. Sarsi, it was up to your master and not to me to think of these movements by which suitable reasons might be rendered for such great elongations of the comet. If any movement lends itself to such a requirement, you ought to have named it and accepted that one alone; you should not have passed it over in silence and adduced with Tycho a simple circular motion about the sun (which is quite unsuitable for saving the appearance in question), and then pretended that not he but we were in the wrong for not divining that he might have given inward assent to ideas very different from those he had written. Moreover, Sig. Mario never said that nature afforded no way to save the elongation of ninety degrees; indeed, if such an elongation took place it is perfectly clear that there was also a way in which it did so. What he said was that in the hypothesis accepted by the Father there could be no such elongation without the comet's touching the earth and even penetrating it. Sarsi's excuse is vain, then, up to this point. But perhaps he pretends that any trivial excuse is allowable for his master, while for me even the strongest one remains invalid. If that is it, I gladly hold my peace and freely concede him the point in question.

In the second place, I am going to produce another excuse on behalf of myself disguised as Sig. Mario. Naively confessing my never having thought of movements along eccentrics or oval lines or any other irregularities, I say that this was because it is not my

habit to give ear to conceptions which have nothing to (p. 41) do with the matter in hand. And why does Sarsi want to introduce an oval motion round the sun in order to have the comet elongate ninety degrees? Does he perhaps believe that by stretching the shape in one direction and compressing it in another he can succeed in his design? Surely not; not even if he could stretch it to infinity.¹³ And the same impossibility exists for the eccentric which has its smaller part beneath the sun. For Sarsi's information, should your Excellency ever meet him, you may suggest to him two straight lines AB and CD, CD being perpendicular to AB, and say to him that supposing

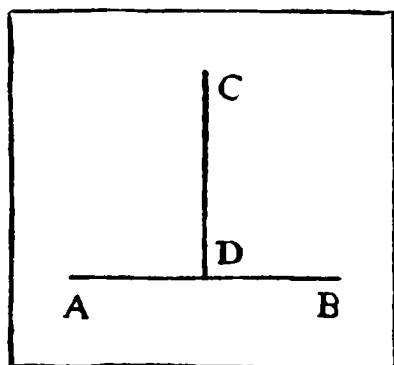


Fig. 17

the straight line DC to be that which goes from the eye to the sun, then that by which the comet would be seen at an elongation of ninety degrees would necessarily be either DA or DB, it being generally conceded that the apparent motion of a comet is in the plane of a great circle. Entreat him then for our edification to describe the eccentric or oval designated by him, along which the comet in its motion may get down far enough to be seen along the line ADB, as I confess I do not know how to do this.

Thus far, two of the proposed means are excluded; there remain the other eccentric (with its center removed to the right or left from the line DC), and the irregular path. As to the eccentric, it is true that it is not entirely impossible to draw this on paper in such a way as to create the elongation sought; but I say to Sarsi that if he would

set himself to draw the sun with the orbs of Mercury and Venus around it, and also the earth encircled by the orb of the moon, both of which must be provided for, and then should attempt to fit in this eccentric for the comet gracefully, I feel certain that confronted with such absurdities and monstrosities he would be afraid to rescue his master by means of such a makeshift even if he could. As to irregular lines, there is no doubt that by them not only this appearance may be saved, but any other; yet I warn Sarsi that the introduction of such lines, far from helping his master's cause in any way, (*p.* 42) would only prejudice it more seriously; not only because he never mentioned them, and on the contrary accepts a circular line, the most regular that exists, but because it would be very frivolous to propose this. Sarsi himself may understand this if he will consider what is meant by an irregular line. Those lines are called regular which, having a fixed and definite description, have been susceptible of definition and of having their qualities and properties demonstrated. Thus the spiral is regular, and its definition originates in two uniform motions, one straight and the other circular; so is the ellipse, which originates from the cutting of a cone or a cylinder; and so on. But irregular lines are those which have no determinacy whatever and are indefinite and casual, and hence indefinable; no property of such lines can be demonstrated, nor in a word can anything be known about them. Hence to say, 'Such events take place by reason of an irregular line' is the same as saying, 'I do not know why they occur.' The introduction of such lines is in no way superior to the sympathy, antipathy, occult properties, influences, and other terms employed by some philosophers as a cloak for the correct reply, which would be: 'I do not know.' That reply is as much more tolerable than the others as candid honesty is more beautiful than deceitful duplicity. Father Grassi was therefore much more prudent in not proposing irregular lines as a means of fulfilling his needs than his pupil was in mentioning them.

If I may say frankly how it looks to me, I certainly believe that Sarsi himself understood well enough the inefficacy of his answer and how ill-founded it was. I judge this from his having hurried quickly over it, though the point was most important in the matter

under consideration, and the difficulty raised by Sig. Mario was quite serious. Sarsi himself serves as my best witness to this when on page 81, in speaking of a certain argument used by his master, he writes: 'Furthermore, what significance we gave to this argument can I believe, be sufficiently understood from that fact that it was presented in few words, and its meagreness was due to the fact that all had previously been explained much more precisely and fully in the two other arguments.' The brevity and simplicity (*p.* 43) with which he has touched upon this may be seen, among other things, from his not having made a diagram of the eccentrics and ellipses introduced to save the whole. Yet farther on, we shall meet with a sea of diagrams introduced into a long discourse designed to corroborate an experiment which in the end fails to achieve the slightest assistance for the main point under consideration there. But without my saying any more, your Excellency may embark upon the following ocean of distinctions, syllogisms, and other logical paraphernalia, and see how high an opinion Sarsi entertains for something which, frankly, I consider to be so much woolgathering.

XII

'Since Galileo . . . [*pp.* 44-45] (*p.* 46) . . . their very clear lights.' (*p.* 75, l. 34 to *p.* 79, l. 33).

Here, as your Excellency sees, in exchange for the equivocation into which Father Grassi had run by following (as Sig. Guiducci remarks) in the footsteps of Tycho and others, Sarsi wants to show that I have erred as much or more in logic when (*p.* 47) I argued in the following way to show that the magnification of the telescope is the same for the fixed stars as for other objects, and is not imperceptible or nonexistent as Father Grassi had written. 'Many stars completely invisible to the naked eye are made easily visible by the telescope; hence this augmentation should be called infinite rather than nonexistent.' Here Sarsi gets up in arms and in a series of long attacks does his best to show me to be a very poor logician for having called such enlargement 'infinite.' At my age, these altercations simply make me ill, though I myself used to plunge into them with delight during my childhood, when I too was under a schoolmaster. So to

all this I answer briefly and simply that it appears to me that Sarsi openly reveals himself to be just what he tries to prove me to be; that is, little cognizant of logic, since he takes as absolute something which was meant relatively. It has never been asserted that the enlargement of the fixed stars is infinite. Rather, the Father having written that it is nil, and Sig. Mario having noted that this is not true because a great many completely invisible stars are rendered visible, it was added that such enlargement ought rather to be called infinite than nil. Now who is so simple as not to understand that if we call a profit of one thousand upon a capital of one hundred 'large,' and not 'nil,' and the same gain upon a capital of ten 'very large,' and not 'nil,' then the acquisition of one thousand upon a capital of nothing should be called 'infinite' rather than 'nil'? When Sig. Mario has spoken of the absolute enlargement, even Sarsi knows (and has written in several places) that he has said it is the same as that of all other objects seen through the same instrument. So when in this place he tries to tax Sig. Mario with a short memory, saying that he ought to remember having said at other times that the same instrument increases all objects in the same ratio, the accusation becomes foolish. And even if Sig. Mario had called it infinite, without any relative term, I should never have expected such carping criticism as this attack, since using the word 'infinite' in place of 'very large' is a manner of speaking which is employed every day. Here indeed Sarsi has a large field for showing himself to be a better logician than all the other (*p.* 48) authors in the world, in whom I assure him that he will find the word 'infinite' chosen nine times out of ten in place of 'very large.' Nor is that all, Sarsi. If the Preacher should confront you and say, 'The number of fools is infinite,'¹⁴ what would you do then? Would you perhaps dispute with him and maintain his statement to be false? You could prove on equal scriptural authority that the world is not eternal, and that having been created in time there cannot have been and cannot be infinite men; and since foolishness reigns only among men, the proposition in question could never be true even if all men, past, present, and future, were fools; for there could never be an infinite number of human individuals even if the world were to endure eternally.

H

Getting back to the point, what shall we say of the other fallacy so subtly detected by Sarsi in our calling 'enlargement' that which takes place when an invisible object is made visible by the telescope? This cannot be called 'enlargement,' he says, because enlargement presupposes some quantity, and enlarging is nothing but making greater from less. I really do not know what to say to this in excuse for Sig. Mario, except that he was speaking everyday language, as it is called. Believing that the property of the telescope by which it shows us these objects which we cannot discover without it was the same as that by which things already visible are shown to us much larger, and, feeling that this is generally called an enlargement of the image or of the visible object, he permitted himself to transfer this name in the same sense to something which Sarsi now declares to us should be called not 'enlargement' but 'passage from nonexistence into existence.' Hence, for example, when the telescope enables us to read from a great distance some writing of which we should see only the capital letters without it, to speak logically one should not say that the telescope enlarges the capitals, but causes the small letters to pass from nonexistence into existence. But if the word 'enlargement' cannot be correctly used when nothing actual is presupposed (*p.* 49) to receive the enlargement, perhaps it is no more justifiable for Sarsi to appropriate the word 'passage' or 'transfer' where there do not exist two end points, one from which something leaves and another to which it passes.

And who knows? Perhaps Sig. Mario held and holds the opinion that from very distant objects the pure images reach us only under such acute angles as to remain imperceptible to our senses, and are as if nonexistent, though they really do exist. If I may give my opinion, I think that if they were truly nothing, all the telescopes in the world would not suffice to make them become something. Thus the images of invisible stars, no less than those of the visible ones, are spread throughout the universe, and hence enlargement may be predicated of them also, with all due respect to Sarsi and without any error in logic. But why should I go on raising questions about something for which I have sensate and conclusive proof? There is a supplemental radiance seen by us which is not really located around the stars, but

is in our eyes; only the bare and sharply bounded image comes from the star. We know positively that a nebula is nothing but an aggregate of many minute stars which are invisible to us; yet the field which they occupy does not remain invisible. It shows itself in the aspect of a little whitened patch, coming from the conjunction of the separate radiances that crown each of these little starlets. But since these irradiations exist only in our eyes, it must be that each image of each starlet exists really and distinctly in our eyes. From this, one may deduce another doctrine—that the nebulae and even the Milky Way do not exist in the sky but are a pure sensation (*affezione*) of our eyes, in the sense that, if our vision were so acute as to distinguish those minute stars, there would be neither nebulae nor a Milky Way in the sky.

These conclusions not having been previously voiced by others, I suppose they will not be admitted by Sarsi, and he would still have it that Sig. Mario sinned by calling 'enlargement' that which according to him should be called 'passage from nonexistence into existence.' Well, be that as it may, I have permission from Sig. Mario (in order not to engage in (*p.* 50) new battles) to concede to Sarsi the entire victory in this skirmish, as well as in the one which follows, in which Sarsi agrees that the revealing of invisible fixed stars may be called enlargement in relation to visibility but not in relation to quantity. All this is conceded to him if only he will grant to us that both the invisible and the visible stars (let them be increased in relation to whatever he pleases) are ultimately increased in such a way as to render completely false the doctrine of his master, who wrote that they were not increased in any way at all. It was upon that statement that he founded the third of those arguments by means of which he undertook to prove the primary thesis of his treatise; namely, that of the location of the comet.

But what shall we reply to still another error in logic which Sarsi would attribute to us? Let us first hear it, and then we may decide on the most appropriate reply. Not content with having shown that the disclosure of invisible fixed stars ought not to be called an infinite enlargement, Sarsi goes on to prove that saying that their disclosure results from the telescope is a grave error in logic, whose laws require

that when one effect may be derived from several causes we cannot correctly infer from it one particular cause. In order to show clearly that many causes may exist (in addition to that of the telescope) for something becoming seen which was not seen at first, Sarsi names these one by one, saying that it is necessary to eliminate them all and show that they have no part in causing invisible stars to be seen through the telescope. Hence, in order to escape Sarsi's imputations, Sig. Mario would have to show that first, bringing the telescope to the eye would not in itself and by itself increase the visual power (for this, indeed, is a cause that might make something visible which was not seen previously); second, he would have to show that this act would not be a removing of clouds, trees, roofs, or other intervening obstacles; third, that it was not a mere use of ordinary spectacles (I am simply enumerating the causes set down by Sarsi himself, your Excellency, without altering anything); fourth, that it was not an improved illumination of the object; fifth, that it was not (*p.* 51) a bringing of the star down to earth or a venturing forth on our part into the sky, by which the interval between would be diminished; sixth, that it did not make the stars swell up so as to enlarge and become more easily visible; seventh and last, that it was not an opening of closed eyes. Each of these—and especially the last—is sufficient to make us see what we did not see before.

I do not know what to say to you, Sig. Sarsi, except that you reason very well. The only thing that dissatisfies me is that all these imputations fall back upon your master without so much as touching Sig. Mario or me. I ask you whether any of these causes which you produce as capable of making us see what we would not see without them (as, for example, bringing things closer, interposing vapors or lenses, etc.)—I ask you, I say, whether any of these causes can produce the effect of enlarging visible objects in the way in which the telescope produces that effect. I believe that you will answer yes. Then I shall add that you have openly accused your master of being a poor logician too. For he, speaking generally to everyone, would recognize the enlargement of the moon and of all other objects by the telescope alone, without bothering to exclude any of the other causes, as in your opinion he would be obliged to do. But that obligation

does not fall upon Sig. Mario at all; he was speaking only to your master, and not to everyone; and he was attempting to show that what your master had said about the effect of the instrument was false. Hence, he considered the matter in the way in which his adversary had considered it, and was not obliged to do so in any other way. Indeed, your charge of being a poor logician falls the more heavily upon your master inasmuch as he has transgressed against this very same rule on another most important occasion—I mean by inferring revolution in a great circle from motion apparently straight. For the same effect might be caused by motion truly straight, or by any other motion in the same plane as the eye. Very acute men might inquire into all three of these cases with good reason, and according to your statement your master himself did not refuse to accept motion along an oval line or even an irregular one. But I do not believe it would ever occur to any but the most simple-minded to raise a question whether any of your seven causes as set forth above could (*p.* 52) possibly bear upon the disclosure of invisible stars when they are observed through the telescope, if I may speak frankly.

I do not mean to place Sarsi in the ranks of the simple-minded because he has allowed himself to get into this position, your Excellency, for it is clear that he has not spoken from inner conviction; ultimately he very nearly concedes that, since nothing but the telescope is being considered, these other causes may be set aside. Yet such an open concession had as its consequence the vacuity of his previous accusation, and nullifies the idea that I am a poor logician which it might have impressed upon some of his readers. To obviate all this, he adds that not even the above suffices for good reasoning, because it is not in one way only that the telescope renders visible what was not seen before, but in two. The first is by bringing objects to the eye under a greater angle, which makes them appear larger; and the other is by uniting the rays and images so that they act more strongly. And since either of these suffices to render visible that which was not perceived, one may not infer a single one of these causes from the effect. Such are his very words, whose inner sense I must admit I cannot fathom, inasmuch as they are a bit too general.

It seems to me that it would have been more to the point for him to have expressed himself more specifically, for his proposition may be understood in several ways. Perhaps the first of these to come to mind is one which contains within itself a manifest contradiction, for to bring objects under a greater angle represents an effect contrary to that of compressing together the rays and the objects. Since the rays conduct the images, it is hard to understand how during their passage they are compressed together and at the same time form a larger angle; an angle is formed by lines running together, so it would seem that by restricting them the angle should become more acute rather than wider. But if Sarsi had conjured up some other way in which (*p.* 53) the rays, uniting, might form a larger angle (and I do not deny that such might be found), then he ought to have stated it and distinguished it from this other, so as not to leave the reader in the midst of doubts and equivocations.

Assuming for the moment that there are two ways for the telescope to work, I should like to know whether it always operates both ways at once, or sometimes one way and sometimes the other, separately, so that when it is used to enlarge the angle it desists from restraining the rays, and when restricting the rays it preserves the angle always at its original size. If it always operates with both these methods, then Sarsi is acting with great simplicity when he criticizes Sig. Mario for not having designated and accepted one and excluded the other. But if it operates with one only, then it is Sarsi who has erred in not designating that one and excluding the other—showing, for example, that the telescope works by enlarging the angle when we look at the moon, which enlarges very considerably, but that when we look at the stars it merely unites the rays and does not increase the angle. So far as I can say with confidence, in the infinite—or I should say in the very many—times that I have looked through the instrument, I have never noticed any difference whatever in its mode of operation, and so I think it works always in the same way. Moreover, I believe that Sarsi thinks so too; and in that case, it must be that the two operations of enlarging the angle and restricting the rays always coincide, which accordingly puts Sarsi's objection completely and entirely beyond the pale. For it is certainly true that an

error is committed by anyone who infers a particular cause from an effect which may depend upon several separate causes; but if the causes are always inseparable, one may at will infer from the effect whichever cause suits him best, inasmuch as every time the effect is present that cause must necessarily be present also. Thus, to give an example, whoever may say: 'So-and-so has lighted the fire, hence he made use of a burning-glass' would err, for fire might be lighted by hammering an iron, by tinder and spark, by rubbing two flints together, or in other ways. But he who says: 'I have heard my neighbor striking fire' and adds (*p.* 54) 'therefore he has a flint' would be criticized unreasonably by anyone who would argue against him that, since in that operation, in addition to flint, there are steel, tinder, and sulphurous material as well, he could not in good logic infer flint with certainty. And thus if the enlargement of the angle and the union of rays always concur in the operations of the telescope, and one of these makes the invisible become seen, then why should one not infer from the effect whichever he likes of the two?

I think I see somewhat into Sarsi's mind; if I am not mistaken, he would have the reader believe what he himself certainly does not believe—that perceiving stars which previously were invisible depends not upon the enlargement of the angle but upon the union of the rays, and that the stars are made visible not because their images become larger but because their rays are strengthened. But he did not want to come out into the open because the other arguments raised by Sig. Mario, of which Sarsi had said nothing, would burden him too heavily—especially the argument that the intervals between stars are perceived to be enlarged in the same ratio as objects here below, which intervals ought not to increase in size unless the stars do, since they are as distant from us as the stars are.

But to sum up, I am sure that if Sarsi would explain himself as to how he understands these two operations of the telescope (I mean of restricting the rays and of enlarging the angle), it would be obvious that not only are they both performed together but that they are one and the same thing, so that the rays are never united without the angle being enlarged. In order to hold a different opinion, he would be obliged to show that the telescope sometimes unites the

rays without enlarging the angle, but that this happens only when the fixed stars are being observed. This he will never show to the end of time, for it is a most foolish fiction, or, to be blunt, a falsification.

I did not expect to spend so many words on this trifling, your Excellency, but since the more has been done the less remains to do. Now for this other charge of violating the laws of logic. In Sig. Mario's division of the effects of the telescope, he is said to have included an effect which does not exist and to have left out one that (*p.* 55) should be given. For when he said, 'The telescope renders stars visible either by enlarging their images or by illuminating them,'¹⁵ Sarsi will have it that he should have said instead, 'by enlarging them or by uniting the images and the rays.' I reply that Sig. Mario never had any intention of dividing what is one, and, so far as he and I are concerned, such is the manner of the operation of the telescope in representing objects to us. When he says, 'if the telescope does not render stars visible by enlarging them, then by some unheard-of means it must illuminate them,' he does not introduce illumination as an effect to which he subscribes, but he counterpoises it against the other as an obvious impossibility, with the intent of making still more evident the truth of this other. This is quite a customary manner of speaking, as when one says: 'If our enemies did not scale the fortress, they must have rained here from the sky.' Now if Sarsi thinks he can win acclaim by impugning this way of talking, then another door is open to him (in addition to the one regarding 'infinite') for winning a logical battle against all the authors in the world. But in trying to show himself off as a great logician, let him beware lest he appear an even greater sophist. I seem to see your Excellency grin, but can I help this? Sarsi took it into his head to write in opposition to Sig. Mario's treatise, and he has been forced to grasp at sky hooks, as they say. For my part, I not only excuse him but praise him, for it appears to me that he has accomplished the impossible.

Returning to our subject, we have already seen that Sig. Mario did not seriously suggest illumination as a property of the telescope. Nor is that all; Sarsi himself confesses that this was put forth as impossible. Hence, illumination is not one of two alternatives;

rather, as I have remarked, so far as we are concerned, there exists no alternative at all. As to the union of the images and rays recorded by Sarsi as an alternative omitted by Sig. Mario in his division, it would be well for Sarsi to specify how this second operation differs from the first. Up to this point, we have understood them to be the same thing, but when we are assured that they are two distinct and different operations, we shall understand how we have erred. Even then, the error will not be one of logic in listing alternatives, but one of optics in not having adequately fathomed all the effects (*p.* 56) of the instrument.

When Sarsi says in conclusion that he does not at present wish to linger and note down other errors than those few found thus casually in a single place, and will omit the others, I first thank him for his compassion toward us. Next I rejoice for Sig. Mario, who may be sure that throughout his entire treatise he has not committed the slightest error in logic; for though Sarsi seems to hint that there are many more, I, at any rate, shall believe that he chose as the worst these which he has singled out and criticized. I leave it to you to judge by implication the quality and importance of the rest.

I now come to consider this final part, in which Sarsi as a special favor to me attempts to ennoble the telescope with the admirable property of illuminating as well as enlarging the objects which are viewed through it. However, before I proceed, I wish to thank him for his courtesy, as I question whether this effect will put me under much obligation once we have considered the strength of the demonstrations brought forward to prove his point. And since in his own explanation it seems to me that for some unknown reason the author rambles around repeating the same proposition over and over again, I shall try to extract the meat of it. This appears to me to be as follows: 'The telescope represents objects as larger because it brings them to us under a larger angle than that under which they are seen without the instrument. Also, restricting almost to a point the images of luminous bodies and their scattered rays, it renders much more lucid that cone (or rather, luminous pyramid) by which objects are seen; and thus shining objects are represented as enlarged and also as illuminated by greater light. That the optic pyramid is rendered

brighter by restriction of the rays is proved by reason as well as by experience. Reason teaches us that light collected in a smaller space must illuminate more, and experience shows us that a glass lens placed beneath the sun will not only burn wood where the rays come together, but will liquefy lead and blind the sight. From all this it may be concluded that it is as correct (*p.* 57) to say that the telescope illuminates the stars as to say that it enlarges them.'

In return for Sarsi's courtesy and good will in exalting and ennobling this admirable instrument, the least I can do at present is to give complete assent to all the above statements and experiments. Yet it grieves me that in a sense they are more prejudicial to him by being true than they would be by being false; for the principal conclusion that they are supposed to prove is utterly false, and I do not think we can maintain that those who draw a false conclusion from true premises do not seriously transgress in logic. It is true that the telescope enlarges objects by bringing them under a greater angle, and it is quite true that they come to us in perspective. No less true is it that the closer uniting of the rays of the luminous pyramid renders them brighter, and consequently does the same for the objects viewed. The reason Sarsi assigns for this is correct, that the same light reduced to a smaller space illuminates better. And finally there is nothing wrong with his lens experiment of burning and blinding by uniting the solar rays. But it is quite false that luminous objects are represented to us as brighter through the telescope than without it; indeed, the truth is that we see them much more dimly. If Sarsi, while looking at the moon through the telescope, had ever opened his other eye and with it looked directly at the same moon, he could without any trouble have made a comparison between the splendor of the huge moon seen through the instrument and that of the small one seen with the naked eye. Having made this observation, he surely would have written that the light of the moon looked at directly appears a good deal brighter than that of the other one.

The falsity of the conclusion is thus quite evident; it now remains for us to show the fallacy in its deduction from true premises. Here, it seems to me that Sarsi did not consider what would happen to a merchant who in looking over his account books should read only

the credit side, and in this way persuade himself that he was solvent or even rich, which would be a true conclusion if there were no debits on the other side. It is true, Sig. Sarsi, that the lens—that is, the convex glass—would unite the rays and thus multiply (*p.* 58) the light and favor your conclusion; but where have you left the concave glass? This counterpart of the lens in the telescope is its most essential part, since it is held near the eye and the ultimate rays pass through it; it corresponds to the final balancing and settlement of accounts in the ledger. Do you not see that though the convex lens unites the rays, the concave glass expands them and forms an inverted cone? If you had tried to catch the rays which pass through both the glasses of a telescope, as you have done in observing those refracted by a single lens, you would have seen that whereas the latter meet in a point, the former go on expanding infinitely (or to speak more properly, through an immense distance). This is very clearly seen experimentally by receiving the sun's image upon a sheet of paper, as when one is tracing the sunspots. In proportion as the end of the telescope is moved farther and farther away, a larger and larger circle is described upon such a sheet by the cone of rays, and the larger this circle is made the less bright it becomes in comparison with the remainder of the sheet which is struck by the direct rays of the sun. And if this and all other experiments had been unknown to you, it would still be hard for me to believe that you had never even once heard of the fact that the larger an object shows itself through a concave glass the darker it appears. Then how can you expect the telescope to brighten anything while enlarging it?

Sig. Sarsi, give up trying to exalt this instrument with these admirable new properties of yours unless you wish to throw it into utter disrepute among those who up to the present have had little enough regard for it. Also, note that in this accounting I have passed over as true one entry which is false; this is that the strengthening of light by the uniting of rays renders more luminous the object seen. That would be true if the light were going out to meet the object, but here it is coming toward the eye and produces a contrary effect. For in addition to offending the eyes, it renders the medium brighter; and a brighter medium, as I believe you know, makes objects

appear darker. It is for precisely this reason that the stars appear more brilliant as the sky (*p.* 59) becomes dark at night and look dimmer when the heavens brighten. These things, as your Excellency sees, are so obvious that I cannot believe Sarsi was unaware of them; he induced himself to prove a paradox in order to show the liveliness of his wit rather than because of any inner conviction. His ultimate conclusion confirms me in this opinion when, I believe to show that he was speaking in jest, he follows with these words: 'I therefore affirm that the telescope may be as truly said to illuminate the stars as to enlarge them.' Your Excellency knows that both he and his master have always said, and go on saying, that it does not enlarge them at all. We shall see that Sarsi is forced to maintain that conclusion in the things which follow.

XIII

Read, therefore, your Excellency: 'I hasten to the third (*p.* 60) . . . than an argument.' (*p.* 79, l. 34 to *p.* 81, l. 17).

Regarding what is here written, I must consider in the first place what reason Sarsi can possibly have for writing that I have been quite unhappy over Father Grassi, inasmuch as there is no trace of any lament of mine in Sig. Mario's treatise, nor have I ever (*p.* 61) complained to anyone else, or in my own heart. I have no reason to complain; it would seem to me very foolish for anyone to lament that men of high reputation were opposed to his position when he had easy and obvious ways of showing it to be correct, as I am sure I have. All that occurs to me is that under this pretense Sarsi has attempted to hide, I know not why, those ulterior motives which have impelled him to be angry with me—a thing I regret, as I had much rather have employed this time in some other study more congenial to me. I wish Sarsi would believe that Father Grassi had no intention of offending me when he criticized as uninformed those people who belittle the argument derived from the slight enlargement of the comet by the telescope. But if I personally have ever been declared to be among these, I should have been permitted to produce my reasons and defend my cause, especially when it is just and true. Also I want to concede to Sarsi that his master's intentions were

good when he set himself to sustain that opinion, thinking to preserve and increase the repute and esteem of the telescope against the calumnies of those who claimed that it was a fraud and a deceiver of vision and thus attempted to rob it of its wonderful value. In this act the Father's intention seems to me as praiseworthy and good as the choice and quality of Sarsi's defenses appear to me poor and harmful, when against the impostures of its maligners he tries to cover the true effects of the telescope by attributing to it false ones. This does not seem to me to be a good place to argue the nobility of the instrument, and meanwhile may Sarsi kindly pardon me if I do not acknowledge my obligation to him in as much profusion as might perhaps suit him, or confess my debt for the new merits and honors showered upon the telescope. He cannot reasonably pretend that I should increase my debt and my affection toward people who make silly and false attributions and who threaten me with loss of their friendship because I reveal their errors by speaking the truth.

Immediately after this, though perhaps not very opportunely, he is induced to (*p.* 62) call the telescope my foster child and to reveal at the same time that it is not my offspring in any other way. Now how is this, Sig. Sarsi? When you are busy placing me under great obligations for benefits showered upon this supposed child of mine, must you go on to tell me that it is only a foster child? Is this rhetorically sound? I should have thought rather that on such an occasion you would have tried to make me believe it my own child even if you had been certain that it was not. The part which I played in the discovery of this instrument, and whether I may reasonably lay claim to it as my offspring, I set forth long ago in my *Starry Messenger*. I wrote that in Venice, where I happened to be at the time, news came that a Hollander had presented to Count Maurice [of Nassau] a glass by means of which distant things might be seen as perfectly as if they were quite close. That was all. Upon hearing this news, I returned to Padua, where I then resided, and set myself to thinking about the problem. The first night after my return, I solved it, and the following day I constructed the instrument and sent word of this to the same friends in Venice with whom I had been discussing the subject the previous day. Immediately afterward, I applied myself

to the construction of another and better one, which I took to Venice six days later; there it was seen with great admiration by nearly all the principal gentlemen of that republic for more than a month on end, to my considerable fatigue. Finally, at the suggestion of one of my friendly patrons, I presented it to the ruler in a full meeting of the Council. How greatly it was esteemed by him, and with how much admiration it was received, is testified by ducal letters still in my possession which reveal the munificence of that serene ruler in recompense for the invention presented to him, reappointing and confirming me for life to my professorship at the University of Padua at double my former salary, which was then more than triple that of some of my predecessors. These acts did not take place in some forest or desert, Sig. Sarsi; they happened in Venice, and if you had been there you would not dismiss me thus as a simple schoolmaster. But (*p.* 63) by the grace of God, most of those gentlemen are still living there, and by them you may be better informed.

Perhaps someone will say, however, that in the discovery and solution of a problem it is of no little assistance first to be conscious in some way that the conclusion is true and to be certain that one is not attempting the impossible; and hence that my knowledge and certainty that a telescope had already been made were of so much help to me that without this I should perhaps not have made the discovery. To this I shall reply by making a distinction. I say that the aid afforded me by the news awoke in me the will to apply my mind to it, and without this I might never have thought about it; but beyond that I do not believe that such news could facilitate the invention. I say, moreover, that to discover the solution of a known and designated problem is a labor of much greater ingenuity than to solve a problem which has not been thought of and defined, for luck may play a large role in the latter while the former is entirely a work of reasoning. Indeed, we know that the Hollander who was first to invent the telescope was a simple maker of ordinary spectacles who in casually handling pieces of glass of various sorts happened to look through two at once, one convex and the other concave, and placed at different distances from the eye. In this way, he observed the resulting effect, and thus discovered the instrument.

But I, incited by the news mentioned above, discovered the same by means of reasoning which, easy as it is, I wish to reveal to your Excellency. Recounting it thus where it is to the purpose, I may by its simplicity render less incredulous those people who, like Sarsi, may wish to diminish whatever praise there is in it that belongs to me.

My reasoning was this. The device needs either a single glass or more than one. It cannot consist of one alone, because the shape of that one would have to be a convex (that is, thicker in the middle than at the edges), or concave (that is, thinner in the middle), or contained between parallel surfaces. But the last named does not alter visible objects in any way, either by enlarging or reducing them; the concave diminishes them; and the convex, while it does indeed increase them, shows them very indistinctly and confusedly. (*p.* 64) Therefore, a single glass is not sufficient to produce the effect. Passing next to two, and knowing as before that a glass with parallel faces alters nothing, I concluded that the effect would still not be achieved by combining such a one with either of the other two. Hence I was restricted to trying to discover what would be done by a combination of the convex and the concave, and you see how this gave me what I sought. Such were the steps in my discovery, in which I was not at all assisted by the conception that the conclusion was true.

But if Sarsi and others think that certainty of a conclusion extends great assistance in the discovery of some means for realizing it, let them study history; there they will learn that Archytas made a dove which could fly, that Archimedes made a mirror which kindled fires at great distances (as well as constructing other remarkable machines), that other men have kindled perpetual fires, and a hundred more conclusions that are equally amazing. By reasoning about these, they may easily discover, to their great honor and profit, how to construct such things. Or if they do not succeed, they will at least derive some benefit in the form of a clarification of their ideas about the help which they promise themselves from foreknowledge of the reality of the effects, which will be much less than they have imagined.

I return now to what Sarsi has to say. He is careful not to reduce himself to asserting that the argument taken from the slight enlargement of very remote objects is worthless, for that is not true, but he

says that he and his master have never held it in great esteem. This he shows by the fact that his master has written very little about it, while his other two arguments are seen to be expanded and amplified without sparing words. To this I reply that it is not the number of words but their effectiveness which allows us to deduce other people's regard for what they say. Everyone knows that there are demonstrations which by their very nature cannot be explained except lengthily, while others exist in which length would be entirely superfluous and tedious. Now if one attends to the words here, the argument is conducted with as many as would suffice for a clear and complete explanation. Besides, Father Grassi himself writes (*p.* 65) that the argument, being necessarily implied by optical principles, has very great force in proving his point. This gives us a clear enough indication of the esteem which he at any rate has tried to establish for it, but which I am willing to believe with Sarsi was very small in his own eyes. I am persuaded of this not by the brevity of the argument but by another and much sounder conjecture. This is, that while the Father put up an appearance of demonstrating the comet to have been very distant because it behaved like the fixed stars in receiving little magnification from the telescope, when he went on immediately to determine this location more specifically he put it below certain objects which receive a very great magnification from that same instrument. As a matter of fact, he puts it beneath the sun, which, as the Father and Sarsi well know, is magnified hundreds or thousands of times. But Sarsi failed to penetrate the clever device by which his master attempted to give courteous approbation to his friends without at the same time allying the pleasure they felt over the invention of the new argument, meanwhile secretly showing himself very acute and well informed to those who are more intelligent and less emotional. In this, he was imitating the gallant act of that great gentleman who discarded his flush in order not to disturb the ecstasies of the young prince, his opponent, occasioned by the prospect of winning heavily with the 'fifty-five' he had just revealed and laid down on the table.¹⁶ But Sig. Mario, a trifle severely perhaps, has tried to state his ideas and to reveal the nullity and falsity of this argument with all his cards on the table. He was governed by a

different end [than courtesy], which was to try to remedy defects and correct errors with some compassion for the afflicted rather than to foment mistakes and magnify them in order not to incur displeasure.

At the end, Sarsi writes that his master had no thought of offending me in criticizing those who caviled at the argument. To this it is not necessary for me to reply further, since I have already said what I think and what I have never heard contradicted. And I wish that Sarsi would believe that neither did I, when proving the argument wrong, have (*p.* 66) any intention of offending his master; rather, I intended helping anyone who was in that error. Nor can I well understand what occasion there is here for my being taxed as one who had rather lose a friend than a jest; I do not even see how saying 'this argument is incorrect' can be regarded as particularly witty.

XIV

Now let your Excellency continue by reading: 'But let us now . . . (*p.* 67) . . . than the moon.' (*p.* 81, l. 18 to *p.* 82, l. 36).

Sarsi, as you see, is preparing here with admirable boldness to maintain by means of acute syllogisms that nothing can be more true than the frequently advanced proposition that objects seen through the telescope are the more enlarged the closer they are, and the less so the more distant; and such is his self-confidence that he almost promises that I shall admit this though at present I deny it. I believe that in the course of weaving this cloth he is going to get himself so entangled in it (more than he supposes now, when he is laying the warp) that in the end he will voluntarily confess himself vanquished. This will be apparent to anyone who will pay attention to the fact that he will end by saying precisely the same (*p.* 68) things that Sig. Mario wrote—though disguised in such a way, and fitted in piecemeal among such a variety of wordy ornaments and arabesques, or displaced and foreshortened at such angles, that perhaps those who consider them less carefully will at first think them somewhat different from what they essentially are.

Meanwhile I add, in order not to discourage him, that if what he is attempting is correct, then the argument supported by this proposition which is utilized by his master and his other astronomer friends

for finding the location of the comet is not only the most ingenious and most conclusive argument of all, but that this effect of the telescope far surpasses all others because of the important consequences which follow from it. And I cannot help being astonished that Sarsi, who knows it to be true, should have said a little while ago that he and his master had less regard for it than for the other two arguments—the one taken from circular motion and the other from smallness of parallax—which, if he will allow me to say so, are not fit to be handmaids to this one. Your Excellency, if this thing is true, the road is paved for Sarsi to the making of admirable inventions which many others have attempted but no one has ever achieved. Not only might any distance on earth be measured from a single place, but the distances of the heavenly bodies may be established without error. For once we had observed a circle one mile away which, when observed through the telescope, appeared to be thirty times as large in diameter as when seen with the naked eye, then as soon as we saw the height of a tower to be magnified ten times we could be sure that it was three miles distant. If the moon's diameter were three times greater than when seen with the naked eye, we could say that it was ten miles away, and the sun would be fifteen if its diameter merely doubled. Or, if we should see the moon enlarged three times by some fine telescope, and it is more than one hundred thousand miles away as Father Grassi well says, then the ball upon a cupola one mile away would increase in diameter more than a million times. Now (*p.* 69) to add what I can to so astounding a venture, I shall proceed to set forth a few trifling questions that arose in me as Sarsi went along. Should your Excellency choose, you may show these to him at some time so that by answering them he may establish his whole position more solidly.

Sarsi, then, wishes to persuade me that the fixed stars receive no appreciable enlargement from the telescope. He commences with objects inside the room, and he asks me whether I need to lengthen the telescope very much when I look at these; and I answer, yes. Now he passes to objects outside the window at a great distance, and he tells me that in order to see them it is necessary to shorten the instrument considerably, which I affirm. Next I concede to him that

this comes about from the nature of the instrument, as he says, which requires a longer tube in order to show objects which are very close, and a shorter one for those more distant. Moreover, I confess that the longer tube shows the objects larger than does the shorter one, and finally I grant him his whole syllogism for the present, the conclusion of which is that, in general, nearby objects increase more, and farther ones less. Adapting this to the particulars named, this means that the fixed stars, which are distant objects, are less enlarged than things placed within a room or in a courtyard—for it appears to me that Sarsi includes within such limits those things which he calls 'close,' he not having specifically removed this boundary to any greater distance.

But the statement thus far seems to me a long way from satisfying Sarsi's requirements. For next I ask him whether he places the moon in the class of nearby objects, or in that of distant ones? If he puts it with the distant ones, he will conclude the same for it that he does for the fixed stars, namely, slight enlargement. But this is a direct contradiction of his master, who in order to situate the comet above the moon requires that the moon be one of those objects which are greatly enlarged, and he even writes that the moon does grow greatly and the comet but little. Yet if Sarsi puts the moon among close objects (which are those that grow greatly), I shall reply to him that he should not at the outset have restricted the boundary of close objects to the walls of a house, but should have extended (*p.* 70) them at least to the sphere of the moon. Now having extended this boundary that far, let Sarsi return to his original question, and ask me whether I need to lengthen the telescope very much in order to see close objects—that is, those which are not beyond the orb of the moon. I answer, no; and here the archer's bow is broken and the shooting of syllogisms is over.

If we go back and consider his argument better, we find it to be defective. It takes as absolute that which must be understood relatively, or takes as bounded that which is unbounded; in a word, an incomplete dichotomy (as the error is called in logic) has been created when Sarsi divided visible objects into far and near without assigning boundaries and limits between closeness and remoteness.

He has made the same mistake as a person who might say, 'Everything in the world is large or small.' In this proposition, there is neither truth nor falsity, nor is there in saying, 'Objects are either near or far.' From indeterminacy of this sort, it comes about that the same things may be called very near and very far; the closer may be called distant and the farther close; the larger may be called small and the smaller large. Thus one might say, 'This is a very small hill,' and 'This is a very large diamond.' A courier calls the trip from Rome to Naples very short, while a great lady grieves that the distance from her house to the church is too great. In order to avoid this equivocation, if I am not mistaken, Sarsi needed to give his division at least three parts, saying: 'Of visible objects some are near, some far, and others are situated at a medium distance,' so that something would stand as a limit between the near and the far. Nor could he stop even there; he should further add an exact determination of the distance of this limit, saying for example: 'I call medium, a distance of one league; far, that which is more than one league; near, that which is less.' I fail to understand why he did not do this, unless it was that he perceived his case would be stronger if he furthered it by cleverly juggling equivocations before the simple-minded than it would be if he reasoned (*p.* 71) it soundly before the more intelligent. And truly it is a great advantage to have one's bread buttered on both sides and to be able to say, 'Because the fixed stars are distant, they enlarge very little, but the moon enlarges very much because it is close,' and then again, as the need arises, to say: 'Objects in the room, being close, increase a great deal; but the moon, because it is distant, does so but little.'

Second, Father Grassi does indeed give only a single reason for objects seen through the telescope appearing now more and now less enlarged; this is the lesser or greater distance of these objects; nor does he touch upon the lengthening or shortening of the instrument by a single word. Now Sarsi says that this is quite true; yet when he is constrained to demonstrate it, the great or little distance of the object is not sufficient for him and he has to adjoin the greater or lesser length of the telescope. He frames his syllogism as follows: 'The closeness of the object is the cause of lengthening the telescope;

but this lengthening is the cause of greater enlargement; therefore the closeness of the object is the cause of greater enlargement.' Here it seems to me that instead of rescuing his master, Sarsi makes him worse off, having him equivocate from *per accidens* to *per se*. It is thus that they err who would put avarice among the rules of hygiene, saying: 'Avarice is a cause of living soberly; sobriety is a cause of health; therefore avarice keeps one healthy.' Here avarice is an occasion, or at best a very remote cause *per accidens*, of health; health ensues outside the main intention of the miser as miser, whose end is thrift alone. What I say is so true that I shall with equal cogency prove avarice to be the cause of illness, thus: 'The miser, from thrifting at home, goes frequently to the festivities of friends and relatives; frequency of festivities causes many illnesses; therefore avarice is a cause of illness.' From such arguments, it ultimately appears that avarice, as avarice, has nothing to do with health. Nor does the closeness of an object have anything to do with its greater enlargement; the reason for lengthening the telescope when looking at nearby objects (*p.* 72) is to remove the confusion with which those objects are tinged, and this is removed by the lengthening. But since the lengthening is followed by a greater enlargement quite outside the main intention, which was to clarify the object and not to enlarge it, the closeness in this case cannot be called anything more than an occasion, or at best a very remote cause *per accidens*, of the greater enlargement.

Third, if it is true that only that may be properly called a cause which is always followed by the effect, and which when removed takes away the effect, then only the lengthening of the telescope may be said to be the cause of greater enlargement. Let the object be at any distance whatever; upon the slightest lengthening there will follow a noticeable enlargement. But on the other hand, so long as the instrument is kept at the same length, you may move the object as much closer as you please, reducing it even from a distance of a hundred thousand paces to but fifty, without the magnification of its naked-eye appearance thereby becoming one whit greater in the nearer position than in the farther. It is indeed true that bringing it closer at very small distances, such as four paces, or two, or one, or

one-half, makes the image of the object always cloudier and dimmer, so that in order to see it clearly and distinctly one must lengthen the telescope more and more; upon this lengthening there then follows a greater and greater enlargement, but, inasmuch as such enlargement depends only upon the lengthening and not upon the approach, it must be governed by that and by that alone. And since at a distance beyond half a mile there is no need to alter the instrument at all in order to see objects clearly and distinctly, no changes occur in their magnifications, and all take place in the same proportion. If the surface of a ball seen through the telescope at a distance of half a mile increases a thousand times, then so will the moon's disc increase a thousand times and no less; so will that of Jupiter, and finally that of a fixed star. Nor is there any such occurrence as that which Sarsi wishes to split hairs about and to review with all the rigor of geometry. (*p.* 73) When he has finished mincing it and reducing it to atoms and has taken every advantage to himself, his gain will not equal that of a man who has carefully gone about inquiring which gate of the city one should use in order to reach India most quickly. In the end, Sarsi will have to confess (as indeed he seems partly to do toward the close of the passage just read by your Excellency) that treating the telescope with every severity it would have to be shortened less than a hair's-breadth in order to look at the fixed stars instead of the moon. But from all this severity, what eventually emerges to uphold Sarsi? Absolutely nothing; all that he can harvest from it is that if, for example, the moon is enlarged a thousand times, the fixed stars are enlarged nine hundred and ninety-nine. But for his defense and that of his master they should not even double, for doubling is not an imperceptible thing, and they speak of the fixed stars as not increasing perceptibly.

I know that Sarsi understood these things very well, even in the reading of Sig. Mario, but he wants to keep his master alive as well as he can by means of a quintessence of syllogisms subtly distilled; we may put it thus because Sarsi will shortly call 'too minute' some points of Sig. Mario's which are actually fat when compared with these of Sarsi's. But to conclude my objections at last, it occurs to me to say something about the example which Sarsi has borrowed from

objects seen naturally. He says that as these are removed farther from the eye they are seen under constantly diminishing angles, but that after they arrive at a certain distance where the angle becomes very small, then even if an object is removed much farther the angle is not thereby sensibly diminished; still, he says, it can be proved that it becomes smaller. If the sense of this example is what it appears to me to be, and indeed as it must be if it is to square with the concept exemplified, I hold a very different opinion from Sarsi. For to me it appears that in effect he wants the visual angle to continue diminishing with the departure of the object, but always in a successively smaller ratio, (*p.* 74) so that beyond some great distance the object may be farther removed with little more diminution of the angle. I am of the contrary opinion, and I say that the diminution of the angle is made in a continually greater ratio the more the object is removed.

To explain this more simply, I note in the first place that the apparent magnitude of visible objects may be correctly determined by the size of the angles under which they are represented to us, [that is,] when we are dealing with parts of the circumference of a circle in whose center the eye is located; but in dealing with all other objects this is incorrect. For then it is the apparent magnitudes not of the visual angles, but of the chords of the arcs subtended by such angles, which must be determined; and these always diminish in just the opposite ratio to the distances. The diameter of a circle seen at a distance of one hundred yards appears to me as just one-half that which it would appear to me from a distance of fifty, and seen at a distance of one thousand yards it will appear twice what it will at two thousand, and so on for all distances, and it never happens that at some enormous distance it will appear so small as not to appear halved at double that distance. But if we still want to determine the apparent magnitudes of the angles, as Sarsi does, the facts will turn out still more unfavorable to him, for such angles diminish not in proportion as the distance increases, but in a smaller ratio. And that which contradicts Sarsi's statement is that, comparing the angles with one another, they diminish progressively in a greater ratio for larger distances than for smaller. If the angle of an object

placed fifty yards away is to the angle of the same object placed at a distance of one hundred yards as 100 is to 60, then the angle at a distance of one thousand will be to that at two thousand as, say, 100 is to 58; at a distance of four thousand it will be to that at eight thousand as 100 is to 55, and at a distance (*p.* 75) of twenty thousand the ratio will be as 100 is to 52. Thus the diminution of the angle will proceed in greater and greater ratios, without ever becoming interchangeable with the ratio of the distances. Sarsi writes that the visual angle, made very acute by great distance, does not continue to diminish at extreme distances in as great a ratio as it did at lesser distances. If I am not mistaken this is utterly false, and the diminution actually proceeds in an increasing ratio.

XV

Read now, your Excellency, 'But he will say . . . and then another?' (*p.* 83, l. 1 to l. 15).

Sarsi, as you see, here represents me as at length convicted through the force of his syllogisms and as snatching for protection at some very feeble support by saying that if it is true that the fixed stars do not receive enlargement as do nearby objects, then at any rate this is not using the same instrument, since for very close objects the telescope must be lengthened; and he adds with a 'get thee hence' that I am seizing at trifles. But it is not I, Sig. Sarsi, who need recourse to 'at any rate' and to minutiae. You indeed have needed these before now, and you will again as we proceed. You needed to say that in the very subtle concepts of geometry, at any rate, the fixed stars (*p.* 76) require more shortening of the telescope than does the moon; from this it later developed, as I have remarked before, that were the moon to increase one thousand times the fixed stars would increase nine hundred and ninety-nine, whereas to maintain your statement you would require that they grow not even twice. This is indeed reducing yourself to an 'at any rate,' Sig. Sarsi; it amounts to taking that to be a serpent which, scotched and trampled, no longer has any life in it outside the tip of its tail, which still goes on twitching in order to make the passers-by believe that it is yet alive and strong.

To say that the lengthened telescope is a different instrument from what it was before is quite essential to the point of what we are saying, and it is as true as can be. Nor would Sarsi have thought otherwise if he had not equivocated from the subject matter of what we meant to the form or figure of our argument, when he passed judgment upon us, as may readily be shown without even forsaking his own example. I ask Sarsi: Inasmuch as the pipes of an organ do not all sound in unison, why do some render deeper tones and some higher? Will he perhaps say that this comes about because they are of different materials? Surely not. They are all of lead, but they sound different notes because they are of different sizes; as to the material, that has no part whatever in the formation of the sound. Some pipes are made of wood, some of pewter, some of lead, some of silver, and some of paper; but all sound in unison when their lengths and sizes are equal. On the other hand, with the same quantity of material—that is, with the same few pounds of lead—there may be made now a larger and now a smaller tube, and different notes may be formed from it. With regard to the production of sound, those instruments are different which are of different sizes, and not those which are of different materials. Now, if by melting down one pipe and recasting the same lead one may make another and longer tube, and consequently one of lower pitch, will Sarsi refuse to say that this is a different pipe from the first? I think he will not. But if someone were to find a way to make the second, longer tube without melting down the first, would this not be the same thing? Surely it would. Now the method will be (*p.* 77) to make it of two pieces, one inserted in the other. This may be lengthened and shortened at will and made into diverse pipes which will produce various notes, and such is the construction of the trombone. The strings of a harp are all of the same material, but they render different sounds because they are of various lengths. A single string on a lute will accomplish what many do on the harp, because in playing this instrument the sound is drawn now from one part of a string and again from another part, and this is the same as lengthening and shortening it, and in a word altering it into different strings so far as the production of sound is concerned. The same may be

said of the tube of the throat, which, varying in length and breadth, accommodates itself to the formation of various voices and may correctly be said to become various tubes. And since a greater or lesser enlargement depends not upon the material of the telescope but upon its shape, in such a way that the longer shows things as larger, just so does the tube constitute different instruments when the same material is preserved but the interval between the lenses is altered.

XVI

Now let us hear the other syllogism which Sarsi frames. 'But let Galileo . . . a different way.' (p. 83, l. 16 to l. 30).

I concede this argument in its entirety, but I do not see that it proves anything detrimental to Sig. Mario's cause or favorable to Sarsi's. It is no advantage to Sarsi that nearby objects seen (*p.* 78) through a long telescope are more enlarged than distant ones seen through a short one. Such is the conclusion of his syllogism, though it is quite different from Sarsi's original project, which was to prove two main points. One of these was that objects as far off as the moon, and not merely those within a room, enlarge very much indeed, while the fixed stars enlarge not merely little but insensibly when viewed through the same instrument. The other was that the difference between these enlargements stems from the difference in distances of these objects and is in proportion to the distance. These things he will never prove in all eternity, since they are false. As to the present syllogism, so far as it concerns the matter in hand, let our evidence of its vapidness be that I shall proceed in his very footsteps and conclusively prove the exact opposite, thus: Objects requiring to be viewed through the same instrument receive from it the same enlargement; all objects from one-quarter of a mile to a distance of thousands of millions of miles require to be viewed through the same instrument; therefore all these receive the same enlargement. Yet Sarsi must not conclude that he has not written anything here that is foreign to the truth or to me; as to me, at any rate, I assure him that he has concluded something contrary to my meaning.

At the end of this section, he says that a telescope which is now long and now short may be called the same instrument though differently applied. If I am not mistaken, there is some equivocation here; to me it seems that matters stand quite the other way, and that it is the instrument which is varied while its application remains precisely the same. The same instrument is said to be differently applied when it is put to different uses without suffering any alteration; thus the anchor was the same when applied by the pilot to secure the ship and by Orlando to catch a whale,¹⁷ but it was differently applied. Our case is just the reverse, for the use of the telescope is always the same, it being always applied to looking at visible objects, while the instrument itself is diversified by its alteration in one essential respect, which is the interval between its lenses. Thus Sarsi's quibble is made manifest. (*p.* 79)

XVII

But let us proceed: 'But he will say . . . the appearance itself.' (*p.* 83, l. 31 to *p.* 84, l. 21).

Here, your Excellency, note first of all how my previous prediction under XIV, above, (*p.* 215) commences to be verified. First Sarsi belligerently maintained that nothing could be more true than that objects seen through the telescope are the more enlarged the closer they are, and the less so the farther off; whence the fixed stars do not increase much, being very distant, but the moon does, being close. Now we begin to see here a full retreat and an open confession. First, diversity of the distances of objects is no longer the real cause of diverse enlargements, but recourse is required to the lengthening and shortening of the telescope, something (*p.* 80) not mentioned or even hinted at (and perhaps not so much as thought of) by them until it was brought up by Sig. Mario. Second, not even this takes place in the present case, inasmuch as no alteration is made in the instrument. So this refuge fails, and the argument based upon it is completely invalid. In the third place, I see that he goes back to causes very remote from those he introduced at the beginning as the true and only ones; he says that the slight enlargement apparent in the fixed stars no longer depends upon either their great distance or

the shortness of the instrument, but that it is an optical illusion. He says that the naked eye sees the stars with a large unreal irradiation, and that because of this they appear large to us, while through the telescope the bare body of the star is seen, and this, though enlarged like all other objects, does not appear very great in comparison with the same star seen by the naked eye, in relation to which the enlargement appears very slight. From this, he concludes that at least as to appearance the fixed stars still exhibit a very slight enlargement, so that I must not be astonished that he and his master have said what they did, as they do not require any cause for such an appearance but merely the appearance itself. Pardon me, Sig. Sarsi; in attempting to remove my astonishment, you not merely fail to take it away from me, but you increase it a good deal with your new reasons.

First, I marvel not a little to see you introduce this discourse in a doctrinaire manner, almost as if you were trying to teach it to me, when you have learned it word for word from Sig. Mario. Moreover, you add that I do not deny these things, with the intention (I believe) of leaving the reader with the idea that I had the solution of the difficulty right in my hands but could not recognize it or avail myself of it. I marvel in the second place that you should say that your master does not seek the cause of the insensible enlargement of the fixed stars but merely the effect itself, whereas he more than once repeated that this cause was the enormous distance. But in the third place, what multiplies my astonishment a hundredfold is that you do not perceive that if this is true, you quite wrongly represent (*p.* 81) your master as lacking even that common sense by virtue of which everybody, even the merest idiot, correctly reasons and reaches his own conclusions.

To make palpable the truth of what I say, take away any consideration of the cause and set forth only the effect, since you affirm that your master requires not the cause but only the effect; then reason thus: 'The fixed stars enlarge insensibly; the comet enlarges insensibly; therefore . . .' Therefore what, Sig. Sarsi? What do you conclude from this? You had better reply, 'Nothing,' if you want to make the best of a bad job, for if you pretend to be able to infer one conclusion, I shall claim to infer a thousand equally well connected

with the premises. If you think one may say, 'Therefore the comet is very distant, because the fixed stars are also very distant,' then I shall say with no less right, 'Therefore the comet is incorruptible, because the fixed stars are also incorruptible,' and next I shall say, 'Therefore the comet twinkles, because the fixed stars twinkle,'¹⁸ and with no less reason I might say, 'Therefore the comet shines by its own light, because the fixed stars do.' But if I should draw these conclusions you would smile at me as a logician devoid of logic, and you would be quite right. You would courteously point out to me that from these premises I cannot infer anything about the comet except such things as have an essential connection with the insensible enlargement of the fixed stars. Since this enlargement does not depend upon and has no connection whatever with incorruptibility or with twinkling or with intrinsic light, none of these conclusions may be drawn with regard to the comet. And he who wishes to infer from the premises that the comet is very distant must first have established conclusively that the insensible enlargement of the stars depends causally upon great distance, for otherwise the converse would not be available; that is, that those objects which enlarge insensibly must necessarily be very remote. Now you see what errors in logic you have unworthily saddled upon your master; I say unworthily because this error is yours, and not his. (*p.* 82)

XVIII

Now, your Excellency, read to the end of this First Weighing: 'But now let Galileo . . . (*p.* 83) . . . Galileo's own conclusions.' (*p.* 84, l. 22 to *p.* 86, l. 8).

In the first place, we have here an argument patched, so to speak, upon the old one. It consists of diverse fragments of propositions designed to prove that the comet was between the moon and the sun. Sig. Mario and I may concede this whole thing to him without prejudice, since we have never affirmed anything about the location of the comet nor denied that it might be above the moon; we have only said that the demonstrations so far introduced by other authors are not immune from objections. Sarsi accomplishes nothing in the way of removing these objections by adducing new demonstrations here

to prove the conclusion true (even if these were conclusive and rigorous), inasmuch as it is possible to argue incorrectly and to commit paralogisms and fallacies even about true conclusions. Still, since I like to have hidden things come to light and since I desire to win to true conclusions, I shall proceed to advance a few considerations touching this discourse, and for clearer understanding I shall first (*p.* 84) reduce it to the shortest exposition I can.

He says, then, that he has it from my *Starry Messenger* that the fixed stars, shining with their own light, are much irradiated with a fulgor which is not real but only apparent. The planets, having no light of their own, are not thus irradiated, particularly the moon, Jupiter, and Saturn, which are seen to be almost barren of any such splendor. Venus, Mercury, and Mars, though lacking any light of their own, are nonetheless irradiated because of their proximity to the sun, whose rays touch them the more brightly. He goes on to say that in my opinion the comet receives its light from the sun, and he adds that he himself, with other reputable authors, has regarded the comet as a temporary planet, and that thus it may be reasoned about as are the other planets, of which those that are closer to the sun are more irradiated and consequently are less enlarged when seen through the telescope. And since the comet was enlarged little more than Mercury and much less than the moon, it would be reasonable to conclude it to be not much farther from the sun than Mercury is, and very much closer to it than the moon. Such is his argument, and it fits so smoothly and assists his needs so neatly that it is as if the conclusion had been framed before the premises and the latter depended upon the former instead of the other way round. It is as if the premises had not come from Nature's bounty but had been prepared for the conclusion by the preciseness of the subtlest art. And now let us see how conclusive they are.

First, it is quite false that I wrote in my *Starry Messenger* that Jupiter and Saturn have little or no irradiation, but that Mars, Venus, and Mercury are grandly crowned with rays; it was the moon alone that I thus sequestered from the rest of the stars, fixed and wandering alike.¹⁹ Second, in order to make the comet a quasi-planet and as such to dress it in the properties of the other planets, I am not so

sure that it is sufficient for Sarsi, his master, and the other authors to have regarded it as one and to have named it so. If their opinions and their voices have the power to call into existence the things they have considered and named, why then I beg them to do me the favor of considering and naming 'gold' a lot of old hardware that I have about my house. But names aside, what (*p.* 85) condition induced them to deem the comet a temporary quasi-planet? That it shone like a planet, perhaps? But what cloud, what smoke, what wood, what wall, what mountain touched by the sun does not shine equally? Has not Sarsi seen it demonstrated in my *Starry Messenger* that the terrestrial globe itself shines more than the moon does? And why do I speak of the comet shining like a planet? For my part I do not hold it impossible that the comet's light might be so weak and its substance so thin and rare that if anyone could get close enough to it he would lose it completely from view, as happens with some fires which emerge from the earth and are seen at night and from afar, but which are lost when close at hand.³⁰ In the same way, we see distant clouds sharply bounded, which later from nearby show only a little misty shadowiness so indefinitely bounded that a person entering into them may almost fail to distinguish their bounds or to discriminate them from the surrounding air. And those projections of solar rays through rifts in the clouds, which so resemble comets, are never seen except by people who are far from them. Or must the comet be included with the planets by reason of motion? But what thing separated from the elemental parts (which would obey the terrestrial condition) is there that does not move in the diurnal motion along with the rest of the universe? And if the longitudinal (*traversale*) motion is meant, this has nothing to do with planetary motion, being neither in the same direction, nor uniform, nor perhaps even circular. And, apart from these matters, does anyone believe that the substance and material of a comet are similar to those of the planets? The latter may be said to be most solid, for the moon in particular convinces us of this quite palpably, as does in general the sharply bounded and unchanging shape of all the planets. The substance of the comet, on the other hand, may be believed to dissolve within a few days, and its shape, which is not circularly

bounded but confused and indistinct, gives us an indication that its material is rarer and more tenuous than fog or smoke. In a word, the comet more resembles a toy planet than the real thing.

Third, I do not know how completely he may have compared the comet's irradiation and (*p.* 86) enlargement with those of Mercury, which he surely did not see and could not have seen during all the time the comet was visible; the occasions Mercury gives for being observed are quite rare, and during that period it was always quite close to the sun. So I think I may safely believe that Sarsi did not make such a comparison at all, difficult and uncertain of accomplishment as it is, but that he speaks thus because it would better serve his cause if it were so. That he did not get round to this test is indicated to me also by the fact that, in referring to observations of Mercury and the moon, with which he compares those made of the comet, he seems to me to be somewhat confused. To prove the comet to have been farther from the sun than Mercury is, he needed to say that it was less irradiated than Mercury and more enlarged when seen through the telescope; yet he wrote this down in reverse, saying that it was not much more irradiated than Mercury and that it received almost the same enlargement, which amounts to saying that it was more irradiated and less enlarged than Mercury. Next, comparing the comet with the moon, he writes the same (although he speaks of writing the opposite); that is, that it was enlarged less than the moon and was more irradiated. Yet later, in concluding, he draws contrary conclusions from identical premises, saying that the comet is closer to the sun than the moon is, and farther from the sun than Mercury is.

Finally, since Sarsi professes to be a very precise logician, I do not know why, in his classification of luminous bodies which are much or little irradiated and which in consequence receive a lesser or greater enlargement when seen through the telescope, he has omitted our elemental lights. For candles shine, and blazing torches seen from a distance, and some pebbles, and bits of wood or other small bodies, and even the leaves of plants or drops of dew struck by the sun; and from certain viewpoints these are irradiated as much as the most refulgent star, and when seen through the telescope they

show the same type of enlargement that stars do. So there is no help from this source for those people who promised themselves that by means of the telescope they could transport the comet to the sky and remove it from the (*p.* 87) elemental sphere. Meanwhile, let Sarsi stop thinking he can rescue his master, and learn that to try to maintain one error is to be forced to commit a hundred, and, what is worse, to be defeated in the long run. I also beg him to stop repeating, as he still does at the end of this section, that these doctrines of his are mine; for I never wrote any such thing, nor said it, nor thought it. So much for the First Weighing.

XIX

Now let us proceed to the Second. 'Although up to . . . (*p.* 88) in any way elemental.' (*p.* 86, l. 15 to *p.* 87, l. 23)

Previously Sarsi went along arbitrarily shaping premises fitted to the conclusion he intended to prove; now it seems to me that he proceeds to shape conclusions in order to oppose them as ideas belonging to Sig. Mario and me, though they are quite different from those set forth in Sig. Mario's *Discourse*, or at least differently construed. That the comet is beyond doubt a mere image and a simple appearance has never been positively affirmed; it has merely been raised as a question, and offered for consideration by philosophers, together with such reasons and conjectures as appear suitable to convince them that such might be the case. Here are Sig. Mario's words in this connection: 'I do not say positively that a comet is formed in this way, but I do say that just as there are doubts about this, so there are doubts about the methods used by other authors; and if they claim to have established their opinion beyond doubt, they will be under obligation to show this and all other positions to be vain and fallacious.'²¹ With similar distortion, Sarsi represents us as having definitely declared that the motion of the comet must necessarily be straight and perpendicular to the earth's surface, something which is not proposed in that way but is merely brought under consideration in that it would explain the changes observed in the comet more simply and in better agreement with the appearances. The idea is so temperately proposed by Sig. Mario

that ultimately he says these words: 'Hence we must content ourselves with what little we can conjecture thus among the shadows.'²² Yet Sarsi has tried to represent these opinions as being the more firmly believed by me as he has imagined himself able the more effectively to annihilate them—which (*p.* 89) if he does, I shall hold myself obliged to him, as in the future I shall have one less opinion to consider when I set my mind to philosophizing about such matters. But since it seems to me that there is still a little life left in Sig. Mario's conjectures, I shall proceed to make a few remarks about the strength of Sarsi's objections.

Attacking the first conclusion with great boldness, he says that to anyone who had once looked at the comet no other argument would be necessary to prove the nature of its light; for compared with other true lights it showed itself only too clearly to be true and not spurious. As your Excellency will note, Sarsi has such confidence in the sense of sight that he deems it impossible to be deceived any time that a spurious object may be set beside a real one. I confess that I do not have such a perfect discriminatory faculty, but resemble the monkey that firmly believed he saw his mate in a mirror, and so live and real did the image seem to him that he did not discover his error until he had run behind the mirror two or three times to catch her. Assuming that what Sarsi sees in his own mirror are not true and real men at all, but are mere images like those which the rest of us see in ours, I should like to know what those visual differences are by which he so readily distinguishes the true from the spurious. For my part, I have countless times been in some room with closed shutters and have seen on the opposite wall the reflection of sunlight coming through some tiny hole, and as far as vision could tell I have judged it to be a star, no less bright than Venus or the Dog Star. When we walk over a field toward the sun, in how many thousands of straws and pebbles, a little polished or moistened, will the reflection of the sun be seen in the aspect of the most splendid stars? Sarsi has but to spit upon the ground, and undoubtedly he will see the aspect of a natural star from that point toward which the sun's rays are reflected. Furthermore, any body placed at a great distance and struck by the sun will appear as a star, particularly if it is placed

so high as to be seen by night as the (*p.* 90) other stars are seen. And who could distinguish the moon seen in daytime from a cloud touched by the sun, were it not for differences of apparent shape and size? Surely, no one. Finally, if simple appearances can determine the essence of a thing, Sarsi must grant that the suns, moons, and stars seen in still water or in a mirror are true suns, real moons, and actual stars. But Sarsi had better change his mind about this, and not suppose that he can improve his position by citing the authority of Tycho, Thaddeus Hagek, or others—except insofar as having such men for company makes his own error the more excusable.

XX

Continue to read, your Excellency: 'Yet at that time . . . aimless and uncertain.' (*p.* 87, l. 24 to *p.* 88, l. 14). (*p.* 91)

Neither Sig. Mario nor I ever wrote that fuming vapors rise from some parts of the earth to the moon and even to the sun, and that having got outside the cone of the terrestrial shadow they are impregnated by solar rays and give birth to the comet, though Sarsi would attribute this to me. What Sig. Mario did write is that he does not consider it impossible that sometimes there may be raised from the earth exhalations and other such things so much subtler than usual that they would ascend even to the moon, and might be material for the formation of a comet; and that sometimes there occur unusual sublimations of the twilight material, as exemplified by the aurora borealis. But he does not say that this is identical with the material of comets, which must necessarily be much rarer and thinner than the twilight mists and the material of the aurora borealis, since the comet shines much less than the aurora. Thus if the comet appeared to the east in the whiteness of dawn while the sun was no more than six or eight degrees from the horizon, no doubt the comet would not be discerned, for it would be less bright than the surrounding field. Likewise, straight motion upward is attributed to the same material only with probability and not with certainty. This is said not to retreat in fear of Sarsi's objections, but merely to let it be seen that we are not departing from our custom of declaring nothing as certain except what we know beyond doubt, as our philosophy and

mathematics teach us. Now, it being granted that we have not said the things with which Sarsi burdens us, let us hear and examine his objections.

His first objection is founded upon the impossibility of vapors leaving by a straight line toward the sky when a swift north wind is distorting the air and everything within it. Such a wind was felt by him for many days after the appearance of the comet. The objection is really very ingenious, but it loses a good deal of its force through our having reliable information that there was not the slightest disturbance of the air in either Persia or China during those days, so I shall believe that the comet's material arose from one of those regions unless (*p.* 92) Sarsi proves to me that it came not from thence but from Rome, where he felt the northerly blast. And even if the vapors had left from Italy, who knows but that they had set out before those windy days? For many days would have passed before their arrival at the orb of the comet, this being some 470,000 miles away according to Sarsi's master. After all, such a journey takes time, and not a little of it. The ascent of vapors as observed from the earth falls a good deal short of the speed of a bird in flight, so that not even a few years would suffice for such a journey. But assuming that such vapors were to move during a windy season, a man who has complete faith in storytellers and poets should admit that the commotion of the wind does not rise more than two or three miles high, for there are mountains whose summits transcend the windy regions.²³ Hence the most that might be concluded would be that within that space the vapors would not go vertically but would fluctuate in their course, while beyond this there would be no interference which would twist them from a straight path.

XXI

Now, your Excellency, continue: (First argument) 'But finally, . . . [*p.* 93](*p.* 94) . . . be carried upward.' (*p.* 88, l. 15 to *p.* 89, l. 34).

Long experience has taught me that with regard to intellectual matters, this is the status of mankind: The less people know and understand about such matters, the more positively they attempt to reason about them, and on the other hand the number of things

known and understood renders them more cautious in passing judgment about anything new. There once lived, in a very solitary place, a man endowed by nature with extraordinary curiosity and a very penetrating mind. He raised many birds as a hobby, much enjoying their songs, and he used to observe with great admiration the happy contrivance by which they would transform at will the very air they breathed into a variety of sweet songs. Close to his house one evening, he chanced to hear a delicate sound, and, being unable to imagine what it could be except some small bird, he set out to capture it. Arriving at the road, he found a shepherd boy who was blowing into a kind of hollow stick and moving his fingers about on the wood, thus drawing from it a variety of notes similar to those of a bird, though by quite a different method. Puzzled, and led on by his natural curiosity, he gave the boy a calf in exchange for his recorder and retired to solitude. Realizing that if he had not chanced to meet the boy he would never have learned of the existence of two methods for forming musical notes and very sweet songs, he tried traveling far from his home in the hope of meeting with some new adventure. The very next day he happened to pass near a small hut, and, hearing a similar tone within, he went inside to find out whether it was a recorder or a blackbird. There he found a boy holding a bow in his right hand and sawing upon some fibres stretched upon a concave piece of wood. The fingers of the left hand (which supported the instrument) were moving, and without blowing the boy was drawing from it various notes, and most sweet ones too. Now, you who are participating in this man's mind and sharing in his curiosity, judge his astonishment! Finding himself to have two unexpected new ways of forming tones and (*p.* 95) melodies, he began to believe that still others might exist in nature. His wonder increased when upon entering a certain temple he glanced behind the gates to learn what it was that had sounded, and perceived that the noise had emanated from the hinges and fastenings as he had opened the gate. Again, impelled by curiosity, he entered an inn expecting to see someone lightly bowing the strings of a violin, and instead saw a man rubbing the tip of his finger round the rim of a goblet and drawing forth from it a very sweet sound. And later he observed that wasps, mosquitoes,

and flies did not form separate notes from their breaths, as did his original birds, but made steady tones by the swift beating of their wings.

In proportion as his amazement grew, his belief diminished that he knew how sounds were created; nor could all his previous experience have sufficed to make him understand or even believe that crickets, which do not fly, could draw their sweet and sonorous shrilling not from breath but from a scraping of wings. And when he had almost come to believe that there could be no further ways of forming notes—after having observed in addition to what has been recounted numerous organs, trumpets, fifes, stringed instruments of various sorts, and even that little iron tongue which when placed between the teeth makes strange use of the buccal cavity as a sounding box and of the breath as a vehicle of sound—when, I say, he believed that he had seen everything, he found himself more than ever wrapped in ignorance and bafflement upon capturing in his hand a cicada, for neither by closing its mouth nor by stopping its wings could he diminish its strident sound, and yet he could not see it move either its scales or any other parts. At length, lifting up the armor of its chest and seeing beneath this some thin, hard ligaments, he believed that the sound was coming from a shaking of these, and he resolved to break them in order to silence it. But everything failed until, driving the needle too deep, he transfixed the creature and took away its life with its voice, so that even then he could not make sure whether the song had originated in those ligaments. Thereupon his knowledge was reduced to such diffidence that when asked (*p.* 96) how sounds are generated he used to reply tolerantly that although he knew some of the ways, he was certain that many more existed which were unknown and unimaginable.

I might by many other examples make clear the bounty of nature in producing her effects by means which we would never think of if our senses and experience did not teach us of them, though even these are sometimes insufficient to remedy our incapacity. Therefore I should not be denied pardon if I cannot determine precisely the manner in which comets are produced, especially as I never boasted that I could, knowing that it may occur in some way far beyond our

power to imagine. The difficulty of understanding how the cicada's song is formed even when we have it singing to us right in our hands is more than enough to excuse us for not knowing how a comet is formed at such an immense distance. Let us therefore stop at Sig. Mario's and my primary purpose, which is to set forth those questions which have appeared to us to throw doubt upon the opinions previously held, and to propose some new considerations. Let us examine and consider whether there is anything that can give us light in any way and can pave the road for the discovery of truth, as I go on to consider the arguments which Sarsi has raised against us and which made our conceptions appear improbable to him.

Sarsi proceeds and grants to us that if there were no dispute about the existence of vapors or other material suitable for the formation of a comet, difficulties would occur in their rising from earth and ascending to the uppermost regions where they could receive the solar rays directly and reflect them to us. How is it that if the whole of them is illuminated, only the reflection of a single part comes then to us? Why is the reflection not made as in those vapors which display the aurora borealis, which are lighted all over and show themselves to us so? He adds that he has beheld a marvelous thing in the middle of night, that is, a little cloud near the zenith, illuminated all over and sending its splendor to us most liberally from every part. And, he continues, all dense and opaque clouds send sunlight to us from all (*p.* 97) [those] parts [which are] seen, while thin ones are penetrated by the light and look bright to us without having any shady part. So if the comet is formed as Sig. Mario says in such misty vapors, widely extended and not collected into a spherical figure, and if these are touched all over by the sun, why do we receive the reflection of only a little globe and not that of the equally illuminated remainder?

Even though the solutions to this objection are fully set forth in Sig. Mario's *Discourse*, I shall nevertheless proceed to answer them here in proper order, with the addition of some further considerations as the counter-arguments occur to me step by step. First, Sarsi should have no difficulty whatever in conceding that in all the material sublimated by a comet, the reflection of the sun's light to

the eyes of a particular observer may be made from only one particular place even though the whole is equally illuminated. In support of this we have a thousand experiences for every one that may seem to contradict it; indeed, among those produced by Sarsi as contradicting this position we shall readily find that the majority favor it. No doubt exists that of any plane mirror exposed to the sun the whole surface is illuminated, and the same is true of every marsh, lake, river, sea, and in a word every smooth surface of whatever material. Yet to the eye of a particular person the reflection of solar rays is made only from one particular place on this surface, and that place changes with the moving of the observer's eye. The external surface of thin clouds diffused over a wide space is all equally illuminated by the sun, yet haloes and mock suns display themselves to a particular eye as only at a particular place, which likewise changes location with every movement of the eye.

Sarsi says, 'That thin sublimated material which sometimes produces the aurora borealis is nevertheless seen as entirely illuminated, as in fact it is.' But I ask Sarsi how he can be sure of this. He can reply to me only that he does not see any part which is not illuminated, (*p.* 98) while he sees all the rest of the surfaces of mirrors, waters, and marble slabs, over and above the small portion which renders the live reflection of the sunlight. True; but I should call his attention to the fact that if the material were similar in color to all its surroundings, or if it were transparent, then he would not distinguish anything except that single bright reflected ray, as sometimes happens when the surface of the sea cannot be distinguished from the air but the reflected image of the sun may be discerned. Thus, placing a thin piece of glass at some distance, we can see only that single part in which a light is reflected while the rest remains invisible on account of its transparency. Sarsi's error here is like saying that no criminal may ever rest assured that his crime will remain hidden, while ignoring the incompatibility which exists between remaining hidden and being known. Doubtless anyone by whom two registries were kept, one for offenses which remain concealed and the other for those which are discovered, would never have anything at all registered and noted in the former. So I say

that one may believe without any inconsistency that the material of the aurora borealis is diffused through a very great space and is uniformly lighted by the sun, but since only that part is revealed and made visible to me from which refractions come to my eyes, the rest remaining invisible, I seem to see the whole. Besides, is not one hemisphere of the twilight vapors which surround the entire earth always equally illuminated by the solar rays? Surely it is; yet that part which lies directly between us and the sun looks much brighter than the more distant parts. This effect, like the others, is but an appearance and an illusion of our eyes, since wherever we are we always see the solar body as the center of a luminous circle which loses brilliance gradually as the distance from this center increases to one side or the other. But to people farther north than we, the part which is brightest for me would appear darker and what looks darkest to me would be brighter for them; thus we might say that there exists around the sun a huge perpetual (*p.* 99) halo traced upon the convex surface which bounds the vaporous sphere. This halo changes place according to the movements of the observer in the same way as do those haloes which are sometimes formed in thin clouds.

As to the little cloud which Sarsi says he has seen so bright in the darkest night, I might ask him how he was sure that it was not likewise larger than what he saw—especially since he says that it was in a sense transparent, as it did not veil even the faintest of the fixed stars. For that very reason, there could be no way for him to be sure that it did not extend invisibly far beyond the limits of its observed lighted part, it being quite transparent; hence it remains in doubt whether this was not also one of the appearances which alter with an alteration of the position of the eye. Besides, there is no reason why it could not all appear luminous and still be a mere apparition, as would happen if it were not larger than the space occupied by the image of the sun. Thus if the image of the sun occupies one finger's-breadth in a mirror, and we cut away the rest of the mirror, there is no doubt that this little bit remaining could appear to us bright all over. Furthermore, if the mirror were smaller than the image, it not only might be seen bright all over, but the image in it would not appear to move with every movement of

the eye, as it does in a large mirror; rather, since the mirror would be inadequate for the sun's entire image, it would follow that upon moving the eye the reflection would be now of one part and now of another part of the solar disc, and thus the image would appear immovable until the eye arrived at some place to which the image was not directed and would therefore be entirely lost. Hence, it is very important to consider the size and quality of the surface in which the reflection occurs, for accordingly as this surface is less smooth, the image of the same object will appear larger and larger, and sometimes the eye must travel a long way before the image completely runs off the mirror, (*p.* 100) and this image will appear fixed although it is actually mobile.

The better to explain a point which is of the utmost importance and which may perhaps give someone (I shall not say Sarsi) a new idea, let your Excellency fancy himself at the seashore at a time when the water is quite tranquil and the sun is already declining toward the west. You will see a very bright reflection of the sun in the surface of the sea near the vertical line passing through the solar disc. This will not spread over a large area; indeed, if the water is very peaceful as I said, you will see a pure image of the solar disc bounded as in a mirror. Next let a light breeze spring up and ruffle the surface of the water, and simultaneously your Excellency will see the sun's image begin to break into many parts and spread and extend into a larger space. If you were close by, you might be able to distinguish the broken pieces of the image from one another, but from a greater distance you would not see this separation because of the narrowness of the intervals between the pieces, or because by their great brilliance the shining parts would mix together and behave as do several fires close together which from afar appear to be one. Now let the ruffling proceed into greater waves, and that multitude of mirrors from which the image of the sun will be reflected to the eye according to the various slopes of the waves will extend into wider and wider intervals. Next, withdraw to a greater distance and climb a hill or some other prominence in order to see the water better, and the lighted field will appear one and continuous. From a very high mountain about sixty miles from the

Bay of Leghorn, on a clear and windy day about an hour before sunset, I have seen a very bright strip spreading to right and left of the sun and occupying a length of tens and perhaps hundreds of miles which was the same kind of reflection of sunlight as those just mentioned.

Now let Sarsi imagine most of the sea on both sides to be removed so that there remains only a breadth of two or three miles (*p.* 101) at the middle, pointing toward the sun. This surely would all be illuminated, and it would not be movable with every motion of the observer to one side or the other—unless perhaps he were to move several miles, for then he would begin to lose the left side of the image (if he moved toward the right) and the bright image would become constricted until it was made very thin and even vanished entirely. But it is not in this way that the image would actually move with the motion of the observer; rather, we should continue to see it entire, though all of it would move so that its center would always correspond with the direction of the sun, which for other people observing it at the same time would correspond with various other points on the horizon.

Here I wish to tell your Excellency something that has occurred to me as the solution of a maritime problem. Experienced sailors sometimes recognize that a wind will come to them from a certain part of the sea before long, and they say that a sure sign of this is to see the air in that direction more bright than it would ordinarily be. Consider, your Excellency, whether this might not arise from a wind disturbing the waves over there already. From such waves, as from many mirrors spread over a wide area, there would originate a much brighter reflection of the sun than would exist if the sea were calm. Then that part of the vapor-laden air may be made brighter by this new light and by the diffusion of that reflection. This air, being high, sends also some reflection of light to the eyes of the sailors, while they, being low and far off, would not be able to receive the primary reflection from that part of the sea which is already being ruffled by the wind twenty or thirty miles away. And that is how they perceive and predict a wind from afar.

But pursuing our original idea, I say that the solar rays do not

impress the image of the sun with the same size in all materials, or I should say upon all surfaces. In some, namely those which are plane and smooth like a mirror, the solar disc is shown to us as bounded and as equal (*p.* 102) to the actual sun; in smooth convex surfaces it appears smaller to us, and in concave ones it appears sometimes smaller and sometimes larger and sometimes even the same, in accordance with the different distances between the mirror, the object, and the eye. If the surface is not even but is wavy and full of elevations and depressions (composed, as we might say, of a great multitude of little mirrors placed at varying tilts and exposed to the eye in thousands of ways), then this same image of the sun, divided into thousands of pieces, will meet our eye as thousands of parts spread over a large space. Thus it will impress us as a collection of many little bright patches. The number of these will make it appear from a distance as a single continuous luminous field, stronger and more vivid in the center than toward the edges. At the edges the light becomes languid, and finally it vanishes mistily when the obliquity of the eye to this surface no longer causes the visual rays to find any waves to reflect them toward the sun. This large image is also capable of moving with a movement of the eye, provided the surface where the reflection takes place continues beyond the limits of the image. If the material occupies a small space (smaller than that of the whole image), it will continue to appear bright until the eye arrives at that boundary beyond which the obliquity of incident rays upon the material causes the reflections to be no longer directed toward the sun, when the light vanishes and is lost.

Now I tell Sarsi that when he saw a cloud hanging in the sky, sharply bounded and lighted all over—and remaining so even though the eye might change its position through some distance—he could not thereby be sure that that illumination was a thing any more real than haloes, mock suns, rainbows, and reflections in the sea. I tell him that its consistency and apparent stability may depend upon the smallness of the cloud, or this may make it incapable of receiving the entire expanse of the sun's image, which would broaden out through a space many times larger (*p.* 103) than the cloud if there were no lack of material. And if it were seen entire at a time when other

fields of clouds extended beyond it, I say that the movement of the eye would also cause it to move in its entirety. A conclusive argument of this for us is our often seeing at sunrise or sunset many little clouds suspended close to the horizon; those nearly in line with the sun show most splendidly and look almost like fine gold. Of those to the side, the ones less distant from the sun are of medium light and are brighter than those more distant, and by degrees they show themselves continuously less lighted, until finally those that are very remote have little or no brilliance—to us, that is; but to anyone so situated that they came between his eye and the place of the setting sun, these would be the brightest. Therefore, let Sarsi understand that if the clouds were not scattered but formed one continuous stretch, it would come about that to each observer that part which was at the center for him would look most lighted, and at the sides it would appear less bright according to the distance from the center; thus what is the summit of splendor for me is its terminus for others.

Here someone might say that since that bit of cloud remained fixed, and the light in it was not seen to move with every change of place by the observer, this should be sufficient to make parallax acceptable for determining its altitude; and, the same thing being possible for the comet, it is proper for those people to utilize parallax who seek to determine its location. To this it is replied that this would be correct if one first demonstrated that the comet was only a part of an entire image of the sun and not a whole one. Then not only all the material in which the comet was formed would be illuminated, but the image of the sun would pass beyond its edges and would suffice to illuminate a much larger area if there existed material there disposed to reflect light. Now this not only has not been demonstrated, but one may very reasonably believe the contrary; namely, that the comet is an entire image and not a mutilated or truncated one, for this is argued by its regular shape which is designed with such beautiful (*p.* 104) symmetry.²⁴ From this may be drawn a simple and apt reply to Sarsi's objection when he asks how one can imagine (in accordance with Sig. Mario's statement) the comet's being in a material extending through a wide space above,

without this material being entirely illuminated, but with only a small circle of it sending the reflection back to us. Let me ask the same question of him or of his master, who will not have the comet be a fire, but (if I am not mistaken) who inclines to the belief that at least its tail is a refraction of solar rays. I ask him whether he believes that the material in which such a refraction is made is cut exactly to the measure of this tail, or that here and there it extends beyond it? If it extends beyond, as I believe he will reply, then why is this material not seen, since it is touched by the sun? One cannot say here that the refraction occurs in the substance of the aether, which being most transparent has no refracting power; still less does it occur in other material, for if there were anything suitable for refracting it would also be suitable for reflecting the solar rays. Besides, I do not see what right he has to call the head of the comet a 'little circle' when his master, by profound calculations, has found it to contain 87,127 square miles. Perhaps no cloud has ever achieved such proportions.

Resembling our man of a short time ago who was of the opinion that sound could be produced in but a single way, Sarsi goes on to say that it is impossible for the comet to be generated by reflection in smoky vapors, and that the example of the rainbow does not reduce the difficulty even though it really is an illusion of sight. For the generation of rainbows and other such things requires a moist material which is already resolving into water and which only then acts essentially like smooth and polished bodies, reflecting light from those parts where there are the proper angles of reflection and refraction required for the production of such an effect, as happens in mirrors, water, and crystal balls, while in thin and dry things not possessing smooth surfaces like mirrors, not much refraction takes place. Thus there is required for this an aqueous material, and (*p.* 105) consequently a heavy one incapable of rising above the moon and the sun (where in my opinion also nothing can go except things which are in the nature of very light exhalations), and hence, a comet cannot be produced in those smoky vapors.

A sufficient reply to all this reasoning would be to say that Sig. Mario has never limited himself as to the precise material in which a

comet takes shape, nor said whether this is moist or smoky or dry or smooth, and I am sure that he would not blush to say that he does not know. But seeing that in vapors, in thin non-aqueous clouds, and in those which do resolve into tiny drops; in stagnant waters, in mirrors, and in other materials there occur various illusions of diverse images which take shape by means of reflections and refractions, he has deemed it not impossible that there exists in nature still another material, designed to render us one more image different from all the rest, and that this image is the comet. Such a reply, I say, is adequate to the objection, if indeed any part of the objection were valid. Yet, as I have said before, my desire to pave the road for the investigation of truth induces me, so far as I am able, to make some remarks about certain particular things in the foregoing reasoning.

First, it is true that the illusion of the rainbow takes place in an effluvium of very tiny drops of water, but I do not believe that conversely no similar illusions can take place without such an effluvium. A triangular crystal prism brought close to the eye shows us all objects as tinted with the colors of the rainbow, and rainbows are often seen in 'dry' clouds without any rain whatever descending to the earth. Are not the same varicolored illusions seen in the feathers of many birds when the sun strikes them in certain ways? And that is not all; let me tell Sarsi something which is perhaps new—if anything new can be said to him. Let him take any material, whether stone or wood or metal, and holding it up to the sun gaze at it most intently. There he will see all colors appearing in most minute specks; and if in looking at this, he will make use of a telescope set for looking at very near objects, he will see what I mean very much more distinctly and without any need of these bodies being resolved into dew or moist vapors. Besides, those little clouds that (*p.* 106) look so bright in twilight and reflect the sun to us so vividly that it almost dazzles us are among the thinnest and driest which exist in the atmosphere, while those which are moist look the darker the more pregnant with water they are. Haloes and mock suns occur without rain and without moisture in the thinnest and driest clouds (or rather fogs) which occur in the air.

Second, it is true that smooth and well-polished surfaces such as those of mirrors render a strong reflection of the sun's light to us, so much so that we can hardly look at them without injury, but it is also true that a reflection is made from surfaces not so smooth, though a less powerful one in proportion as the polish is less. Now your Excellency may see whether the splendor of the comet is among those which dazzle the vision or among those which by reason of their feebleness do not offend at all. From this you may judge whether there is required for its production a surface resembling a mirror, or whether a surface much less smooth suffices. I should like to show Sarsi a method of representing a reflection very similar to that of a comet. Take a very clean carafe, your Excellency, and holding a lighted candle not far from it you will see in its surface a tiny image of this light, very bright and sharp. Then take upon the tip of your finger a small quantity of any slightly oily material that will adhere to the glass, and oil it as thinly as you can at the place where the image is seen, so that the surface becomes a little dimmed. Immediately you will see the said image become dim. Next turn the glass so that the image emerges from the oily spot and just touches its edge, and rub once with your finger straight across the oily part. Instantly you will see a straight ray formed in imitation of the comet's tail, and this ray will cut transversely and at right angles across the rubbing you have made with your finger; if you rub across this once more, the same ray will be led away in another direction. This happens because the skin on the ball of the finger is not smooth, but is marked with certain twisted lines utilised by the sense of touch for feeling the most minute distances in (*p.* 107) tangible things, and in moving the finger over the oily surface these leave some very thin tracks, in the edges of which the reflections of light take place. These being many and being regularly disposed, they form a light stripe at whose head may be placed, by moving the glass, that first image which was made in the unoiled part. The head will then be seen as brighter than the tail, and the tail as somewhat less brilliant. The same effect will be seen if the glass is fogged by breathing upon it instead of using oil. If you should ever suggest this little diversion to Sarsi, and if he protests specifically and at length, I beg your

Excellency to tell him that I do not mean to imply by this that there is a huge carafe in the sky, and someone oiling it with his finger, and that a comet is thus formed, but that I merely offer this example (and I could offer others, while perhaps there are still others unthinkable to us) as an illustration of nature's bounty in her variety of methods for producing effects.

Third, I say that it is not true at all that reflections and refractions can be made only in meteorological materials and forms which inherently contain much water (because only then are the surfaces smooth and polished), or that these are necessary conditions for producing those effects. As to the necessity of polish, I say that even without it there would occur unique and distinct reflections of the image. I say so because the image occurs, broken and confused, when the entire surface is as rough and uneven as you please; for the image of a colored cloth, which is distinctly perceived in a mirror facing it, is seen to be confused and broken when thrown back to us by a wall, from which there is only a certain hint of the color of the cloth. But if your Excellency will take a stone or a wooden ruler not smooth enough to render images to us directly, and expose this obliquely to the eye as if you wished to determine whether it is flat and straight, then you will distinctly see upon it the images of objects that are brought close to the other end of the ruler—and so distinctly that if you held a manuscript there you might easily read it. And if you will place your eye close (*p.* 108) to the end of some very long straight wall, the first thing you will see is a perpetual concourse of exhalations toward the sky, especially when the wall is struck by the sun, so that all objects seen through them appear to tremble. Next if you will have someone approach the wall slowly at the other end, you will see that when he is very close to it his image will come out toward him from these ascending vapors, which are not in the least moist and heavy but rather very dry and light. Nor is this all. Hasn't Sarsi ever heard all the talk that was spread, especially by Tycho, about refractions that are made in exhalations and vapors that surround the earth, even when the air is most calm, most dry, and farthest from any rain or moisture? And he need not cite to me, as he does, the authority of Aristotle and of all teachers of perspective,

because all he will accomplish is to declare me to be a more careful observer than they—something which I believe is diametrically opposed to his intention. So much in reply to Sarsi's first argument; let us get on to his second.

XXII

'Perhaps someone . . . (*p.* 109) . . . that they did not exist.' (*p.* 89, l. 35 to *p.* 90, l. 26).

There are many things to be considered in this argument which in my opinion vitiate it a good deal.

First, neither Sig. Mario nor I ever went so far as to say that aqueous and dense vapors were attracted on high to produce a comet; hence all objections founded upon the impossibility of this idea fall to the ground and disappear.

Second, that bodies appear illuminated less and less in proportion as they are rarer and more transparent, and more and more as they are denser, which Sarsi says he has observed from long experience, I hold to be quite false. A single experience convinces me of this; namely, seeing a cloud illuminated as brightly as if it were a mountain of pure marble, though the material of the cloud is somewhat thinner and more transparent than that of the mountain. So I do not see the necessity of Sarsi's saying that the material of the comet is denser and more opaque than that of the planets, and it is this that he seems to be saying if I have correctly construed his words. The more so as I do not think it evident that the comet was brighter than first-magnitude stars and the planets. But if indeed it was, then what is the point of bringing up this density of the material, when we see twilight mists shine much brighter than the stars and the comet? And when one small golden cloud is a hundred times brighter?

Third, assuming that the comet was generated in a thick and dense vapor, it need not follow that because of refraction within this vapor there would be a noticeable discrepancy in the intervals between one star and another, as he pretends. That would be in discord with Tycho's measurements, no difference having been observed by him although he measured them with great accuracy. Frankly, (*p.* 110) there are two things here that displease me very

much. One is that I can find no way to credit what Sarsi says without refusing to credit his master, since the one speaks of having measured the distances between stars with great precision, while the other ingenuously excuses himself for not having had the opportunity to make such observations with the required accuracy, not possessing large and accurate instruments like those of Tycho. For that reason, he even hoped that others would not be overly concerned with his instrumental observations. And the other thing is that I find no way to inform your Excellency with due modesty and reserve how I fear that Sig. Sarsi failed to understand exactly what sort of thing refractions are, when they take place, and how they accomplish their effects. Therefore, please tell him some time (for I know you can do so with infinite courtesy) how the rays which come from the object to the eye at right angles to the surface of the transparent material in which refraction must occur are not refracted at all; their refraction is nil, and since stars near the zenith send their rays to us perpendicular to the spherical surface of the vapors that surround the earth, these suffer no refraction. But in proportion as they incline more and more toward the horizon and consequently cut the said surface more and more obliquely with their rays, they are more and more refracted and show their position with greater and greater error. Next tell him that because the boundary of this material is not very high, since the vaporous sphere is not much larger than the terrestrial globe on whose surface we are located, the incidence of the rays coming from points close to the horizon is quite oblique. This obliquity becomes always less the more the surface of the vapors is raised, so that if it were many terrestrial radii distant, the rays coming to us from any point in the sky might cut the said surface with very little obliquity and behave as if tending toward the center of the sphere, which is the same as saying that they would be perpendicular to the surface.

Now since (*p.* 111) Sarsi places the comet very high above the moon, no perceptible refraction need occur in vapors extended to such a height, and in consequence no sensible appearance of difference in position would exist in the fixed stars. Hence, Sarsi had no need to thin out the vapors in order to excuse the lack of refraction,

much, less in such a way as to remove it entirely. Others have fallen into this same error when they thought they could demonstrate that the celestial substance is no different from the nearby elemental substance, and that the former cannot be given a multiplicity of orbs because in that event great differences would arise in the apparent places of the stars from refraction in so many different transparent substances. This argument is idle, since even if all these orbs were of differing diaphanous materials, their very size would prevent any refraction for our eyes which repose right at the center of such orbs.

XXIII

Now let us pass to the third argument. 'Furthermore, Galileo . . . (p. 112) . . . rather than mathematical.' (p. 90, l. 27 to p. 91, l. 23).

Sarsi continues, as remarked before, to form conclusions at will and attribute them to Sig. Mario and me in order to refute them and in this way to make us the authors of absurd and false opinions. Sig. Mario speaks of the aurora borealis in order to illustrate the possibility that tenuous and thin materials might rise very far from the earth, but Sarsi pretends that he means the comet also to be of this same material. Not content with this, since he himself holds the opinion that reflection of light could not occur in any meteorological substances except moist and aqueous ones, he attributes a little farther on to Sig. Mario and me the statement that aqueous and heavy vapors would rise into the sky to form a comet. Now he will have it that we have stated that the material of the comet is the same as that of the sunspots, which were mentioned by Sig. Mario only to explain why he thinks that some materials may possibly be generated, move, and dissolve amidst the celestial substance, and not at all to affirm that comets are produced from these. Hence, your Excellency may see more clearly that my protest a short time ago was not entirely unnecessary or beside the point when I disavowed the idea that the comet took shape on an immense oiled carafe. [In the first place, in order to reply to everything, I say that Sarsi did not need to repeat so often his reproach against us for abhorring poetry, for as I have already said we do not abhor it at all. Rather,

we might add that it is not unknown to us that because of the interwoven relationship which exists among all the arts, not only may the philosopher be permitted sometimes to intersperse poetic fancies in his treatises (p. 113) as Plato and many others have done, but it is also occasionally permissible for the poet to seed his poems with scientific speculations, as has been done by Dante among our ancient poets in his *Divine Comedy* and by the noble Stigliani among the moderns in his *New World*. Next I say, getting back to our dispute, that]²⁵ I have never declared comets and sunspots to be of the same material, but I surely realize now that if I had not needed to fear any stronger opposition than what Sarsi produces here, I should not shrink from affirming this and I daresay I could make it stand up.

Sarsi discovers a great contradiction in the possibility of a thin material going straight toward the body of the sun and then being set in rotation upon its arrival there. But why does he refuse to pardon this assumption on the part of Sig. Mario, as well as in Aristotle and all his school who make fire ascend straight upward to the lunar orb and there transform its straight motion into circular? How can Sarsi maintain it to be impossible that a piece of wood cast from a high place vertically into a swift river should commence being carried round the terrestrial globe as soon as it strikes the water? His other objection would truly be more valid; namely, that while all other materials similar to the comet strive eagerly to embrace the sun, the comet alone has fled, withdrawing toward the north. This objection would be more binding, as I say, had he himself not untied it a short time before—if, in having Apollo wash his face and then throw out the water for the comet to spring from, he had not declared to us his opinion that the material of the sunspots departs from the sun and does not return there.

XXIV

Now let us hear the fourth argument. 'Now I come . . . [p. 114-115] . . . it had first entered.' (p. 91, l. 24 to p. 94, l. 14). (p. 116)

We have already seen how much significance there is in the last three arguments, and I think that even Sarsi did not esteem them highly, because the arguments drawn from physics, as he calls the

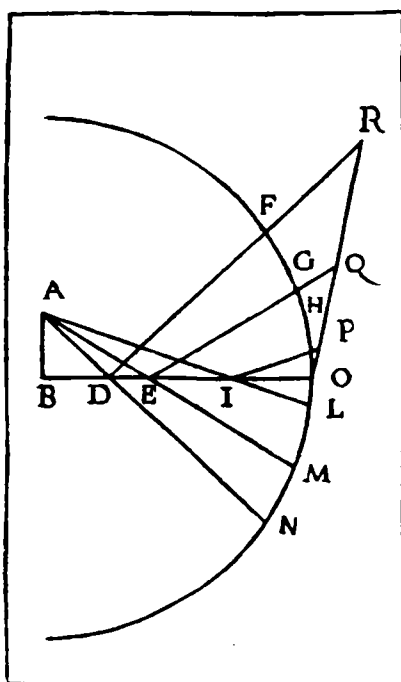
following ones taken from optical demonstrations, he considers to be far more conclusive and effective than what has gone before; a clear indication that he has not received complete satisfaction from the reasoning drawn from nature. But he had better pay attention to his cause, and consider that to a person who wants to convince others of something which, if not false, is at least very questionable, it is a great advantage to be able to use probable arguments, conjectures, examples, analogies, and other sophisms, and to fortify himself further with unimpeachable texts, entrenching himself behind the authority of other philosophers, scientists, rhetoricians, and historians. To reduce oneself to the rigor of geometrical demonstrations is too dangerous an experiment for anyone who does not thoroughly know how to manage these, for just as there is no middle ground between truth and falsity in physical things, so in rigorous proofs one must either establish his point beyond any doubt or else beg the question inexcusably, and there is no chance of keeping one's feet by invoking limitations, distinctions, verbal distortions, or other fireworks; one must with but few words and at the first assault become Caesar or nobody. This geometrical strictness will enable me more briefly, and with less tedium for your Excellency, to disentangle the ensuing proofs—which I shall call optical or geometrical more to pacify Sarsi than because I find in them, or in his diagrams, much of perspective or of geometry.

As your Excellency sees, Sarsi's intention in this fourth argument is to prove that the comet does not belong among merely apparent, simulacra caused by reflection and refraction of solar rays through the relation existing between the comet and the sun, because this relation to the sun (*p.* 117) is different from those preserved by things known certainly to be mere appearances, such as rainbows, haloes, mock suns, and reflections from the sea. All these, he says, move with the sun's movement in such a way as to go always in the same direction as the sun, whereas the opposite happened in the case of the comet, which therefore was not an illusion. It would be quite adequate here to reply that there is no obvious necessity for the comet to follow the pattern of haloes, rainbows, or the other illusions named, since it differs from a rainbow, a halo, and the

others. Nevertheless, I wish to concede here something more than I am obliged to, if only Sarsi will not try to take advantage of me later by calling futile for me some mode of argument which must be accepted as conclusive for him. I therefore ask Sarsi whether or not he considers as conclusive the argument derived from the contrariety of patterns in the comet and in mere images, the former opposing and the latter copying the sun's motion? If he says not, then all his argument is vain and I have nothing further to say. But if he says it is conclusive, then justice demands that in proving the comet to be an illusion it shall be just as valuable to me to prove that it does follow the pattern of some empty appearance in regard to following or opposing the sun's motion. But in order to find such an appearance, I need not even stray from one of those produced by Sarsi himself in order to show us simply and clearly that the comet goes opposite to this image; for to me it appears not opposite but precisely the same.

Therefore, take his third diagram,²⁶ your Excellency, in which he draws an analogy between the comet and the reflection of the sun made on the surface of the sea. When the sun is at H, its image is seen by the eye at A along the line AI; when the sun gets to G, the image will be seen along the line AE; and being at F, its image will appear along the line AD. Now as the sun appears to us (*p.* 118) to move in the sky along the arc HGF, it remains to be seen in what direction its image likewise appears to us to move with respect to the sky, which is where Sarsi observed the motions of the comet and the sun. For this, we must produce the arc FGHLMN and extend the lines AI, AE, and AD to L, M, and N; then we say: When the sun was at H, its image was seen along the line AI, which corresponds in the sky to the point L; when the sun came to G, its image was seen along the line AE and it would appear to be at M; finally, when the sun reached F, its image would appear at N. Hence, the sun moving from H to F, its image would appear to move from L to N. But this is to appear to move contrary to the sun, Sig. Sarsi, and not in the same direction, as you believed—or rather, as you attempted to make other people believe.

I put it this way, your Excellency, because I cannot believe that

*Fig. 18*

he should have equivocated in so obvious a matter. Besides, one may see also that in explaining himself he makes use of some very inappropriate and unusual ways of speaking, merely to accommodate something to his requirements which cannot be so accommodated, as there is nothing in it. Thus, he sees that as the sun passes from H to G and from G to F, its image comes from I to E and from E to D. This progress IED is a true and real approach toward the eye at A, and since Sarsi needs to be able to say that the image and the sun move in the same direction, he decides to say flatly that the motion of the sun along the arc HGF is an approach toward the point A, and that going toward the zenith is the same as going toward the center! Moreover, he has to pretend not to notice another very grave absurdity which he would be (*p.* 119) adopting by maintaining that the image imitates the movement of real

objects. If it did, it would likewise be true conversely that the object should copy the image; let your Excellency consider what would follow from this. From the end of the diameter O, draw the straight line OR which falls outside the circle and forms some angle with BO, and produce DF, EG, and IH to this at the points R, Q, and P. It is obvious that if the real object moved along the line PQR the image would go along IED, and since this is an approach toward the eye at A, and according to Sarsi the image does whatever the object also does, then the object in moving from the point P to the point R approaches the point A. But it is actually departing, and hence the absurdity.

Notice, moreover, that when Sarsi here considers what would happen as between a real object and its image, he is assuming that the material in which the image must be formed remains motionless and that only the object moves; for if he meant that the material also moved, then various other consequences would follow with regard to the appearances of the image. And therefore from what Sarsi adds about the comet's not having returned upon the return of the sun, nothing can ever be inferred unless one determines first the movement or rest of that material in which the comet is produced.

XXV

I pass on to the fifth argument. 'Furthermore, if the comet . . . (p. 120) . . . similar to them.' (p. 94, l. 15 to p. 95, l. 27).

My astonishment remains and even grows at seeing how frequently Sarsi pretends not to see things that are right before his eyes, perhaps in the hope that his pretense will give birth in others not to a feigned but to a real blindness. At present, he wants his argument to prove that if the comet were a mere appearance it would show itself in the shape of a circle, or part of a circle, because such is the case with rainbows, haloes, coronae, and various other images. I do not know how he can maintain this, having been reminded a hundred times of the sun's reflection in the sea, as well as of those projections of rays through apertures in the clouds that appear as straight stripes very similar to a (p. 121) comet. But perhaps he persuaded himself that without further trouble his point for the comet is

arrangement could occur only in diametrical opposition or in lineal conjunction of the vapors with the sun, and thus rainbows are always seen in opposition to and haloes or coronae in conjunction with the sun so as to appear circular; but as to comets, I do not know if they are ever seen either in opposition to or in conjunction with the sun. If in composing this demonstration it had ever occurred to Sarsi to think of that material which he represents around the point A as sea water instead of as vapors, (*p.* 122) he would have perceived that his own argument, conducted in the same way and with the same words, would prove that the reflection in the sea would necessarily have to extend in a circular line. Then, thanks to the fact that his senses would show him the contrary, he would have discovered the fallacy in his syllogism.

XXVI

Now let us hear the sixth argument. 'I hold . . . the greatest astronomers.' (*p.* 95, l. 28 to *p.* 96, l. 21).

It is quite false that either Sig. Mario or I ever wrote or said that (*p.* 123) images produced by the sun maintain the same parallax as the sun, as Sarsi here declares as a foundation for his syllogism. Rather, Sig. Mario names and discusses many such images, and afterwards adds this: 'Now inasmuch as parallax is nil in some of the aforesaid images, while in others its operation is very different from its operation on real objects . . .'²⁷ Nor is it to be found in Sig. Mario's essay that he declares the parallax to be the same as that of the sun or moon except in haloes; in the others, including rainbows, it is assumed to be different. Therefore, the first proposition of the syllogism is false; now let us see how true and conclusive the second is, assuming that the parallax of all mere images does have to be equal to that of the sun.

Sarsi wishes to prove on the authority of Tycho and of his master, as indeed he must, that the parallax observed in the comet is greater than that of the sun; but he then abstains from producing the specific observations made by Tycho and many other reputable astronomers upon the comet's parallax, so he does not let the reader see how widely these vary among themselves. Well, whatever they

may be, they are either correct or incorrect; if correct, so that complete faith may be placed in them, one must conclude either that the comet was simultaneously below the sun and above it, and even in the firmament, or else that, because it was not a fixed and real object but a vague and empty one, it was not subject to the laws of fixed and real things. If the observations were incorrect, then they lack authority and nothing whatever can be determined from them. Tycho himself, among so many disparities, chose those observations which best served his predetermined decision to assign the comet a place between the sun and Venus, as if these were the more reliable. As to the other observations supplied by Sarsi's master, they differ so much among themselves that the latter judged them to be unfit for establishing the position of the comet, saying that they were taken with inexact instruments and without due consideration of the time, of refraction, and of other circumstances. (*p.* 124) Thus he himself does not oblige others to set much store by them, and he reduces himself to a single observation which requires no instrument and is capable of being carried out quite accurately by simple vision alone; and he puts this ahead of all the others. It relates to the exact conjunction of the head of the comet with a certain fixed star, which conjunction was seen simultaneously from places very distant from one another. Well, Sig. Sarsi, if that happened, it is completely contrary to your requirements, since from this it may be deduced that the parallax was nil, whereas you adduced this authority to confirm your proposition that this parallax was greater than that of the sun. See now how the very authors you have called up testify against your cause.

As to what you say about our having confessed the observations made by great astronomers to have been most precise, I reply to you that if you will better consider where and when that was said, you will understand that these might be called 'precise' even if they differed among themselves much more than they did. They were called precise and sufficient to refute Aristotle's opinion when he would have the comet be a real object very close to the earth. Did you not know that your master himself proves that the mere distance between Rome and Antwerp could cause in a real object located in

the highest region of the air a parallax greater than fifty, sixty, one hundred, or even one hundred and forty degrees? And if so, could not those observations be called exact and conclusive which, being each less than a single degree, differ among themselves by a few minutes?

XXVII

Now read the last argument, your Excellency: 'Finally, we ought not . . . (p. 125) . . . to oppose Galileo.' (p. 96, l. 22 to p. 97, l. 4).

Sarsi thinks so highly of this argument that to him it seems alone sufficient to prove his point; yet I see in it no efficacy for persuading me when I consider that in producing these vain images the sun enters as the efficient principle and the clouds or vapors as the material principle. Now since the efficient is everlasting, then when the material is not lacking, rainbows and haloes and mock suns and all other appearances should be perpetual; hence, attention should be paid to the long or short duration of the establishment and position of the material. Nothing dissuades us from thinking that above the elemental regions there might be some material of longer duration than clouds or mists or rain falling in small drops, or other elemental materials; so that a reflection or refraction of the sun in this would show itself to us for a longer time than do rainbows, mock suns, or haloes. But without leaving the elements, do we not have perpetual appearances in auroras, which are refractions of solar rays in the vaporous region, and in those reflections on the surface of the sea? For if the observer, the sun, the vapors, and the surface of the sea remained always in the same arrangement, the aurora or the brilliant stripes in the water would be seen perpetually.

Besides, from less or greater duration, one does not conclusively infer an essential difference. Even among comets, without our going further afield, some are seen to last ninety days or more while others are dissipated on the third or fourth day. And since it is observed that the more enduring comets show themselves much larger than the others even when they first appear, who knows but that there are some, and perhaps many, which (p. 126) last not merely less than a few days, but even not many hours, and which are not observed by

reason of their smallness? And comets themselves give us assurance that in the place where they are formed material is generated which is better suited to maintain itself than are our elemental clouds or mists, for they are produced from material (or in material) that is neither celestial and eternal nor yet necessarily dissipated in very brief times. Hence the question remains whether that which is produced in this material is a pure and simple reflection of light (and in consequence an empty image) or something else which is fixed and real. Hence Sarsi's argument is inconclusive, and will so remain until he demonstrates that the cometary material is unfitted to reflect or refract the sunlight. For as to its being fitted to endure many days, the duration of the comets themselves more than amply proves this to us.

XXVIII

Now let us pass on to the second question of this Second Weighing. (First argument) 'I now come . . . acquainted with prophecy.' (p. 97, l. 8 to l. 21).

Here, before I go on, I cannot suppress some resentment toward Sarsi for his completely unmerited imputation of hypocrisy on my part. This is far from justified, for I frankly confess (as I always have) that I am vanquished and almost totally blind when it comes to penetrating the secrets of Nature. (p. 127) I certainly desire to achieve some small knowledge of those secrets, but nothing is more contrary to such an intention than hypocrisy and feigning. Sig. Mario in his essay has never feigned anything, nor has he needed to do so, since whatever he has adduced that is new has always been set forth conjecturally and with reserve. He has not sought to make others accept as proved that which he and I take as questionable or at most as probable, and which we have arrived at and opened to the consideration of people more knowing than we in order to have their help either in drawing confirmations of some true conclusion or in completely excluding some false one. And just as Sig. Mario's writing is honest and sincere, Sig. Lothario, yours is equally filled with pretense. In order to clear the path for your opposition, nine times out of ten you have feigned not to understand what Sig. Mario

wrote; you prepare your material arbitrarily, giving his writing meanings very far from what he intended, and frequently adding to it or deleting from it. The reader, crediting what you then put forth by way of rebuttal, remains in the belief that we have written a lot of foolishness which you have acutely exposed and refuted. This has already been noted by me in the foregoing, and in what is left it will be no less so.

Let us come to the point. What is Sarsi's motive in writing that we have failed in what we wanted above all; namely, in our pretense that the comet moves in a straight line; for in that case it would have to go always toward the zenith without any deviation? Now who apprised you of that consequence but Sig. Mario himself, who mentions this? He might indeed have perpetrated a pretense on you, and you in your grace would then have pretended not to detect his pretense. Nor is that all. Only two lines above, you wrote that I have naively confessed my inability to solve (or my disinterest in solving) the very argument which I introduced, and you add that what I chiefly wanted was to feign ignorance of it. Now what kind of self-contradiction is this? A man brings forth and naively publishes a proposition, and then you charge him with having greatly (*p.* 128) desired to hide it and to feign ignorance of it! Really, Sig. Lothario, you require of your readers great simple-mindedness and small attention to business.

Now let us see whether this statement, in which no pretense of ours is to be found, contains any pretense on the part of Sarsi. And there is certainly more than one such to be found. First, he makes me say things which were never said or written, in order to open the road for showing me to be so ignorant a geometer as to have failed to understand consequences whose proof requires no higher knowledge than certain trite propositions from the first book of Euclid. When we say that if the comet moves with straight motion, it will appear to us to move toward the zenith, he tries to make us say that by so moving it will arrive at the zenith. Here Sarsi must confess either that he did not understand the meaning of the phrase 'to move toward a place' or that he tried by feigning and dissimulation to attribute a false proposition to us. I do not believe the first is

possible, for otherwise he would have to suppose that saying to 'sail toward the pole' or 'throw a rock toward the sky' meant that a ship arrived at the pole and a rock at the sky. Hence it follows that he pretended not to understand the true proposition which we wrote, and attributed to us the false one in order to have a chance to subject us to his unwarranted remarks.

Moreover, he does not quote Sig. Mario's words faithfully in another regard either, for where the latter says that it is either necessary to abandon the straight motion attributed to the comet or else, retaining that, to add some other cause for its apparent deviation, Sarsi wilfully changes the words 'some other cause' to 'some other motion' in order to deduce, quite apart from any intention of mine, a motion of the earth, and to produce more fireworks and foolishness here. He finally concludes by saying that he is not one of those who can read minds; yet he often sets himself to penetrate the inner meanings of other people.

XXIX

Now let your Excellency continue. 'Therefore, I ask . . . (*p.* 129) pious and religious.' (*p.* 97, l. 22 to *p.* 98, l. 20).

Here, as you see, he is at some pains to show that no other motion which may be attributed to the comet or to any other world body can suitably support the straight motion adduced by Sig. Mario and at the same time explain its apparent deviation from the zenith. This argument is superfluous and vain, since neither Sig. Mario nor I ever wrote that the cause of such deviation depended upon any other motion, either of the earth or of the heavens or of any other body. Sarsi (*p.* 130) has introduced this as a caprice, so let him answer it himself. He should not pretend that others are obliged to support what they have not said or written, nor perhaps so much as thought—and indeed Sarsi admits this himself when he plainly says that he does not believe it ever entered my head to introduce motion of the earth to rescue this deviation, he having always known me to be a very pious and religious man. Now if that is so, why did you bring this up? To what end did you try to prove it unsuited to this requirement? But we had better proceed.

XXX

Read on, then, your Excellency. 'But lest I . . . (p. 131) . . . never reach R.' (p. 98, l. 21 to p. 99, l. 15).

As your Excellency sees, Sarsi once more alters Sig. Mario's writing, pretending that he has said that motion perpendicular to the earth would finally conduct the comet to the zenith point; this is not to be found in the book, though Sig. Mario does indeed say that such motion would be toward the zenith. In my opinion, Sarsi does this in order to create an occasion for bringing in his geometrical proof, based upon no more profound a foundation than the mere understanding of the definition of parallel lines. From this action, some people might deduce a proposition not very complimentary to Sarsi, for he regards this conclusion of his and his proofs of it either as very ingenious, and not for ordinary men, or as trifling and suited for mere children. If he thinks this a childish thing, he may be sure that neither Sig. Mario nor I happen to be in such a sad state of ignorance as to have fallen into error from lacking such information. But if he considers it a subtle and important matter, then I must judge his state sad indeed, and him in need of returning to his master for further instruction.

It is true, then, that a thing moved perpendicular to the earth's surface would never arrive at the zenith (unless that thing should depart from the exact place of the observer, which Sarsi may have overlooked), (p. 132) but it is also true that we did not say it would arrive there.

XXXI

(Third argument) 'Furthermore, since . . . (p. 133) . . . rules of trigonometry.' (p. 99, l. 16 to p. 101, l. 9).

I should have thought from Sarsi's foregoing proof that he might at least have seen, and possibly have understood, the first book of the *Elements*, but what he writes here makes me gravely doubt whether he has any mathematical skill at all. From the diagram drawn by himself to his own taste, he wants to deduce the rate of retardation of speed attributed to the comet by Sig. Mario. First he shows that he has not noticed that in all books by mathematicians no attention

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be in different ratios when the observer is near the beginning of the line of motion and when he is far from it. Yet, finding that in his diagram the arcs GF, FI and IL (which are apparent motions) decrease rapidly and much more than what is observed in the movement of a comet, Sarsi is persuaded that on no account could a similar motion be adapted to the latter, and he has not noticed how these decreases may appear less and less unequal accordingly as the altitude of the moving body is assumed to be greater. Yet he does know that the proper proportions are not preserved in the diagram,

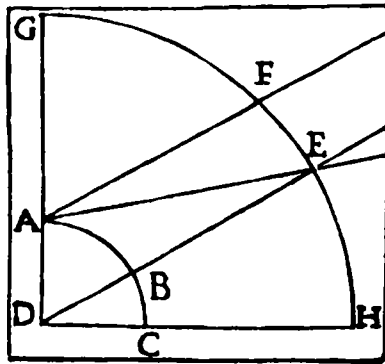


Fig. 21

though it does not matter that they are not. He assures us of his having noted this by his ensuing diagram, in which he proves the angle DEA to be only one and one-half degrees, though in the diagram it is more than fifteen degrees; and the radius of the moon's orbit DE is scarcely three times the terrestrial radius DB, though he calls it thirty-three times as large. This alone was surely enough to show him how simple-minded anyone would be who tried to deduce the mentality of a geometer by measuring his diagrams with a pair of compasses. Therefore, Sig. Lothario, I say that in the same straight and uniform motion there might be a great diminution, a middling one, and also a small, a tiny, and an imperceptible one; and if you wish to prove that none of these would correspond to the motion of the comet, you will have to do something a little more artistic than measure the picture. I assure you that in writing thus,

you will not secure applause from anyone except those (and here I do not mean either Sig. Mario or yourself) who crown with victory the one who talks the most and has the last word.

Now let us hear, your Excellency, what Sarsi produces at the end. I believe (*p.* 135) that from what he adds about the smallness of the apparent motion he wants to prove that straight motion, so often mentioned, does not in any way belong to the comet. I say that I believe this, and not that I am sure of it, for the author, after all his demonstrations and calculations, does not draw any conclusion at all. He assumes for this purpose that the comet at its first appearance was more than thirty-two terrestrial radii distant from the earth, and that the observer was located sixty degrees from that point on the earth's surface which lay perpendicularly under the line of the comet's motion. These two things assumed, he determines the quantity of apparent motion which would bring it to the sky at an angle of scarcely one and one-half degrees. Here he stops, without applying the statement to any purpose or drawing from it any conclusion. Since Sarsi did not bother, I shall draw two conclusions from it; the first is that which Sarsi would have had the simple reader infer by himself, and the other is that which may be deduced from it as a true consequence—but not casually and by the simple-minded.

Here is the first: 'Therefore, O reader, in whose ear still resounds that which was written above (that is, that the apparent motion of our comet crossed the sky by scores of degrees), form now the idea and hold it as certain that Sig. Mario's straight motion does not help him at all, for by it only a single degree and one-half may be crossed, and that with difficulty.' This is the consequence for the simple-minded; but whoever possesses the flower of natural logic will, by uniting Sarsi's premises to the conclusion to which they lead, form this syllogism: 'Assuming that the comet at its first appearance was at an altitude of thirty-two terrestrial radii, and that the observer was sixty degrees removed from its line of motion, the quantity of its apparent motion could not exceed one and one-half degrees; but it actually exceeded scores of degrees; therefore (and now comes the true conclusion) at the time of the first observation our comet was

not at an altitude of thirty-two radii from the earth and the observer sixty degrees distant from its line of motion.' Let Sarsi frankly admit this conclusion, which destroys his very assumptions. Yet (*p.* 136) his syllogism is defective in another respect, too, for it has no validity against Sig. Mario. He openly wrote that a simple straight motion could not be sufficient to satisfy the apparent deviation of the comet, and that one would have to add some other cause for that; this condition, omitted by Sarsi, completely vitiates the latter's inference.

I note further another error in logic, and not a trifling one, in this discourse of his. From the great change in location of the comet, Sarsi wants to prove that Sig. Mario's straight motion could not be suitable to it because the alteration which follows from such motion is small. The truth is that there might follow from such motion small alterations, middling ones, or quite large ones, accordingly as the moving body was higher or lower and the observer farther from or closer to the line of this motion. Sarsi, without asking his opponent at what altitude and distance he assumes the moving body and the observer to be, puts them both at places suited to his requirements and ill-assorted for those of his adversary, saying: Assume that the comet at the beginning was thirty-two radii high and that the observer was sixty degrees away. But my dear Sig. Lothario, if the adversary shall say that it was not at that distance by many thousands of miles, and that the observer likewise was much closer, what will you do then with your syllogism? What will you conclude from it? Nothing. It would be necessary that we, and not you, should have attributed such distances to the comet and the observer; then you might have run us through with our own weapons. Or, if you had still wished to transfix us with your own, you would first need to prove them actually to have been at such distances (which they were not in fact), and not assume this arbitrarily and choose the distances most prejudicial to your opponent's case.

This circumstance alone inclines me somewhat to believe that something may be true which until now I have refused to believe; namely, that you may have been a pupil of him whom you now impersonate. For if I am not mistaken, he falls into the same error

when he attempts to prove false Aristotle's opinion and that of others who have believed a comet to be elemental and to be domiciled in the elemental region. (*p.* 137) He raises against them, as a great difficulty, the enormous bulk which a comet would have to possess, and the incredibility of the earth's furnishing it with food and nourishment. Then, to show it to be an enormous framework, he places it (without permission from his opponents) in the very highest part of the elemental sphere, that is, at the very arc of the moon's orbit. Then, from its appearance to us, he sets about calculating its volume and finds this to be little less than five hundred million cubic miles—and let the reader note that one cubic mile is a space so large that it would hold more than one million ships, which is perhaps more than there are in the world. Such a framework would truly be indecent and dishonest, and it would be too great an expense for the human race to send it alms for its food and nourishment. But Aristotle and his adherents would reply, 'Good Father, we tell you that the comet is elemental, and that it may be fifty or sixty miles distant from the earth, or perhaps less; not one hundred twenty-seven thousand seven hundred and four, as you make it upon your own authority alone. So its body is not, by thousands of miles, as large as you believe, nor is it insatiable and incapable of being pastured.' Here there is nothing for us to do except shrug our shoulders and remain silent. If Sarsi is to convict his adversaries, he must confront them with their own most favorable assertions, and not with those most prejudicial to them; otherwise he allows them to escape every time, leaving their enemies feeling stunned and foolish—as Ruggiero felt at the disappearance of Angelica.²⁸

XXXII

Now let us hear what follows; read this fourth argument, your Excellency. 'Now although the earth . . . (*p.* 138) . . . motion of the comet.' (*p.* 101, l. 10 to *p.* 102, l. 2).

Here he wants to prove that not even by assuming that motion of the earth which Copernicus assigned to it can this straight motion and that deviation from the zenith be explained and maintained. For if, indeed, from the earth's motion there follows an apparent declination

of the sun now to the south and now to the north, still, in the space of forty days during which the comet was observed, this declination did not amount to more than three degrees, so that of the comet could not appear to be much more. Adding to this the mere degree and one-half which might come from its own straight motion, we still remain far from that very great motion which was seen in it.

Since we did not affirm that the cause of that apparent deviation was any movement of any other body, and least of all of the terrestrial globe (which Sarsi himself admits he knows we hold to be false), it would clearly appear that he has capriciously introduced it in order to swell the size of his book. No obligation falls upon us to reply in support of what we have not said. But I do not wish for that reason to omit remarking that I much question whether Sarsi has even formed for himself any correct idea of the motions attributed to the earth, or of the many and varied appearances which would have to be perceived in other world bodies as a result of it. I perceive that he thinks that any deviation which might appear in the solar body located in the center of the ecliptic should be perceived equally (or little differently) in any other visible object wherever (*p.* 139) situated in the universe, whether inside or outside the ecliptic, within or without the *orbis magnus*,²⁹ to the north or south, or near or far with respect to the earth. All this is far from correct, nor is there any contradiction in this change appearing to be three degrees in the sun while it might appear to be ten, twenty, or thirty degrees in other objects because of their different positions. And in conclusion I may add that if the movement attributed to the earth (which I, as a pious and Catholic person, consider to be most false and vain) lends itself to yielding explanations for so many widely diverse appearances observed in the heavenly bodies, then I should not be sure that so false a thing might not deceptively correspond with the appearances of the comet, unless Sarsi will descend to more specific considerations than those which he has thus far produced.

XXXIII

Now, your Excellency, read the fifth argument. 'This is if the motions . . . (*p.* 140) . . . motion of the comet.' (*p.* 102, l. 3 to *p.* 103, l. 13).

Here Sarsi thinks he can confirm his statement by various other changes occurring in other comets and described by other authors. But what I have written above answers this too, nor do we need more. First, Sarsi's generalities are too broad, and he does not get down to specific considerations of particular states of these comets as to their being high, low, southern, northern, or their having appeared at the time of the solstices or that of the equinoxes. These conditions omitted by him are most necessary in such judgments, as he himself will be able to understand whenever he gets round to giving closer attention to this theory.

XXXIV

I now pass on to the last question of the present weighing: 'Whether the curvature . . . (p. 141) . . . left undecided.' (p. 103, l. 14 to p. 103, l. 33).

Only too clearly does Sarsi reveal himself desirous of stripping me and entirely despoiling me of every ornament of glory. Not content with revealing as false and inconclusive the reason produced as mine by Sig. Mario for the tail of the comet sometimes appearing to us as bent in an arc, he remarks here that nothing new was achieved by me in this, it having all been published long before and refuted as false by Johann Kepler. Hence in the reader's mind (if he stops at Sarsi's recital), there will remain an idea of me as not only a thief of other people's ideas, but a petty miserable thief at that, who goes about pilfering even things that have been refuted. But who knows; perhaps in Sarsi's eyes the pettiness of the theft does not render me more culpable than I would be if I had more hardily set myself to search out books by some noble author not so well known in these parts; and if having found some I had attempted to suppress his name and attribute his entire work to myself, perhaps that enterprise would appear to Sarsi to be as grand and heroic as the other seemed to him cowardly and abject. Since I do not have the nerve for this, I freely confess my cowardice. But though I am poor in both bravery and power, I am at least an upright man, Sig. Lothario, and I do not want to remain with this undeserved scar (p. 142) on my face; I wish to write frankly and reveal what you left out, and, since

I cannot divine from what passion it may have originated, I leave it to you to specify this later in your apology.

Tycho indeed tried to assign the cause of this apparent curvature, reducing it to some propositions proved by Vitellio, but Sig. Mario showed that this did not exist among the things written by that author, which were far from serving the purpose of this curvature. Sig. Mario then added something that appeared to him and to me to be its true cause and demonstrable reason. Up jumps Sarsi, and in his attempt to refute this and to show in addition that it belonged to Kepler, he falls into the same pit as Tycho and reveals himself as having understood nothing of what Kepler and Sig. Mario wrote. At any rate, he pretends not to have understood either of them, and would have it that both write the same thing when they say quite different things. Kepler wants to give a reason for the curvature such that the tail is really curved, and not just apparently; Sig. Mario supposes it to be really straight, and seeks a cause for its apparent bending. Kepler reduces this to a diversity of refractions of rays, themselves solar, occurring in the same celestial material in which the tail itself is formed. He says that this material (in that part of it only which serves for the production of the tail) is more dense in proportion to its proximity to the comet, and since thus various refractions are made, from the final composition of all these there results an overall refraction which is extended not in a straight line but in an arc. Sig. Mario introduces a refraction not of the sun's rays but of the image of the comet itself, and not in the celestial material belonging to its head but in the vaporous sphere surrounding the earth. Thus, for each, the cause, the material, the place, and the manner of production are all different, and no correspondence exists between these two authors beyond this single word 'refraction.' Here are Kepler's exact words: *Non refraction potest esse causa inflexionis huius; nisi nescio quid monstri confingamus, materiam aethericam certis gradibus propinquitatis ad hoc sidus magis magisque crassam, nec nisi ex una sola parte in quam cauda vergit.*³⁰ ('Refraction cannot be the cause of this bending; nor am I unaware what monsters we construct, the aethereal material becoming denser and denser at a certain rate in the vicinity of that star, but only from a single place in which the tail bends.')

(*p.* 143) Is it possible, Sig. Lothario, that you have allowed yourself to be so transported by the desire to obscure my name in the field of science, whatever it may amount to, as to disregard not only my reputation but even that of many of your friends? With errors and fictions, you have attempted to make them believe your teachings to be sound and sincere; by such means you have acquired their applause. But later, if they should ever happen to see this writing of mine and thereby come to understand how often and by what tricks you have treated them as simpletons, they will consider themselves to have been shabbily dealt with by you, and the esteem and grace which you hold in their hearts will change its state and condition. The argument introduced and refuted by Kepler is something quite different, and, as he has always been known to me as a person no less frank and honest than he is intelligent and learned, I am certain that he would admit our statement to be completely different from the one he refuted, and would say that just as that one deserved rebuff, so this other deserves assent. For it is true and demonstrable, even though Sarsi may do his best to refute it.

XXXV

But let us examine the strength of his refutations. (First argument) 'But now let us see . . . (*p.* 144) . . . display any curvature.' (*p.* 104, l. 1 to l. 27).

Sarsi's attack here is too ineffective to confute a necessarily conclusive proof from optics. He wants others to neglect the latter in favor of assertions of his which are capable of being freely altered and accommodated to his purposes—let Sarsi forgive me this suspicion, since he himself gives us frequent occasion for suspecting the credibility of what he sets forth. Indeed, what faith may be reposed in a man's reports of things long past, of which nothing is any more to be seen, when in speaking of permanent things which are present, public, and printed, the same man does not refrain from altering nine out of ten of these in the telling, transforming them into a contrary sense?

I return to my remark that the demonstration published by Sig. Mario is purely geometrical, complete, and logical. First Sarsi needs

to understand it thoroughly, and then if it does not appear conclusive to him he must point out its fallacy either in the falsity of the assumptions or in the steps of the demonstration; but of this he has done little or nothing. Our demonstration proves that if the object seen is extended in a straight line and is situated outside the vaporous sphere in the vicinity of and oblique to the horizon, it must necessarily be seen as curved by an eye placed far from the center of this vaporous sphere; but if it is vertical to the horizon or elevated high above it, then it will be seen as quite straight or as imperceptibly curved. During the first few days the recent comet was seen, it was low and oblique, and was also seen as curved; afterward, having risen high, it was straight, and so it remained because it continued to stay at a great (*p.* 145) elevation. The comet of 1577, which I saw continually,³¹ remained always low and much tilted, and therefore was always seen noticeably curved. Other smaller ones which I have seen very high have always been quite straight, and thus the effect will be found to conform to the demonstrated conclusion, provided one has a correct account of the latter. But let us hear what Sarsi opposes to our demonstration, and see how significant his objections are.

XXXVI

(Second argument) 'Furthermore, I do not see . . . the remaining matters.' (*p.* 104, l. 28 to *p.* 105, l. 20).

As your Excellency sees, Sarsi begins opposing the demonstration as being established upon a false (*p.* 146) foundation—that the surface of the vaporous region is spherical—which he thinks he can disprove in several ways. First he says what we ourselves repeatedly affirm, that such vapors rise much higher in one place than in another. But this proposition is found in no different form in Sig. Mario's book; indeed, some vapors do happen to rise higher than usual, but this is rare and of brief duration, so that in this regard it is quite arbitrary for Sarsi to say that the shape of the vaporous region is not round. He then adds another falsehood—that we have said the comet is formed of those same vapors which rise above the cone of the earth's shadow to produce the aurora borealis, a thing which is not to be found in Sig. Mario's book. In the third place, he goes on to

say: 'If that vapor were elevated three miles in one place and a thousand leagues in another, may I ask whether you take the shape to be spherical even under those conditions?' No indeed, Sig. Sarsi; anyone who would say such a thing would in my opinion be a great fool. But I do not find that anybody ever did say it, and I doubt if anyone has ever dreamed of doing so; suppose you name an author who has.

In the fourth place, he says that those who teach the first rudiments of spherical astronomy say that this vaporous region is oval rather than round. To this I respond that Sarsi need not be surprised that he knew this and I did not, for I learned astronomy not from those rustic teachers but from Ptolemy, and so far as I can remember he states no such conclusion. But finally, if it were definitely true that this shape is oval and not round, what would you deduce from that, Sig. Lothario? Merely that the curl of the comet would be bent in an oval line and not in a circular arc. I might concede this to you without the slightest prejudice to our meaning and our method of demonstrating the cause of this apparent curvature, but not what you attempt to deduce when you conclude: 'If therefore this vaporous region is not spherical, nor everywhere equidistant from the earth, nor uniformly dense in all its parts'—a proposition repeated three times in different words in order to overwhelm the simple-minded—'then the (*p.* 147) curvature of the curl cannot derive from a kind of sphericity which does not exist in nature.' I say that this conclusion does not follow in good logic, and that the most that might follow is that the curvature is not circular but oval, so this would be the extent of your miserable and unhappy gain even if you could be sure that the vaporous region is ovate and not spherical. Whether in fact the curvature has the shape of a circular arc, or an ellipse, or a parabolic line, or a spiral, or something else, I believe could not be determined in any way, as in an arc of at most two or three degrees the differences of such curvatures would be entirely imperceptible.

It remains for me to consider his final words, from which I shall oracularly deduce various considerations and hidden senses of Sarsi. First, it is very clearly understood that he puts them into Sig. Mario's writing not with an impartial mind as to criticizing or praising them,

but with a firm resolution to impugn them, as I also remarked at the beginning. He says, 'And how have I been able to refute things set forth by him which I could not divine?' The truth is quite the opposite; namely, that he has for the most part refuted other things which Sig. Mario did not state and which Sarsi has assigned himself the task of divining. At the same time, he says that Sig. Mario wrote in obscure and tangled words, and that in quite a long disputation his meaning is incomprehensible. To this I reply that Sig. Mario had a different intention from that of Sarsi's master, who (as may be inferred from the introduction to Sarsi's treatise) wrote for the common people and with the purpose of teaching them by his replies what they could not have fathomed for themselves. But Sig. Mario wrote for those better informed than ourselves, and not to teach them but to learn; therefore he always set forth his statements questioningly and never with magisterial certainty, and left the decisions to those who are more learned. And if our writing appeared obscure to Sarsi, then before he censured it he ought to have had it explained to him, instead of setting out to contradict what he did not comprehend, at the peril of finishing without a leg to stand on. If I may say frankly (p. 148) what I think, I really believe that the reason Sarsi did not challenge most of the things Sig. Mario wrote was not a failure to understand them, but on the contrary that they were all too clear and true, and he considered it more judicious to say he did not understand them than to approve and praise them against his wishes.

Now I come to the Third Weighing. Here Sarsi does his best to make us appear poorly informed by means of four propositions taken here and there from more than a hundred which appear in Sig. Mario's *Discourse*. He cloaks in silence all the rest, which are much more important than these, and he accommodates these to his purpose by adding to or deleting from them, or twisting them into senses other than those in which they were proffered.

XXXVII

Now see, your Excellency: 'THIRD WEIGHING . . . (p. 149) . . . polished and smooth.' (p. 105, l. 21 to p. 107, l. 13).

Here, right at the outset, are Sarsi's usual tricks. First of all, it is not to be found in Sig. Mario's essay that we have ever said that neither the air nor fire adhere and attach themselves to smooth and polished bodies. Sarsi imputes this falsehood to us (*p.* 150) at his own caprice in order to pave the way for speaking of a certain glass plate a little later on. Besides, Sarsi pretends not to see that when we say the hollow of the moon's orb is a perfectly polished and smooth spherical surface, this is not because that is our opinion but because Aristotle and his followers would have it so, and we would argue against them *ad hominem*.³² Pretending to find in Sig. Mario's book that which is not there, he feigns not to see what is written there repeatedly and plainly; that is, that we would not admit the multiplicity of solid orbs formerly believed in, but we suppose a very subtle aethereal material to be diffused through the vast reaches of the universe, through which solid world bodies go wandering with their own movements. But what am I saying? I have at this moment recalled that he saw and criticized this before, on page 98, where he writes: 'For since Galileo does not hold with the celestial orbs of Ptolemy, and in Galileo's system there is nothing to be found of solid sky . . .' Here, Sig. Sarsi, you can never hide your having understood perfectly well that we said the hollow of the moon's orb is perfectly spherical and smooth not because we believed it, but because Aristotle considered it so, and against him we would dispute *ad hominem*. Had you believed it to be uttered as our own sentiment, you would never have pardoned us such a contradiction as first to deny the distinctness of orbs and their solidity, and then to admit both the one and the other—an error of much greater consequence than all the others taken together which you attribute to us. Quite vain, then, is all the remainder of your argument in which you do your best to prove that the lunar hollow must be wavy and rough rather than smooth and polished. It is idle, and does not oblige me to make any reply; still I prefer to

Let gentlemanly strife prevail between us,³³

as the great poet says, and I shall consider the strength of your proofs.

You say, Sig. Sarsi: 'If anyone should deny that the surface of the lunar hollow is smooth and polished, by what method or with what reasons would the contrary be proved?' You then add as a proof brought forth by your opponent, an argument fabricated in your own way (*p.* 151) and easily resolvable. But suppose the opponent were to reply to you, 'Sig. Lothario, assuming that the material of the celestial orbs is solid, and is distinct from that contained within the lunar hollow, I say to you with assurance that this concave surface would have to be called as polished and smooth as any mirror. For if it were wavy, the refractions of the visible images of the stars would execute an infinity of vagaries in coming toward us, precisely as happens when we look at external objects through a glass window in which some panes are flat and smooth while others are crudely made. Whether we or the objects move, their images receive no alteration either as to place or as to shape in passing through the smooth panes, but in coming through the crude panes their extravagant alterations beggar description. That is how it would be if the lunar concave were wavy, and it would be a wonderful thing to behold the many transformations of shape, movement, and situation which would appear from one moment to the next in planets as well as fixed stars, accordingly as their images passed to us now through one and now through another part of the lunar orb beneath them. But no such distortion is perceived; therefore the hollow is quite smooth.' What would you say to this, Sig. Sarsi? You would have to work hard to persuade people that there was nothing new for you in this argument, that you had passed it by as superfluous, that it belonged not to me but to somebody else, that it had already been dismissed as stale and mouldy, and in addition that you could ultimately overthrow it.

Then let the above be my argument for proving the lunar concave is smooth and not wavy. Now let us hear what you say to prove the contrary, remembering that our dispute is whether the uppermost elements are swept round by the celestial motion (for the title you give to the conclusion you are opposing is: 'Air and exhalation cannot be moved by the motion of the sky'), and that I have denied this, proving my point by the uniformity of the refractions. You, to show

the contrary, write: 'If the hollow is assumed wavy, the connection of all moving bodies is much better preserved, for thus (*p.* 152) the uppermost elements are moved by the sky's motion.' But Sig. Lothario, this is the error which logicians call 'begging the question,' for you are taking as granted that which is in question and which I deny; that is, that the uppermost elements are moved. Here are four conclusions, two being mine and two yours. Mine are 'the hollow is smooth,' and 'therefore the elements are not swept along.' I prove that the hollow is smooth by means of stellar refractions, and my conclusion is perfectly valid. Yours are, first, 'the hollow is rough,' and second, 'therefore the elements are swept along'; then you prove that the hollow is rough because thus the elements are swept along by its motion, and you leave your opponent right where he was in the first place; you have accomplished nothing, and he will be neither more nor less insistent that the hollow is not rough and the elements are not swept along. In order to escape from this circle, you must prove one of your two points by some other means. And don't tell me that you have sufficiently proved the irregularity of the surface when you say that lower things are in this way better linked with the higher, for simple contact is sufficient to connect them, and you yourself later admit the same adherence and union to exist even if the hollow is smooth and not rough, so your proof is quite frivolous. Nor would it be any more cogent if you were to pretend to have proved the sweeping along of the higher elements because generations and corruptions are produced down here by such a motion, or perhaps because the higher air and the element of fire would be driven down by this motion, these being mere fantasies and airy nothings. It would take us a long while to get warm if we had to wait for the expulsion of fire toward the earth, especially as you yourself are shortly to say that fire is forced upward, and that for this reason it thrusts, and, by thrusting, becomes overcharged in some way so as to adhere more solidly to the celestial surface. Surely these are puerile ideas and arguments, which now support and now refute the same things in accordance with the childish inconstancy of what they say.

XXXVIII

But let us hear by what other means he proves the same conclusion in the second argument, which follows. 'But why do I . . . (p. 153) . . . as the elemental sphere.' (p. 107, l. 14 to p. 108, l. 9).

Before resuming, I once more reply to Sarsi that it is not I who want the sky to have the noblest shape on account of its being a most noble body; it is Aristotle himself, against whom Sig. Mario argues *ad hominem*. As for me, never having read the pedigrees (p. 154) and patents of nobility of shapes, I do not know which of them are more and which less noble, or which more and which less perfect. I believe that all of them are, in a way, ancient and noble; or, to put it better, that there are none which are noble and perfect or any that are ignoble and imperfect, except in so far as for building walls the square shape is more perfect than the circular, while for rolling or for moving wagons I deem the circular more perfect than the triangular.

Now let us get back to Sarsi. He says that abundant arguments have been supplied by me for proving the roughness of the concave surface of the sky, because I will have it that the moon and other planets (bodies which are also celestial, and much more noble and perfect than the sky itself) are of mountainous, rough, and uneven surfaces, and, if this is so, why should that unevenness not be found also in the shape of the sky? Here Sarsi may put down for an answer whatever it is that he would reply to a man who tried to prove that the sea ought to be bony and scaly because whales, tunas, and other fish which inhabit it are.

To his interrogation concerning the reason for the moon not being smooth and polished, I reply that the moon and all other planets being inherently dark, they shine only by illumination from the sun. Hence it is necessary for them to have rough surfaces, for if they were as smooth and polished of surface as a mirror, no reflection of light would reach us from them and they would remain entirely invisible to us. Consequently they would have no actions upon the earth and reciprocally upon each other, and, in brief, each of them being in itself as if nonexistent, they would be for others as if they did not exist in the universe. And on the other hand, almost as much

disorder would ensue if the heavens were of a solid substance and bounded by a surface which was not most perfectly polished and smooth; for, as I have remarked before, on account of the refractions being continually disturbed in a wavy surface, neither the movements of the planets, nor their shapes, nor the projections of their rays toward us (and consequently their appearances to us) would be found to be anything but most confused and irregular. There, Sig. Sarsi, is a sufficient (*p.* 155) reason in response to your question. In payment for it, please cancel in your writings those words where you say that I have often written that the stars are of various and angular shapes. You know very well that this is not true, and that I have never written any such proposition; the most that you may have read is that the fixed stars are of such live and brilliant light that their tiny bodies cannot be perceived as distinct and bounded among their splendid rays.

As to that which Sarsi writes in conclusion about the sun and the smokiness that is generated and dissolved in it and in its surroundings, I have never stated positively whether the latter revolves because of the motion of the former or vice versa, because I do not know. Perhaps neither the surrounding smokiness nor the solar body is swept round, but both naturally possess equally that rotation of which I am certain because I see that these spots revolve in about four weeks. But even if our knowledge about this were complete, I do not see of what use it would be in the present contest; here we are arguing *ad hominem* and *ex suppositione* on a basis of other assumptions which are certainly false, about matters quite different from the sun and its surroundings, and it is being asked whether the lunar hollow, if smooth and hard (which it certainly is not in actuality), rotating (which is still another falsity), sweeps along with it the element of fire (which perhaps does not even exist). Add also this other fiction to which Sarsi says he cannot see any objection, and which he even calls an identity—that one might equally well say, with the same ease and naturalness, that a fluid body contained within the hollow of a spherical solid might be swept along by the latter if it were set in rotation, as to say that this would take place when the thing contained was a solid sphere and the surrounding substance was a liquid. This amounts to believing that just as a ship

is carried and swept along by the motion of a river, so the water in a stagnant pond ought to be swept along by the motion of a boat, which is utterly false. First, as to experience, we see the ship—or even a thousand ships filling a whole river—to be moved by the river's motion, while on the other hand the course of a ship driven at any speed (*p.* 156) is not followed by any water at all. The reason for this should not be very abstruse; no force can be brought against the surface of a ship which is not similarly brought against its whole framework, whose parts, being solid, are firmly attached together and cannot be separated or parted in such a way that some parts yield to the impetus of the external surroundings while others do not. Such is not the case with water or any other fluid, whose parts have hardly any perceptible cohesion or adherence to one another, and are thus readily separated. Hence a very thin veil of water which touches the body of a ship is forced to obey its motion, but other more distant parts forsake the closer ones, and those in turn the ones they touch, so that at a very short distance from the ship's surface they are entirely freed from its force and will. Moreover, the impetus and the motility impressed are preserved much more strongly and for a longer time in solid and heavy bodies than in light and fluid ones. Thus we see a great weight suspended by a cord conserve for many hours the impetus and motion communicated to it by a single push; but you may agitate the air enclosed within a room as much as you like, and no sooner will the impetus which caused the commotion cease than the air will be totally stilled, nor does it retain any of that agitation. So when the surrounding and moving thing is liquid, and it exerts its force against a solid, massive and heavy thing contained in it, it is impressing its mobility upon a subject inherently suited to retaining and conserving this for a long time. Hence the second supervening impulse meets with the motion already impressed by the first; the third impulse finds the impetus conferred by the first and the second; the fourth adds to the operations of the first, second, and third; and thus, step by step, the motion of the movable body is not merely conserved but is augmented. But when the body to be moved is liquid, light, and subtle, and in consequence is impotent to conserve the impressed movement

but loses what is impressed as fast as it gets it, then trying to impress speed upon it is as vain a task as trying to fill the sieve of the Danaïdes,³⁴ for it empties out as fast as it is repoured therein. Behold, Sig. Lothario, (*p.* 157) the enormous difference proved to exist between these two operations which seemed to you to be the same.

XXXIX

Now let us pass on to the third argument. 'But, finally, let us . . . the same motion.' (*p.* 108, l. 10 to *p.* 109, l. 14).

Sarsi continues with this fantasy of his, still pretending that I have said that the air does not adhere to smooth and polished bodies, a thing which is not to be found written either by Sig. Mario or myself. Besides, I do not quite understand what sort of thing he means by this 'adherence.' If he means a coupling which completely resists separation and parting between the surfaces so that they no longer touch, I (*p.* 158) say that such 'adherence' exists and is very great. The surface of water, for instance, may be detached only by immense force from that of a plate of copper or other material, and in this case it does not matter whether the latter surface is polished and smooth or not; a close contact suffices. And this holds bodies united so firmly that perhaps the particles of solid and hard bodies have no other cement than this to attach them together. But this 'adherence' is of no use to Sarsi. Now if he means a conjunction such that the two bodies (I mean that of the solid and that of the liquid) cannot move upon each other even by slipping—which would fit in with his requirements—then I say that there is no such 'adherence' not only between a solid and a liquid, but not even between two solids. Thus we see that in two slabs of marble which are quite level and smooth, the first 'adherence' is such that if we lift up one slab the other follows, while the second 'adherence' is so weak that, if the surfaces in contact are not perfectly horizontal but are tilted even a hair's-breadth, the lower slab will immediately slip away in the direction of the tilt.

To sum up, the moving of the one surface upon the other will meet with no resistance, however great may be the resistance to an attempt to detach and separate them. Thus, in the contact of water

with the boat, though the resistance to detaching and separating one surface from the other may be great, the resistance is only a minimum in the moving of one surface upon the other as they rub together. And as I have remarked above, the swiftly moved ship does not take along with it anything but that veil of water which is in contact with it, and perhaps it is continually robbed even of this and reclothed successively with one after another. And I know that Sarsi will grant me that if a ship is put into the sea moistened with wine or with ink, it will not have plowed the waves more than half a mile before there will remain not the least trace of the fluid which originally encased it. This makes it very likely that the same thing may reasonably be imputed to the water which touches the ship; namely, that this is continually changing. And certainly the tallow with which a ship is smeared, though very firmly attached, (*p.* 159) is nevertheless carried away in a short time by the water that slips over it in its course, which would not happen if the water that touched it remained always the same without changing.

As for the glass plate which stays afloat between the little ridges of water, I say that these ridges do not sustain themselves because of the adherence of the air to the plate preventing the water from running over the plate; if this were so, the same thing ought to happen if one placed in the water a similar plate somewhat moistened, for it is incredible that the air would adhere less to a wet surface than to a dry one. Yet we see that when the plate is wet no ridges form, and the water runs swiftly over it. The maintenance of the ridges, therefore, has some other cause than the adherence of air to the surface of the plate. And quite frequently we see large particles of water sustain themselves in convex shape and in relief very much higher than the little ridges which surround the floating plate; for instance, upon the leaves of cabbages.

In his final proof, Sarsi would have it that pressure or weighing down is sufficient to make one body follow another without any other 'adherence.' He illustrates this by two well-polished slabs of stone, placed one upon the other. The upper one, pressing down, follows the motion of the lower when that is pulled in any direction. I concede him the experiment, but I do not see that it has anything

to do with our problem, first because we are dealing with a thin and liquid body whose parts are not connected together; otherwise the motion of one part would move the whole, as happens in a solid body. And second, Sarsi is too careless about his proof that fire, air, and the exhalation contained within the lunar hollow push and bear against the surface of this hollow. He sets forth as the cause of this compression a continuous rarefaction of these substances which, dilating, and therefore seeking always greater space, exert force against their container and thus in a way attach themselves to it so that they follow its motion. This reasoning is really very careless, for where Sarsi positively affirms that the contained substances (*p.* 160) are continuously rarefying and dilating, his opponent with no less reason (I say 'no less' because Sarsi adduces none at all) might say that they are continuously condensing and shrinking. But even assuming that they continually rarefy, and that from this rarefaction there originates an attachment to the hollow and eventually a sweeping along, it may be believed that hundreds and thousands of years ago, before the rarefaction had reached anything like the limits it has today according to Sarsi's doctrine, this sweeping along would not have existed, there having been lacking any cause for its taking place. Indeed, nothing prevents my saying to Sarsi that this continually occurring rarefaction of his has not yet reached that degree at which it would exert force against and would press upon the hollow of the moon, and that it might attain this some two or three years hence, at which time I concede that the highest elemental sphere will commence to be moved, but in the meanwhile let him concede to me that it is not being moved today. I do not want Sarsi to burst out laughing, if perchance he thinks these and other similar replies are really ridiculous, for it is he who gave me occasion for producing this by having let it slip from his tongue and later from his pen that some material substances do perpetually rarefy and dilate. But in order to assist him I wish to teach him a point in his favor by telling him that this eternal rarefaction with pressure against the lunar hollow is superfluous, since he is able to show that the air is swept along by a bowl upon which it does not press or weigh at all, being located in the very region of the air.

XL

(Fourth argument) 'But let us see . . . [pp. 161-164] (*p.* 165) . . . denied these things.' (*p.* 109, l. 15 to *p.* 115, l. 14).

Sarsi now enters into a very abundant apparatus of experiments in order to confirm his statement and refute ours; these, since they were performed in your Excellency's presence, I submit to you as something in which I ought rather to defer to your judgment than to interpose my own. Therefore, if you will, reread the remainder up to the end of the passage, and I shall touch only piecemeal upon some details concerning various minor points.

First, what Sarsi attempts to attribute to me at the beginning of his experiments is quite false; namely, that I have said that the water contained in a bowl remains as motionless as air when the vessel is moved. I do not marvel that he has written this, however, since it may be permissible for one who continually reverses the sense of things written and printed by others to alter that which he merely says he has heard. Still, it does not seem to me to be entirely within the bounds of good breeding for a man thus to print and publish what he has heard from his neighbor; the more so when (either from not having understood very well, or by his own choice) his report is very different from what was actually said, as happens in the present case. It is my affair to print my things and publish them to the world, Sig. Sarsi, and not yours or anyone else's. And when at times in the course of arguing, a man may have said something foolish (as indeed sometimes does happen), why must you immediately rush into print with it and thus deprive him of the benefit of time for thinking it over better and amending his error by himself, changing his opinion (*p.* 166) and, in a word, preserving his own mastery over his mind and pen? What Sarsi may have heard (but, from what I see here, did not understand very well) is a certain experiment which I exhibited to some men of culture there in Rome, perhaps in the very chambers of your Excellency, partly in explanation and partly in refutation of a third motion attributed by Copernicus to the earth. To many people, this third motion which he assigned to the terrestrial globe round its own center and contrary to all other celestial

movements seemed a most improbable thing, and one which greatly disturbed the whole Copernican system. For all the celestial motions, whether of epicycles or of eccentrics—including Copernicus' diurnal and annual motions of the earth itself—are made in the *orbis magnus* from west to east, and this one alone was made by the earth from east to west, counter to its own two other movements and against all motions of the planets. I used to remove this difficulty by showing that such an event was not only not improbable, but was in accord with nature and was indeed almost necessary; for any body placed and supported freely in a tenuous and liquid medium will, when transported along the circumference of a large circle, spontaneously acquire a rotation about itself which is contrary to its larger motion. This effect was seen by taking in one's hands a bowl full of water and placing in it a floating ball, turning about on the toe with one's arm extended; the ball was seen immediately to turn upon itself in the opposite direction and to complete its rotation in the same time as one's own was finished. Thus the wonder ceased, and one would marvel rather how it could happen otherwise with the earth, if, being a body suspended in a thin and fluid medium and being carried about the circumference of a large circle in the space of one year, it should not have naturally acquired a similar annual rotation about itself opposite to the other. I spoke thus to remove the improbability attributed to the system of Copernicus, and I later added that anyone who would reflect more profoundly would recognize that Copernicus had falsely attributed to the earth a third motion which (*p.* 167) would not be a motion at all, but a non-motion and a state of rest. For it is certainly true that to him who holds the bowl, the ball placed within it will appear to move with respect to himself and to the bowl, and to be turning. But the same ball compared with the wall of the room and with external things does not turn at all and does not change its tilt; whatever point upon it was originally directed toward some external mark on the wall, or some other more distant place, points always in that same direction. Now this is what had been asserted by me, and as your Excellency sees it is very different from what Sarsi relates. This experiment, and perhaps some other, might give occasion for someone who was often present

at our discussions to attribute to me that which Sarsi recounts in this place; namely, that by a certain natural talent of mine I used sometimes to explain by means of a few simple and evident things some others which were more difficult and abstruse; this praise Sarsi does not entirely deny to me, as you see, but partly grants. I must acknowledge this concession as coming rather from his courtesy than from his inner feelings, since so far as I can see he is not one to be easily persuaded of my talent—for deeming Sig. Mario's essay to be mine, he said at the end of the previous weighing that it was written in very obscure terms whose sense he had been unable to divine.

As I said, your Excellency, I submit these experiments to you, who have seen them; and my only reply against them all will be something already written by Sig. Mario in his *Letter*.³⁵ But first I shall pay a little attention to a certain argument which Sarsi combines with his experiments, but which I should really have much preferred had not been mentioned, for the sake of his own reputation as well as that of his master—if it is true that he was a pupil of the teacher he pretends to. Alas, Sig. Sarsi, what is this extravagance that you write here? Unless there has been a ghastly printer's error, these are your words: 'Thence you may see that whenever the mover was greater than the moved, the motion was far easier; for the lid AB having been placed on the vessel, then it was by the motion of the inner surface of the dish and of the lid at the same time that the air had to be moved, (p. 168) and this was greater than the neighboring air which was being moved, for that surface was the container and the air was the contained.' Please tell me, Sig. Sarsi, with what you are comparing this surface of the bowl and of its cover; with the surface of the contained air, or with the air itself—that is, with the body of the air? If with the surface, then it is false that the former is larger than the latter; rather, they are precisely equal, as you are taught by the Euclidean axiom, *Quae mutuo congruunt, sunt aequalia* ('Things mutually congruent are equal'). But if you mean to compare the containing surface with the air itself, as your words actually say, then you are making two enormous errors. The first is in comparing two quantities which differ in kind and are therefore not comparable, as said in one of Euclid's definitions:

Ratio est duarum magnitudinum eiusdem generis ('Comparison is of two magnitudes of the same kind'). Do you not know that a man who says, 'This surface is larger than that volume' is no less wrong than one who might say, 'A week is larger than a tower,' or 'Gold is heavier than the key of C'? The other error is that if one could compare a surface with a solid, the case would be quite the contrary of what you say; not only would the surface fail to be greater than the solid, but the latter would be more than a hundred million times greater than the former. Do not let yourself be taken in by such monstrosities, Sig. Sarsi, nor yet by the general proposition that the container is greater than the thing contained, even when the two partake of mutually comparable quantities. Otherwise, you will have to believe that the outermost winding of a ball of yarn is larger than the yarn within it, since the latter is the contained and the former is the container; and that since they are of the same material, the husk will have to weigh more, being larger. I much suspect that you have taken with some such equivocation a statement that is quite true when accepted in its plain sense, namely, that the container is greater than the contained whenever 'container' is taken to mean both it and its contents. Thus a square described about a circle is greater than the circle, provided one takes the whole square; but if one wished to take only the remainder of the square after deducting the circle, it would not be larger at all (*p.* 169) but would be very much smaller than the circle, even though it surrounded and included it. But alas! time flies, and I am wearing myself out over these puerilities.

Now to counter all Sarsi's experiments, let your Excellency make the bowl rotatable upon its axis, and in order to make sure whether the air which it contains goes along when it is spun rapidly, take two lighted candles; attach one inside the bowl an inch or two from its top, and hold the other in your hand inside the bowl at a like distance from the top. Then make the bowl rotate rapidly; if after a while the air also goes round with it, then since the bowl doubtless moves the contained air and the attached candle with equal speed, the little flame of the first candle will not bend at all, but will remain as if the whole were stationary. This is what happens when a man

runs with a lantern which encloses a flame—this is not blown out or even bent, inasmuch as the air which surrounds it moves with equal speed. The same effect is seen even more clearly in a rapidly sailing ship, for lights which are placed below decks make no movement whatever, but remain in the same state as if the ship were at rest. But the little candle which is held fixed will give a sign of the circulation of the air which, blowing upon it, will cause it to bend. If the result is the opposite—that is, if the air does not follow the motion of the bowl—the fixed flame will remain straight and quiet while the other one, which is carried rapidly by the bowl, will bend by being pushed against the quiet air. Now whenever I have seen the experiment, it has always happened that the little stationary candle has remained burning with a straight flame while the one attached to the bowl has always been much bent and frequently blown out. And surely your Excellency will see the same, or anyone else who wishes to put it to the proof. Thus you may judge what must be said about the action of the air.

From Sarsi's experiments, the most that may be deduced is that a very thin layer of air no more than one-quarter of an inch thick and contiguous to the bowl is carried round. This suffices to produce all the effects of which he writes, (*p.* 170) and a sufficient cause for them would be the roughness of the surface, or any little cavities or prominences greater in one place than another. But finally, if the hollow of the moon should carry along the contained exhalations up to a thickness of one inch, what does Sarsi expect to accomplish with this? Let him not believe that if a bowl carries along half an inch of air, then a larger bowl must carry more; for I think it likely that this would transport less. Neither do I believe that the high speed with which the lunar hollow completes its rotation in twenty-four hours need have any greater effect. Indeed, I shall go so far as to say that I rather suspect it would achieve less rather than more than what would be accomplished by a bowl which made but a single revolution in twenty-four hours. Yet let us assume and grant to Sarsi that the lunar hollow sweeps along with it as much of the contained exhalations as mentioned above; what then? What follows from this that is unfavorable to Sig. Mario's main point? Does the material of the

comet perhaps take fire by this motion, or is it true that this will not take fire, whether moving or non-moving? The latter is what I believe, for if the whole remains fixed it will not excite fire, for which Aristotle requires motion; and if the whole moves, then there will be no friction or rubbing, without which there will be kindled no heat, let alone fire.

Now here you see a great expenditure of words by both Sarsi and me to determine whether the solid hollow of the lunar orb (which does not exist in nature), moving round (which it has never done), sweeps along with it the element of fire (which we do not know to exist), and thereby the exhalations, which accordingly kindle and set fire to the material of comets (of which we do not know whether it is there situated, and are certain that it is not combustible). Here Sarsi reminds me of the saying of that most witty poet:

By Orlando's sword, which they have not,
And perhaps which they never shall have,
These blows of the blind have been given.³⁸

It is now time for us to come to the second proposition; yet before we pass on to that, since at the end Sarsi repeats that I have constantly denied that water moves with the motion of the bowl and that air and other tenuous bodies adhere (*p.* 171) to smooth ones, let us reply once more that he does not speak truly, as neither Sig. Mario nor I have ever said or written any such thing. Sarsi, finding nothing to hold on to, goes along manufacturing handles for himself.

XLI

Now let your Excellency pass on to the second proposition. 'Aristotle says that . . . do not adhere.' (*p.* 115, l. 18 to *p.* 116, l. 22).

At the outset of this dispute Sarsi wants to make Sig. Mario and Aristotle agree, and to show that both have announced the same conclusion when one says that motion is the cause of heat and the other says that it is not motion, but the brisk rubbing of two hard bodies. And since Sig. Mario's proposition is true and needs no comment, Sarsi interprets the other one by saying that if indeed motion, as motion, is not the cause of heat, (*p.* 172) but rubbing is,

still such friction is not made without motion, so that we may say at least derivatively motion is the cause. But if that is what Aristotle meant, why did he not say friction? When one may say definitely what he means with a simple and very apposite word, I cannot see why he should make use of an inappropriate one requiring limitations and being finally transformed into something quite different. Besides, assuming that this was Aristotle's meaning, it is still different from Sig. Mario's; for to Aristotle the rubbing of any bodies is sufficient, even of thin and tenuous ones or of the air itself,³⁷ while Sig. Mario requires two solid bodies and considers that trying to pulverize and thin out the air is as great a waste of time as the proverbial grinding of water in a mortar. I am of the opinion that the original proposition might be quite true, taken in the simplest sense of its words; perhaps it came from some good school of antiquity, but Aristotle, not having properly fathomed the minds of the ancients who set it forth, later deduced from it a false conception. And this may not be the only proposition which in itself is true but which is understood in the Peripatetic philosophy in a sense which is not true. I shall touch on this somewhat later.

Now let us follow Sarsi, who (against Sig. Mario's statement) will have it that heat is excited without any consumption of the bodies that are rubbed together and heated, and who goes about proving this first by reason and then by experiments. So far as reason is concerned, I can dispose of all his instances very briefly; for he addresses interrogations to Sig. Mario and then replies for him and refutes these replies. Hence I say that if Sig. Mario does not respond in that way, Sarsi will have to fall silent.

And really, taking this first reply for example, I do not believe that Sig. Mario would say that in order to become hot, bodies would first have to be rarefied, and that rarefaction diminishes them, and the thinner parts would fly away, as Sarsi writes; from such a reply I think I may say that he is out of accord with Sig. Mario's thinking. In this action, (*p.* 173) one must consider the body which is to generate the heat and that which is to receive it. Sarsi thinks that Sig. Mario would require the excitation and the consumption of parts to take place in the body which is to receive the heat, while I

believe he would have the body which generates heat be the one that is diminished. In a word, it is not the receiving but the conferring of heat which causes diminution in the conferring body. I should therefore like very much to have seen explained more specifically how it is possible to rarefy bodies without any separation of their parts, and how this affair of rarefaction and condensation proceeds, of which Sarsi seems to speak with great confidence; for to me, it is one of the most recondite and difficult questions of all physics.

It is also manifest that Sig. Mario would not have made the second reply—that such consumption of parts is necessary so that these more minute parts may first be heated, and then by these the rest of the body may become warmed. For in this way, the diminution would still affect only the body which is to be heated, whereas Sig. Mario gives this property to that body which does the heating. One must note, however, that it often happens to be the same body which produces heat and receives it; thus, in hammering a nail, those parts of it which are violently compressed excite heat, and that which is heated is this same nail. What I have tried to say thus far is that the consumption of parts depends upon the act of producing heat and not upon that of receiving it, as I shall explain more in detail farther on. Meanwhile, let us hear the experiments by which Sarsi thinks he has revealed the possibility of producing heat by friction without anything being consumed.

XLII

‘But if one desires . . . (*p.* 174) . . . than from hammering.’ (*p.* 116, l. 23 to *p.* 117, l. 13).

I can well believe that Sarsi has not detected any diminution of weight in a little piece of copper beaten and heated many times, even by using the most precise balance. But I do not think for that reason that there was no diminution, since it may very well have happened that the reduction was so little as to remain imperceptible in any balance whatever. I ask Sarsi, first, whether he believes that any difference would be noticeable if a button of silver were weighed, gilded, and then weighed again. He must say no, because we see gold reduced to such thinness that it will even sustain itself upon the

quiet air, dropping with extreme slowness, and with such gold leaf one may gild any metal. Moreover, this same button may be used two or three months before the gilding is consumed, but since it is ultimately consumed it is clearly diminishing every day and even every hour. Again, take a ball of amber, (*p.* 175) musk, or some other odoriferous material; carry it with you for a fortnight, and it will fill with odor a thousand rooms and a thousand streets, and in fact every place that you visit. This will not happen without some diminution of material, for without that there would doubtless be no odor; yet if you weigh it after this time, no sensible diminution will be found. Thus there are revealed to Sarsi insensible diminutions of weight occurring from consumption over a period of months on end, which is quite a different matter from the few minutes that he may have kept hammering at his bit of copper—and just that much more accurate is the assayer's balance than the philosopher's steelyard. Notice also that it might very well be that the attenuated material which produces heat should also be still more subtle than the substance of odors, for remember that the latter can be enclosed in glass or metal through which it does not leak out, but the substance of heat makes its way through every body.

Here Sarsi raises an objection, saying: 'If a test with the balance does not suffice to show us a consumption so small, how will you have it known?' The objection is ingenious, but not so ingenious that a little natural logic has not revealed the solution to us, and here are its steps. Of bodies which are rubbed together, Sig. Sarsi, there are some which are absolutely and certainly not consumed at all, others which are quite sensibly much consumed, and still others which are consumed indeed, but insensibly. Our senses show us that those which are not consumed at all by being rubbed, such as two very smooth mirrors, are not heated. We are certain that those which are noticeably consumed are heated, as is iron upon being filed. Therefore, for those of which we stand in doubt whether or not they are consumed upon being rubbed, if we find by the senses that they are heated, we ought to believe that they are also consumed; and only of those which are not heated may it be said that they are not consumed either.

Before going farther, for Sarsi's instruction I should like to add something to what I have said above. To say, 'This body has not lost weight in the balance, and therefore no part of it has been consumed,' (p. 176) is very fallacious reasoning, since it is possible for some part of it to be consumed and yet for it not to lose weight but to gain it. This will always be the case when that which is consumed has less specific gravity than the medium in which it is being weighed. Thus, for example, it may happen that a very knotty piece of wood taken from near the root will sink to the bottom when it is placed in water. Under water let it weigh four ounces, say. Upon filing away some of it—not the knotty parts, but the lighter part which has less specific gravity than water and thus which partly sustained the whole mass—it may happen, I say, that the remainder will weigh more than the original mass did in the same medium. Likewise, it may be that in filing or rubbing together two pieces of iron, or two stones, or two sticks, some particles of matter less heavy than air are parted from them, and if this alone is removed it will leave the body heavier than before. What I say here is said with some probability, and not merely as a flight and retreat to leave the adversary some trouble in refuting it. For let your Excellency make careful observations in breaking glass or stones or some other material, and at every fracturing you will see some easily perceptible fume emerging which will ascend high in the air, a necessary argument that it is lighter than air. I first observed this in glass, when with a key or some other piece of iron I was knocking the corners off it and rounding it. Besides the many little pieces of various sizes that flew off and fell to the ground, a subtle fume was seen always ascending. The same thing is seen to occur upon breaking any stone in a like fashion. Besides what is evident to sight, the odor gives us a very clear argument and sign that in addition to the said fumes there may be ascending other more subtle sulfurous and bituminous parts which are invisible but which make themselves evident by their smell.

Let Sarsi perceive now how superficial and how far from profound, apart from appearance, is his philosophizing. He need not think that he can reply with limitations, distinctions; with *per*

accidens, per se, and mediate; with primary, secondary, (*p.* 177) and such idle talk; for I assure him that in maintaining one error he will commit a hundred more serious ones and will produce ever greater follies in his camp. Greater, that is, than these which remain for me to consider at the end of the present matter, where he first marvels that what is consumed, being imperceptible to the balance, could nevertheless produce heat, and then he adds that much iron is consumed when filed—much more than when beaten with a hammer—and yet filing produces no more heat than hammering. This reasoning is quite vain, for thus one would measure by weight the amount of something which has no weight, or which is so light as to rise rapidly in the air. And if that which is converted under strong friction into hot material was but a small part of that solid body, it should not be astonishing that a very small quantity of this might be rarefied and expanded into a very great space. Let Sarsi consider into how great a bulk of hot and blazing material a little stick is resolved; and of this, the visible flame is the smaller part, since that which is insensible to sight but perceptible to touch is a great deal larger. As to his other point, this objection would have some plausibility if Sig. Mario had ever said that all the iron consumed in filing would become calorific material, for then it would appear reasonable that the iron consumed by the file would heat up much more than that beaten with a hammer. But it is not the filings which accomplish the heating; it is another substance, incomparably subtler.

XLIII

Let us proceed. 'Therefore I believe . . . (*p.* 178) . . . suited to receive fire.' (*p.* 117, l. 14 to l. 30).

Here it seems that Sarsi is preparing to bring forth a sounder doctrine and a better explanation of the difficulties being dealt with, but I do not see that much new is introduced, or anything very prejudicial to Sig. Mario. Sarsi says that the hot or cold quality of bodies rubbed together corresponds closely with their heating much or little, and that the business depends upon many other things which are not so evident. This I know only too well, but it does not seem to get us any farther. The second clause is too mysterious and

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the first is too obvious and well known, inasmuch as he tells me in effect nothing except that those bodies which are more calorific (or more disposed to heat up) become more heated, and those which are more frigific become less so. I likewise believe what comes next—that a fire is lighted more readily by rubbing together lighter and less dense sticks than hard and dense ones, even if the latter are rubbed together more briskly and for a longer time; but I fail to see that this accomplishes anything against Sig. Mario, who has not said the contrary. And I knew before now that a distaff of hemp will take fire more quickly in a slow fire than a piece of iron will in the most ardent furnace.

He next adds (and supports with the testimony of Seneca) that more lightning occurs during the summer when the air contains a greater quantity of dry exhalations. I admit this, but I question whether such exhalations take fire together with the air, and whether this happens by friction caused by some motion. I should accept as true what (*p.* 179) Sarsi has written if he had first assured me that nature had no other way of starting a fire besides these two—namely, by touching combustible material with an actually burning fire, as when we set fire to a torch with a lighted candle, and by the friction of two non-burning bodies. But since other ways exist, such as the reflection of solar rays in a concave mirror or the refraction of the same rays in a ball of crystal or of water, and when straws and other thin bodies are sometimes seen to take fire in the street by reason of excessive heat without any commotion or agitation, even when the air is quite tranquil (and if it were agitated and blown by wind perhaps no fire would occur)—since, I say, there are these other methods, then why should I not believe that there may be still others different from these, and that exhalations in the air and between the clouds take fire because of them? Why must I attribute this to vehement motion when I see first that fire is not excited without the rubbing of solid bodies, which do not exist among the clouds, and when in addition no commotion is discovered in air or in clouds when lightning and heat lightning are most frequent? I think that saying this has inherently no more truth than when the same philosophers attribute the great rumbling of thunder to a tearing apart of

clouds, or to their knocking against one another, though in the brilliance of the greatest lightning flashes, while the thunder is being produced, not the slightest movement or change of shape is discovered in the clouds, when this ought to be very great in such a rending. I neglect the fact that when the same philosophers are dealing with sound they require the percussion of solid bodies for its production, and say that the striking of wool or hemp will not make any noise, while later on, when they need to do so, they say that mists and clouds striking together will render the loudest sound of all. A tractable and benign philosophy this, which so complaisantly and readily adapts itself to our needs and wishes! (*p.* 180)

XLIV

Now let us pass on to examine the experiment of the arrow shot by the bow, and of the leaden ball thrown by the catapult, these being set on fire and melted by the air as confirmed by the authority of Aristotle, many great poets, philosophers, and historians. 'Although Galileo laughs . . . (*p.* 181) . . . in the air.' (*p.* 117, l. 31 to *p.* 119, l. 7).

It is quite false that Sig. Mario or I would laugh and joke at the experience adduced by Aristotle, there being not the slightest hint of derision in Sig. Mario's book. He merely wrote that we do not believe that a cold arrow shot from a bow can take fire; rather, we believe that if a burning one were shot it would cool down more quickly than it would if held motionless. Now this is not derision, but simply the statement of an opinion. Sarsi next adds, since this experience has failed to convince us, that not only Aristotle but many other great men have believed and written the same. To this I reply that if it is true that in order to refute Aristotle's statement we are obliged to represent that other people have not believed it (*p.* 182) or written it, then neither Sig. Mario nor I nor anyone on earth will ever refute it, since it can never be the case that those who have believed it and written it did not believe it or write it. But I must say that it is news to me that anyone should actually wish to place the testimony of men ahead of what is shown by experience. Adducing such witnesses serves no purpose, Sig. Sarsi, since we

have never denied that many people have written and believed such things. We did indeed say that the thing is false, but as to authority yours alone is as cogent as that of a hundred men in making the effect true or false. You take your stand upon the authority of many poets, and against the experiments which we produce. I reply by saying that if those poets were to be present at our experiments, they would change their opinion, and without any revulsion whatever they might say that they had written hyperbolically or might confess themselves to have erred.

Although it is not possible to have present your poets (who, I say, would yield to experience), yet we have at hand archers and catapulters; and you may see for yourself whether by citing your authorities to them you are able to strengthen them to such an extent that the arrows and lead which they hurl will take fire and melt in the air. In this way you will find out how great is the force of human authority upon the facts of nature, deaf and inexorable as she is to our vain wishes. You tell me that there is no longer an Acestes or a Mezentius or a similar skillful paladin. I shall be content if you shoot an arrow not with a simple longbow but with a very stout steel crossbow and catapult drawn with levers and windlasses which could not be bent by the strength of thirty Mezentiuses; shoot ten arrows, or a hundred; and if it happens in a single instance that the steel tip of one takes fire or its shaft is burned, nay if its feathers so much as tan slightly, I shall have lost the battle—and your respect, which I so highly regard, into the bargain.

Now then, Sarsi, I do not want to keep you in suspense; you shall not take me to be so wayward as not to wish to yield to the authority and testimony of so many admirable poets, or to disbelieve that the burning of arrows and the (*p.* 183) melting of metals have not sometimes occurred. But I do say that the cause of such marvels is very different from that which philosophers have attempted to maintain when they have reduced it to attritions of air, or of exhalations, and such chimeras which are all foolishness. Do you want to know the real cause? Then listen to the peerless poet commenting upon the encounter of Ruggiero and Mandricardo, and the breaking of their lances:

The shafts, ascending, to the heavens came;
 And Turpin, writing here with truth entire,
 Says two or three fell back again in flame,
 These having mounted to the sphere of fire.³⁸

Do you think that the great Ariosto raised no question about the truth of this when he bolstered it with Turpin's testimony? Everyone knows how veracious Turpin is, and how far one is obliged to believe him.

But let us leave the poets in their proper place and return to those who reduce this cause to friction of the air. I deem that opinion false, and I shall consider what you, Sarsi, set forth in trying to show that very hard bodies may be consumed by friction with others that are softer. You say that this is seen plainly in water and in wind too, the latter wearing away and consuming the corners of the most solid towers, while the former by a continual distillation and constant beating can dig away marble and most solid reefs. I concede all this to you, for it is quite true, and I add more—I do not doubt in the least that arrows and not only leaden balls, but balls of stone and even of iron ejected from a cannon, are consumed more by striking the air at their consummate speed than are reefs or walls by the striking upon them of water and of wind. And I say that if to make a perceptible corrosion or peeling of reefs or towers there are required two or three centuries of beating by water and wind, then for the erosion of arrows or cannon balls it would suffice for them to remain in the air only two or three months. But I still do not see how a time of two or three pulse beats could have any noticeable effect on them.

In addition to this, there are two other difficulties for me in applying this truly ingenious conception of yours to our purpose. One (*p.* 184) is that we wish to discuss liquefaction and melting by heat, and not consumption by percussion. The other is that in making your case you require that not the solid body but the soft and subtle body be the one which is pulverized and made thinner—that is, the air, which must take fire thus. Now the experiences put forth prove that rocks, and not air or water, receive the friction; and truly I believe that the air and water, let them strike as they may, are not thereby made any thinner than they were in the first place. So I

conclude that there is little aid and assistance for your cause to be derived from these things or from that which you add about hail and raindrops. I concede to you that these become smaller in falling from on high. I grant this to you not because I believe that the opposite of what you say might not also be true, but because I do not see that in either case this has anything to do with the matter under discussion. That the sling with its hissing and popping is an argument of the air being condensed by its agitation, I leave to you to do with as you please; but note that it will be a contradiction of your position and a disaster to your case, since until now you have always said that strong agitation and commotion make friction, rarefaction, and finally an ignition of the air, whereas now you want condensation in order to give reasons for the whistling of the sling—or else to make sense of the very obscure words of Statius. Thus, that same commotion which rarefies the air in order to serve for melting and burning now condenses it in order to serve the slingers and Statius. But let us pass on to hear the testimony of the historians.

XLV

‘But lest the testimony . . . (p. 185) . . . at those learned men.’ (p. 119, l. 8 to p. 120, l. 2).

I cannot help wondering that Sarsi still wishes to persist in proving to me by means of witnesses that which I can see at any time by means of experiment. Witnesses are examined in doubtful matters which are past and impermanent, not in those which are factual and permanent. Thus it is necessary for a judge to seek by means of witnesses to determine whether it is true that Peter injured John last night, but not whether John is injured, for this he is able to see and to make the *visu reperto*. Furthermore, I say that even in conclusions of which one may attain a knowledge only by reasoning, the testimony of many men is worth little more than that of few, it being certain that the number of those who reason well in difficult matters is much smaller than the number of those who reason badly. If reasoning were like carrying burdens, where several horses will carry more sacks of grain than one alone, I should agree that several reasoners (p. 186) would avail more than a single one; but reasoning

is like running and not like carrying, and one Arab steed will outrun a hundred pack horses. So when Sarsi brings in such a multitude of authors, it does not seem to me that he strengthens his conclusion, but that he ennobles Sig. Mario's cause and mine by showing that we have reasoned better than many men of great reputation.

If Sarsi wants me to believe from Suidas that the Babylonians cooked eggs by whirling them rapidly in slings, I shall do so; but I must say that the cause of this effect is very far from that which he attributes to it. To discover the truth I shall reason thus: 'If we do not achieve an effect which others formerly achieved, it must be that in our operations we lack something which was the cause of this effect succeeding, and if we lack but one single thing, then this alone can be the cause. Now we do not lack eggs, or slings, or sturdy fellows to whirl them; and still they do not cook, but rather they cool down faster if hot. And since nothing is lacking to us except being Babylonians, then being Babylonians is the cause of the eggs hardening.' And this is what I wished to determine. Is it possible that Sarsi in riding post has not observed the coolness brought to his face by the continual change of air? And if he has indeed felt this, can he prefer to believe things which happened two thousand years ago in Babylon, as related by others, rather than present things which he himself experiences? I beg your Excellency to have him see some time how wine is frozen in midsummer by means of rapid agitation, without which it would not freeze at all. And I leave it to you to judge the reasons which Seneca and others arrive at for this effect, when the effect itself is false.

I willingly yield to Sarsi's invitation to me to listen attentively to Seneca's conclusion, and to his, then demanding whether anything could be said more clearly or more subtly; and I confirm that no falsehood could be told more subtly or plainly. But I should not like to have him place me beyond the bounds of good breeding (as he tries to do) by being compelled to believe what (*p.* 187) I deem false lest by denying it I give the lie to men who are the flower of literature—and what is more dangerous, to valorous soldiers. For I think that they believed they were speaking the truth, and hence their falsehood carries no dishonor. But when Sarsi says that he does not want

to be one to give affront to wise men by contradicting what they say and refusing to believe them, I say that I do not want to be among those ignoramuses and ingrates toward nature and toward God who, being given senses and reason, should wish to defer such great gifts to the mistakes of one man, or to believe blindly and stupidly what I wish to believe and subject the freedom of my intellect to anyone who is just as liable to error as I am.

XLVI

'I shall not . . . (*p.* 188) . . . with weakened force.' (*p.* 120, l. 3 to l. 29).

Sarsi continues in the style in which he began, trying to establish by the stories of others what actually exists and may be seen every day by experiment. And as he has found very distinguished men to confer authority upon ancient archers and slingers, so he has found a modern historian no less worthy of credit (nor a whit less authoritative than any ancient) to render credible the melting of modern cannon balls and musket shot. But it does not detract from the credit or dignity of an historian to submit a wrong reason for a physical effect, the historian being concerned only with the effect while the cause is the province of the scientist. So I believe with Sig. Homer Tortora that a cannon ball covered with lead had little effect in breaking down the enemy's walls, but I have courage enough to deny his reason for this borrowed from common philosophy. I hope that this same historian, just as he has previously believed what he has found written by many other great men whose authority has been sufficient to win his faith in everything they have said, will now listen to my reasoning, whether he changes his opinion or merely grasps the idea of wishing to see experimentally what the truth of the matter is. I believe Sig. Tortora in that the iron ball covered with lead in the Corbel battery had little effect, and that the iron core was found to be spoiled by the lead. I do not believe the other part, pertaining to science—that the lead would be liquefied and it is on that account that the iron ball was found to be stripped of it. I believe that attaining the extreme impetus given by a cannon to a ball shot at a wall, the lead covering on that side which was

compressed between the external wall and the internal iron ball was crushed and torn to pieces, and that practically the same thing was true of the opposite side, this being flattened against the iron; and that all the lead, tattered and distorted, (*p.* 189) blew in all directions. Afterward, covered with rubbish and therefore resembling the other fragments in the ruins, it was not easy to find. And perhaps it was not even looked for with that care which curiosity would demand of anyone who wanted to find out whether it was destroyed or merely lacerated. The lead would serve almost as a protection and a pillow for the iron ball, as a result of which the ball would give a weaker percussion; and in ungrateful payment for this the lead was left torn and so badly ruined that not even its corpse was found among the slain. Now since I understand that Sig. Homer is there at Rome, should he ever chance to meet your Excellency, I pray you to read him this little that I have written here—and the remainder which I am about to write on this subject—because I should highly value gaining the concurrence of a person who is justly esteemed by our age.

I say, then, that if we consider how much time elapses while the ball is going from the cannon to the wall, and what would have to take place within such a short time to cause lead to melt, it will be remarkable if anyone persists in the opinion that such an effect nevertheless does ensue. This time is much shorter than one pulse beat, within which time the friction of the air must occur which in turn is to set fire to it and finally to liquefy the lead. But if we imagine the same lead ball to be put in the midst of a fiery furnace, it will not be destroyed in even twenty pulse beats. It now remains for Sarsi to persuade people that the air ignited by rubbing had a heat incomparably greater than that of a furnace. Besides, experience shows us that a ball of wax shot from a musket will pass through a plank, which argues that such a ball is not destroyed while in the air, so Sarsi will have to furnish reasons for air liquefying lead but not wax.

Moreover, if the lead does liquefy, it will surely strike but a slight blow upon arriving against a corselet; hence it is very astonishing to me that musketeers have not thought of making iron pellets which

would not be so easily destroyed. Instead, they go on shooting with lead pellets which few iron plates can resist, and in those which do stand up to them there is found (*p.* 190) quite a deep dent—and the shot, too, is found, flattened, but not liquefied. In birds slain with small shot, there are found grains of lead of unaltered shape; it is up to Sarsi to supply a reason why pieces of lead weighing fifteen or twenty pounds are melted, but not those which run thirty thousand to the pound. Since pellets of distorted shape are to be found any day among the clothing of soldiers, I shall believe that some were shot into breastplates and remained among the cloths, others may have run slantwise into a helmet and have been lengthened in that way, dropping harmlessly into someone's clothes and remaining there without injuring him, and in sum a thousand accidents may happen in a skirmish without any melting taking place. For if melting did occur, it would be necessary (as Sarsi is aware) that lead falling very rapidly from high places should disperse itself into smaller drops than those of water, and be lost entirely, so that nothing of it might be found. I say nothing of the fact that the arrow and the ball accompanied by fiery air ought to show a brilliant track like that of a rocket in flight, especially at night, as Virgil writes that the arrow of Acestes marked its path in flame. This effect is not to be seen except in poetry, though nocturnal events such as lightning and shooting stars are made quite conspicuous by their bright light.

XLVII

'But we do not see . . . (*p.* 191) . . . poets and philosophers.' (*p.* 120, l. 30 to *p.* 121, l. 24).

This melting of the lead ball, which a few lines above Sarsi says is confirmed by daily examples, he next says happens very rarely; for it is so unusual that it has become regarded almost as a miracle. Now this sudden retreat of his assures us still further that he recognizes his own need of defenses and escapes, and this he proceeds to confirm by his own inconstancy, now wanting this and now that. He says that the disturbance of a little breeze suffices to ignite the air; even the mere arrival of a live man at a cemetery of the dead. At

another time (as he has said above and repeats at the end of this passage), he wants a vehement motion, a copious amount of exhalations, great attenuation of matter, and if anything else it would be something akin to these. To this last requirement, I subscribe more than I do to all the others, being quite certain that not only these ignitions but any other effect, however marvelous and recondite, would follow if such requisites for it were to exist.

I should like to know *apropos* of what it is that Sarsi, after having told (*p.* 192) of the flame which is lighted over a cemetery by the mere arrival of a man or a light breeze, asks me what friction of hard bodies there is here. I have said that friction capable of exciting fire is derived only from solid bodies. Now I do not know by what logic Sarsi pretends to deduce from that statement the conclusion that I would have it that wherever there is fire there can be no other cause for it than such friction. Hence I reply to Sarsi that fire may be excited in many ways, among which is the vigorous rubbing of two hard bodies; and since such friction cannot be produced by subtle and fluid bodies, I say that comets, lightning, shooting stars—and now I add the cemetery flames—are not lighted by friction of the air nor of winds nor exhalations, but indeed that each of these most frequently occurs during the greatest tranquillity of air and when the wind is quite still. Perhaps you ask me what, then, is the cause of these burnings? In order not to enter into new battles, I reply that I do not know; but I do know that neither water nor air is ever ground apart or catches fire or burns, being material incapable of being ground, and being incombustible. Lighting a single straw or one strand of flax is followed by a fire which does not stop until all the straw or flax is burned even if there are a hundred million cart-loads of it. If a little stick takes fire the whole house will burn—and, unless protection is quickly given, the whole city and all the wood in the world that may touch the first burning stick. Then what would become of the air, so subtle and with all its parts in contact, if ever a particle of it were to catch fire? Would not the whole of this burn, too?

Sarsi is ultimately reduced to saying with Aristotle that if the air ever happened to be abundantly filled with those warm exhalations in the presence of these requisites mentioned, then balls of lead

would be melted; and not only those from cannons, but those from muskets and those thrown by slings. This, then, must have been the state of the air at the time the Babylonians were cooking their eggs; thus it was, (*p.* 193) to the great good fortune of the besiegers, when they razed the city of Corbel; and when it is thus, things go very pleasantly for people who are being shot at. But since finding such a condition is a matter of chance, and one that does not occur any too frequently, Sarsi says that one must not have recourse to experiment, these miracles being performed not at our will but at a chance which is very difficult to meet with.

So, Sig. Sarsi, even if experiments are performed thousands and thousands of times at all seasons of the year, and in any place you like, without ever once producing what these poets, philosophers, and historians have mentioned, this will mean nothing; and we ought to believe their words rather than our own eyes. But if I find for you a condition of the air having all these requisites which you say are necessary, and still the egg is not cooked nor the leaden balls destroyed, what would you say then, Sig. Sarsi? But alas! I should be offering you too much, for there would always remain a retreat by your saying that some necessary condition was lacking. You are extremely prudent in taking a secure position when you say that there is needed for this effect a violent motion, a great quantity of exhalations, a well-attenuated material, 'and whatever else conduces to it'; this 'whatever else' is what dismays me and gives you a heaven-sent anchor, a safe harbor, a sanctuary wholly secure.—What I had in mind was to suspend the discussion and sit quietly by until some new comet should come along, imagining that while it lasted you and Aristotle would concede to me that just as the air was then properly disposed for setting it afire, so it would be also for melting lead and cooking eggs; for it seemed to me that you would require the same disposition for both results. Then I would have had us set to work with slings, eggs, bows, muskets, and cannons to clear up the truth of this affair for ourselves. And even without waiting for a comet, the time ought to be right in midsummer when the air is flashing with lightning, since all such ignitions are assigned to the same cause. But when (*p.* 194) you did not see a melting of

balls or even a cooking of eggs at such a time, I question whether you would not nevertheless refuse to give in, saying that there was lacking this 'whatever else conduces to it.' If you will tell me what sort of thing this 'whatever else' is, I shall endeavor to provide it; but if not, I shall relinquish my idea. Yet I believe it would undoubtedly go against you; if not completely and without exception, at least to the following extent: When we seek the natural cause of some effect, you are reduced to an attempt to satisfy me with an event so rare that you yourself finally name it among the miracles. But neither by whirling a sling nor by shooting a bow nor by means of muskets or cannons do we ever see these effects which you mention so often; or, if the event ever does take place, it is so rare that we ought to consider it a miracle and accordingly believe in the stories of others rather than seek to test it by sight. And since matters stand thus, I say you should be content to concede that ordinarily comets are not ignited by friction of the air and be willing to accept such an event as a miracle, if people grant to you in return that once in a thousand years a comet may be ignited by such friction in the presence also of all those circumstances which you require.

As to the objection raised and resolved by Sarsi, that some may perhaps say that musket balls and cannon balls are destroyed not by the friction of the air but by the fire with which they are so violently shot, in the first place I am not one of those who would make that objection since I say that they are not destroyed either in one way or in the other. Then as to the reply to this objection, I do not know why Sarsi has not put it in the clearest way, saying that balls and arrows driven by a sling or a bow—where there is no firing—plainly show the emptiness of this objection. This seems to me a much more direct answer than Sarsi's, which is that the time during which the ball is accompanied by the fire is too short for melting so large a chunk of lead. This is true enough, but it is also true that a still briefer time (*p.* 195) is spent in its journey through the air where it is to be melted by friction.

I do not know what to reply to the ultimate conclusion which he draws from this, since I simply do not understand what he means

when he says that it is enough for him to have shown that in these instances I have no retreat whatever for escaping from the testimony of the poets and philosophers. This testimony being written in a thousand books, I have never attempted to escape from it, and indeed anyone who should undertake such a thing would appear to me to be quite bereft of reason. I have certainly said that these attestations are false, and they still appear to me to be so.

XLVIII

‘But he objects . . . conduces to the same.’ (p. 121, l. 25 to p. 122, l. 15). (*p.* 196)

Now, your Excellency, read on to the end of this section, with regard to which there remains little for me to say as I have said a good deal already. I shall merely bring into consideration the way in which Sarsi tries to maintain that the burning of the comet may endure for many months, while other things in the air such as lightning, shooting stars, and the like are momentary. In order to do this, he invents two kinds of combustible materials, some light, rare, dry, and without any accompanying humidity; others viscous, glutinous, and allied with moisture. Of the first, he would make the momentary combustions; of the second, enduring fires such as comets.

Here I detect a manifest contradiction; for in that case, lightning, being made of rare and light material, should occur in the highest regions, while comets, being ignited in a more glutinous, and denser and consequently in a heavier material, would occur in lower places. Yet the contrary happens, since lightning does not occur more than a third of a mile above the earth, as we are assured by the small interval of time between our seeing the lightning and hearing the thunder when the latter reaches us from directly overhead. That comets are undoubtedly incomparably higher we know by their daily movement from east to west, similar to that of a star, even were there no other evidence. This is enough concerning these experiences.

It now remains, in accordance with the promise made above to your Excellency, for me to tell you some of my thoughts about the proposition, ‘Motion is the cause of heat,’ and to show in what sense

this may be true in my opinion. But first I must give some consideration to what we call 'heat,' for I much suspect that in general people have a conception of this which is very remote from the truth, believing heat to be a real attribute, property, and quality which actually resides in the material by which we feel ourselves warmed.³⁹

Therefore I say that upon conceiving of a material or corporeal substance, (*p.* 197) I immediately feel the need to conceive simultaneously that it is bounded and has this or that shape; that it is in this place or that at any given time; that it moves or stays still; that it does or does not touch another body; and that it is one, few, or many. I cannot separate it from these conditions by any stretch of my imagination. But that it must be white or red, bitter or sweet, noisy or silent, of sweet or foul odor, my mind feels no compulsion to understand as necessary accompaniments. Indeed, without the senses to guide us, reason or imagination alone would perhaps never arrive at such qualities. For that reason I think that tastes, odors, colors, and so forth are no more than mere names so far as pertains to the subject wherein they reside, and that they have their habitation only in the sensorium. Thus, if the living creature (*l'animale*) were removed, all these qualities would be removed and annihilated. Yet since we have imposed upon them particular names which differ from the names of those other previous real attributes, we wish to believe that they should also be truly and really different from the latter.

I believe I can explain my idea better by means of some examples. I move my hand first over a marble statue and then over a living man. Now as to the action derived from my hand, this is the same with respect to both subjects so far as the hand is concerned; it consists of the primary phenomena of motion and touch which we have not designated by any other names. But the animate body which receives these operations feels diverse sensations according to the various parts which are touched. Being touched on the soles of the feet, for example, or upon the knee or under the armpit, it feels in addition to the general sense of touch another sensation upon which we have conferred a special name, calling it *tickling*; this sensation

belongs entirely to us and not to the hand in any way. It seems to me that anyone would seriously err who might wish to say that the hand had within itself, in addition to the properties of moving and touching, another faculty different from these; (*p.* 198) that of tickling—as if the tickling were an attribute which resided in the hand. A piece of paper or a feather drawn lightly over any part of our bodies performs what are inherently quite the same operations of moving and touching; by touching the eye, the nose, or the upper lip it excites in us an almost intolerable titillation while in other regions it is scarcely felt. Now this titillation belongs entirely to us and not to the feather; if the animate and sensitive body were removed, it would remain no more than a mere name. And I believe that many qualities which we come to attribute to natural bodies, such as tastes, odors, colors, and other things, may be of similar and no more solid existence.

A body which is solid and, so to speak, very material, when moved and applied to any part of my person produces in me that sensation which we call *touch*; this, though it pervades the entire body, seems yet to reside principally in the palms of the hands and especially in the fingertips, by means of which we sense the most minute differences of roughness, smoothness, and hardness, these being not so clearly distinguished by other parts of the body. Of these sensations, some are more pleasant to us and some less so, and they are smooth or rough, acute or obtuse, hard or yielding, according to the differences of shape which exist among tangible bodies. This sense being more material than the others and having its rise in the solidity of matter, it seems to be related to the earthy element. And since some bodies continually dissolve into minute particles of which some are heavier than air and descend, while others are lighter and rise on high, perhaps herein lies the origin of two other senses, accordingly as these particles strike us upon two parts of our bodies which are very much more sensitive than the skin and which feel the invasion of such subtle, tenuous, and yielding matter. The tiny descending particles are received upon the upper surface of the tongue, penetrating and mixing with its moisture; and their substance gives rise to tastes, sweet or unsavory accordingly as the shapes of these

particles differ, as they are few or many, and as they are fast or slow. The other particles, ascending, enter by our nostrils and strike (*p.* 199) upon some small protuberances which are the instrument of smell; and here likewise their touch and passage are received to our liking or our dislike accordingly as they have this or that shape, move quickly or slowly, and are few or many. The tongue and the nasal passages are indeed seen to be providently arranged for the above as to location; the former extends from below to receive the incursions of descending particles, while the latter are accommodated for those which ascend. Perhaps the excitation of tastes may be likened with a certain analogy to fluids, which descend through the air, and odors to fires, which ascend.

Next the element of air remains available for sounds; these come to us indifferently from below, from above, and from all sides, since we are situated in the air and its movements within its own domain displace it equally in all directions. And the situation of the ear is most fittingly accommodated to all positions in space. Sounds are created and are heard by us when—without any special ‘sonorous’ or ‘transonorous’ property—a rapid tremor of the air, ruffled into very minute waves, moves certain cartilages of a tympanum within our ear. External means capable of producing this ruffling of the air are very numerous, but for the most part they reduce to the trembling of some body which strikes upon the air and disturbs it; waves are thereby very rapidly propagated, and from their frequency originates a high pitch, or from their rarity a deep sound.

I do not believe that for exciting in us tastes, odors, and sounds there are required in external bodies anything but sizes, shapes, numbers, and slow or fast movements; and I think that if ears, tongues, and noses were taken away, shapes and numbers and motions would remain but not odors or tastes or sounds. These, I believe, are nothing but names, apart from the living animal—just as tickling and titillation are nothing but names when armpits and the skin around the nose are absent. And as the four senses considered here are related to the four elements, I believe that vision, the sense which is eminent above all others, is related to light, but in that ratio (*p.* 200) of excellence which exists between the finite

and the infinite, the temporal and the instantaneous, the quantity and the indivisible; between darkness and light. Of this sense and the matters pertaining to it, I pretend to understand but a trifle, and since a long time would still not suffice for me to explain that trifle—or even to hint at its explanation in writing—I pass this over in silence.

Returning now to my original purpose, and having already seen that many sensations which are deemed to be qualities residing in external subjects have no real existence except in ourselves, and outside of us are nothing but names, I say that I am inclined to believe that heat is of this character. Those materials which produce heat in us and make us feel warmth, which we call by the general name *fire*, would be a multitude of minute particles having certain shapes and moving with certain velocities. Meeting with our bodies, they penetrate by means of their consummate subtlety, and their touch which we feel, made in their passage through our substance, is the sensation which we call *heat*. This is pleasant or obnoxious according to the number and the greater or lesser velocity of these particles which thus go pricking and penetrating; that penetration is pleasant which assists our necessary insensible transpiration, and that is obnoxious which makes too great a division and dissolution of our substance. To sum up, the operation of fire by means of its particles is merely that in moving it penetrates all bodies by reason of its great subtlety, dissolving them more quickly or more slowly in proportion to the number and velocity of the fire-corpuscles (*ignicoli*) and the density or rarity of the material of these bodies, of which many are such that in their decomposition the major part of them passes over into further tiny corpuscles (*ignei*), and the dissolution goes on so long as it meets with matter capable of being so resolved. But I do not believe at all that in addition to shape, number, motion, penetration, and touch there is any other quality in fire which is 'heat'; I believe that this belongs to us, and so intimately that when the animate and sensitive body is removed, 'heat' remains nothing but a simple (*p.* 201) vocable. And since this sensation is produced in us by the passage and touch of the tiny corpuscles through our substance, it is obvious that if they were to remain at

rest their operation would remain null. Thus we see that a quantity of fire retained in the pores and narrow channels of a piece of quicklime does not warm us even when we hold it in our hands, because it rests motionless. But place the quicklime in water, where the fire has a greater propensity to motion than it has in air—because of the greater gravity of this medium, and because the fire opens the pores of water as it does not those of air—and the little corpuscles will escape; and, touching our hand, they will penetrate it and we shall feel heat.

Since, then, the presence of the fire-corpuscles does not suffice to excite heat, but we need also their movement, it seems to me that one may very reasonably say that motion is the cause of heat. This is that motion by which arrows and other sticks are burned and by which lead and other metals are liquefied when the little particles of fire penetrate the bodies, being either moved by themselves or, their own strength not sufficing, being driven by the impetuous draught of a bellows. Of these bodies, some resolve into other flying particles of fire and some into a most minute powder; some liquefy and become as fluid as water. But I hold it to be foolish to take that same proposition from the common point of view—that a stone or a piece of iron or a stick must heat up if moved.

The rubbing together and friction of two hard bodies, either by resolving their parts into very subtle flying particles or by opening an exit for the tiny fire-corpuscles within, finally set these in motion and, upon their encountering our bodies and penetrating and coursing through them, our conscious mind (*anima sensitiva*) feels that pleasant or obnoxious sensation which we have named heat, burning, or scalding. And perhaps when the thinning and attrition stop at or are confined within the tiniest particles (*i minimi quanti*), their motion is temporal and their action is calorific only, but, when their ultimate and highest resolution into truly indivisible atoms is reached, light is created⁴⁰ which has instantaneous motion—or let us say instantaneous expansion and diffusion (*p.* 202)—and is capable of occupying immense spaces by its—I do not know whether to say by its subtlety, its rarity, its immateriality, or yet some other property different from all these, and nameless.

I do not wish, your Excellency, to engulf myself inadvertently in an infinite ocean from which I should never get back to port, nor do I desire in trying to remove one difficulty to give rise to a hundred more, as I fear may in part have occurred by my sailing but this little distance from shore. Therefore, I shall desist until some more opportune occasion.

XLIX

‘THIRD PROPOSITION . . . [*pp.* 203–205] . . . they are not rejected.’ (*p.* 122, l. 16 to *p.* 127, l. 20).

(*p.* 206) Let your Excellency now pass on to the third proposition. Read and reread it attentively; I say attentively so that you may the more plainly recognize afterward how craftily Sarsi still continues in his style of altering, deleting from, adding to, and even diverting the argument and mixing with it things that are alien to the subject, trying thus to confuse his readers’ minds so that in the end there may remain with them, among things but obscurely understood, the opinion that Sig. (*p.* 207) Mario did not establish his doctrine in such a way that others have been unable to find means of opposing it.

Many people have held the opinion that a small bright flame appears much larger at a distance because it sets fire to, and thus renders of equal brilliance, the surrounding air; from afar the ignited air and the actual flame both appear then to be a single light. Sig. Mario, refuting this, says that the air is neither ignited nor illuminated, and that the irradiation which makes the enlargement is located not around the flame but upon the surface of our eyes. Sarsi seeks to oppose this correct doctrine, and instead of thanking Sig. Mario for having taught him something he had not previously known, he proceeds to try to prove that the air is illuminated in contradiction to Sig. Mario’s assertion. In this enterprise, he seems to me to have erred in several ways.

First, where Sig. Mario in countering what is said by these philosophers declares that the air is neither ignited nor illuminated, Sarsi suppresses all mention of ignition and deals only with illumination. Sig. Mario might reasonably say to Sarsi that while he was talking about one thing, Sarsi undertook to refute something else; that is,

Sig. Mario spoke of the air surrounding the flame and of the illumination which might arise from this air taking fire, whereas Sarsi has spoken of the illumination which occurs in vaporous air in the absence of fire, when such air is located at a distance from the illuminating object. Besides, Sarsi says at the outset that transparent bodies are not illuminated, and at first he places the air among these; then later he adds that the air is indeed illuminated when it is mixed with gross vapors capable of reflecting light. Well then, Sig. Sarsi, it is the vapors which are illuminated and not the air. You remind me of a man who says that some wheat has made him dizzy and light-headed, when in fact it was mixed with cockleweed; now it is the cockleweed that makes one suddenly ill, and not the grain. Are you trying to teach us that it is vaporous air which illuminates the dawn, which thousands of others have said, and as Sig. Mario himself had written in half a dozen places (*p.* 208) before you? And this is not all. You yourself, in this very passage, say that I admit this phenomenon with regard to the moon and Jupiter. Then all your proofs and experiments relating to dawn, haloes, mock suns, and the moon hidden behind some wall are quite superfluous, since we have never questioned (let alone denied) that vapors diffused through the air, or clouds, or mists are illuminated. But what is it that you wish to do next with this illumination, Sig. Sarsi? To say, perhaps, as in effect you do, that on its account the primary illuminating objects appear larger? Surely you should perceive that if such were the case the sun and moon would have to look as large as the whole dawn—or as their entire haloes—since the vaporous air which is made to participate in their light would be of such dimensions. You have found it written (I say this because you cite philosophers and authorities on optics to confirm and certify these propositions) that the vaporous region is brightened, and that the sun and moon appear larger by reason of that vaporous region when they are close to the horizon than when they are high in the sky, and hence you are convinced that their apparent enlargement depends upon that illumination. Both propositions are true—the vaporous air is illuminated, and the sun and moon do appear larger when close to the horizon as a result of the vaporous region—but it is false that there is any

connection between the two propositions or that the enlargement depends upon the illumination of that region. You are much deceived; put away this mistaken opinion, for these objects appear larger than usual not because of the light in the vapor but because of its spherical outer surface, and because of the greater distance of this surface from our eyes when those objects are closer to the horizon. Not only luminous bodies, but any bodies placed beyond this region will behave in the same way. Place between your eye and any object a convex crystalline lens, at various distances. You will see that when this lens is close to the eye it increases but little the size of the object viewed, but moving it farther away you will see that enlargement (*p.* 209) progressively grow. And since the vaporous region terminates in a spherical surface not much elevated above the convexity of the earth, all the straight lines drawn from our eyes to that surface are unequal, the one toward the zenith being the shortest of all and the others being gradually longer the more they point away from the zenith and toward the horizon. Let it be remarked in passing that here one may easily deduce the cause for the apparently oval shape of the sun and moon when they are near the horizon by considering the great distance of our eyes from the center of the earth, which is also the center of the vaporous sphere. Whole books have been written about this appearance, as I believe you know, though the entire mystery requires no doctrine more profound than to understand why a circle seen straight on looks round to us, while looked at in foreshortening it appears oval.

Now getting back to our subject, I fail to see Sig. Sarsi's purpose in saying that it is ridiculous to assert that dawn and twilight and other similar splendors are generated in the moisture spread over our eyes, and that it would be no more ridiculous to say that when we look toward the zenith our eyes are drier than when we look at the horizon, and that it is for that reason that the moon and sun should look smaller to us in the former position than in the latter. Not having discovered anyone who has ever said such a thing, I do not know why Sarsi introduces this foolishness. But as Sarsi represents us as too simple-minded, let us see whether that accusation would not apply better to him than to us.

We are dealing now with that adventitious irradiation, crowned with which the stars and other lights look much larger than they would if their little bodies were seen without such rays. These rays are much less lucid than the true and primary flame, and in the midst of them the little body remains indistinct so that both it and the irradiation appear to us as a single, large, and resplendent object. Sarsi wants to relate this irradiation and enlargement to the light which is produced by refraction in the vaporous air, and he will have it that for this reason the sun and moon look (*p.* 210) larger near the horizon than on high. What is worse, he pretends that many other philosophers have believed the same, which is false; they never made such a bad mistake. Sarsi should see at once that this is a great error by considering the enormous difference perceived between the light of the sun or moon and the diffused brightness which surrounds either, and within which each is discerned as incomparably brighter and more sharply defined. This is not true of the irradiation of the stars, in the midst of which the little body of a star remains indistinct and is clothed with a brightness identical to its own.

But I hear Sarsi answer by saying that this large sun or moon is not actually a bare and plain body, but an aggregate composed of a small real body and an irradiation of equal brilliance which crowns it and encloses it in the center, resulting in the large and uniformly bright apparent disc. Now if this is so, Sig. Sarsi, why does not the moon appear equally large when it is in the center of the sky, too? Is there no vaporous air above, suitable for being illuminated? I do not know what you would reply; I cannot even guess; for a truth may be assailed only with fallacies and fictions, and since these (as you know) are infinite, I cannot divine what your choice would be. But to put an end to the matter and if possible to lift you and others out of error, it suffices to make it clear to you that that huge moon which you see on the horizon is plain and bare and is not enlarged by any adventitious circumfused light. For this purpose, it should be enough to note that the spots spread out over its whole surface, right out to its extreme edges, are seen precisely the same as when the moon is in the center of the sky. If matters stood as you believe,

then the spots in the low, large moon should be seen collected in its central parts, leaving the surrounding crown bright and spotless. Therefore the sun and moon appear larger when low than when high not because of an additional bright region, but from an enlargement of the whole image by refraction at the distant vaporous surface.

Behold, Sig. Sarsi, how easy it is to defeat the false (*p.* 211) and to sustain the true. Abundant evidence of the falsity of many propositions to be read in your book prevents my believing that you remain ignorant of it, and I think it may be that you recognized in your heart the validity of the correct reasons, but were ultimately reduced to trying whether the opponent might not be forced to retreat and yield by being taken in by things you yourself knew to be false. You have thus boldly introduced these things in imitation of the gambler who, seeing from the cards on the table that he has lost the stakes, attempts by offering a larger bet to make his opponent believe that his hand is much stronger than what he himself can see it to be, hoping he will be overcome with fright and will yield.

Since I see that you have got rather tangled up among these primary lights refracted or reflected in vapors or in the eye, it is appropriate for you as a pupil that I, as a professor and an old hand at teaching, should somewhat disentangle you. Know, then, that the sun, the moon, and the stars—all of them brilliant bodies located far beyond the vaporous region—send out a brilliance which perpetually lights one half of that region. Of this illuminated hemisphere, the western extremity sends to us the morning dawn, and the opposite edge gives us the evening twilight; but neither of these illuminations increases or enlarges or in any way alters the apparent size of the sun, the moon, or the stars, which lie always in the center (or let us say at the pole) of this vaporous hemisphere which they illuminate. The parts which lie between our eyes and the sun or moon look brighter to us than those other parts which are by degrees removed from there, and whose brightness gradually diminishes, and it is this light which gives sign of the moon's approaching appearance when it is hidden behind some roof or wall.

A similar illumination occurs near a little flame which is located within the vaporous sphere, but in this case it is so weak and languid

that if we hide a light behind some wall at night and then start moving (*p.* 212) so as to see it, we shall perceive scarcely any circumfused splendor, nor do we see any light until the flame itself is discovered. Now this very weak light does not enlarge the visible image of the little flame at all. But there is another illumination here, made by refraction in the moist surface of the eye, and by this the real object appears to us to be surrounded by a luminous circle very inferior in brightness to the primary light. It is seen to extend through a greater or lesser space, not merely in proportion to the greater or lesser quantity of this moisture, but also according to the condition of the eye itself. I have observed this in myself, for as a result of a certain affliction I began to see a luminous halo more than two feet in diameter around the flame of a candle, capable of concealing from me all objects which lay behind it. As my malady diminished, so did the size and density of this halo, though more of it has remained with me than is seen by perfect eyes. This halo is not hidden by interposing the hand or any other opaque body between the candle and the eye, but remains always between the eye and the hand so long as the light of the candle itself is not cut off. But it does not enlarge the image of the little flame, and it is much less bright than that is.

Now there is a third very vivid splendor here, almost as bright as that of the original light itself; this is produced by reflection of the primary rays in the moisture at the edges of the eyelids, and it extends over the convexity of the pupil. That it is thus produced is clearly shown by changes in the position of the head, for accordingly as we incline the head, raise it, or hold it straight toward the luminous object, we see the latter irradiated only in the upper part, only in the lower, or in both. But we never see rays appear from left or right, for reflections made near the angles of the eye can never arrive over the pupil, as these angular parts of the eye lie beneath the horizontal line across the pupil on account of the bending of the eyelid over the sphere (*p.* 213) of the pupil. And if by pressing with a finger upon the eyelid one widens the eye and removes the edges of the eyelid from the pupil, one will no longer see rays above or below, since the reflections made in the edges do not then alight upon the pupil.

This, and this alone, constitutes that irradiation by which small lights appear large and radiant to us, and within which the little real flame remains swallowed up and indistinct. That other illumination has nothing, Sig. Sarsi, nothing whatever, to do with enlargement, for it is so far inferior in brightness to the primary light that one would indeed be completely blind not to see the boundaries, confines, and distinctions between the one and the other. Besides, as I have said before, if the discs of the sun and moon were enlarged by this illumination, they ought to appear as large as the immense circles of their dawning. If you would say that you do not deny this radiant crown to be a sensation of the eye, but that I have not thereby proven that some part of it does not depend upon the illumination of the surrounding air, I beg you to desist from such miserable begging for niggardly assistance. What is it that you want to do with that very feeble light, mixed with those extremely bright rays reflected from the eyelid? You would be adding about as much as a torch at noon would add to the sunlight. Of this light diffused through the vaporous air, I am happy to concede to you not only that little which you ask for, but all that is embraced in the entire dawn and twilight and the whole vaporous hemisphere. Nor do I ask that the luminous body be robbed of this by the telescope or by any other means; for your complete satisfaction I even want to have it enlarged by the telescope as are all other objects, so that it will not merely equal the whole dawn but will occupy a thousand times as much space—if so much space may be embraced by the telescope. But none of these things will help you or your master one bit, for in order to maintain your chief conclusion (which is that because of their distance the fixed stars do not receive any enlargement from the telescope), you would need to have the star and its (*p.* 214) irradiation be one and the same thing, or at least you would need to have the irradiation located around the star. Neither of these things is the case, for the irradiation is in the eye, and the stars receive just as much enlargement as any other object seen through the same telescope, as Sig. Mario very correctly writes and proves.

These other little diversions of yours about illuminated vaporous air and about high and low suns and moons are mere palliatives, so

to speak; they are attempts to get away from school and to divert the reader from the main point. Among these various diversions, two are quite correct—this one in which you attempt to show by a long argument that the interposition of a finger does not impede one's view of a candle, and that other one which deals with a thin thread and the interposition of a body smaller than the pupil; but I note that they have nothing to do with the matter in hand. I see that in your heart you too have recognized this, inasmuch as, when the time came for applying these things to the subject and drawing some conclusion, you stopped short, and, leaving them dangling, you passed on to another proposition, still trying to prove by reasoning something that is contradicted by a hundred evident experiences. And although you may see by looking through the telescope that Saturn is sharply bounded and is of a very different shape from the other stars, the discs of Jupiter and Mars (especially when close to the earth) are perfectly round and well defined, Venus is from time to time horned but is very exactly delimited; the little globes of the fixed stars, particularly the larger ones, are quite distinct, and finally a thousand little candle flames situated at a great distance are outlined as if they were close at hand. On the other hand, the naked eye distinguishes none of these shapes without the telescope, and all are seen encumbered with alien rays in the same radiant shape. Yet despite all this, you insist that the telescope does not show them shorn of rays, being persuaded by certain of your own arguments. Well, I am not obliged to reveal all the fallacies in these, having experiences which contradict them; still, for your benefit, I shall briefly hint at them.

In order to throw the most light upon my meaning, I ask you, Sig. Sarsi, how (*p.* 215) it comes about that Venus, so strongly surrounded by these extra alien rays among which its true shape is lost, has been crescent-shaped thousands of times since the world began without ever once being seen so by any living person, and had always appeared constant in shape until I first discovered her alterations by means of the telescope? It did not happen thus with the moon, whose changes of shape are displayed to the naked eye with no perceptible interference from adventitious irradiation. Do not

say that this occurs because of the great distance of Venus and the proximity of the moon, for I shall tell you that what happens with Venus takes place also with the flames of the candles, the shape of which becomes confused among the rays and gets lost at a distance of only a hundred yards, no less than does that of Venus. If you wish to reply correctly, you must say that it comes about from the smallness of Venus' body in comparison with the apparent size of the moon, and you must picture to yourself the length of these rays produced in our eyes as, say, four diameters of Venus, which would not be one-tenth the diameter of the moon. Now imagine the tiny crescent of Venus crowned with a wig which spreads and extends out around it a distance of four of her diameters, and at the same time think of the huge crescent of the moon with a wig no larger than one-tenth of the moon's diameter. It should not be very difficult to understand how the shape of Venus is completely lost under this coiffure while that of the moon will be altered very little. What happens here is precisely what would happen if you were to dress an ant in the hide of a lamb—the ant's tiny members would be completely lost in the length of the hairs, and the resulting appearance would be merely that of a fleece of wool, but the lamb is large enough so that its members may be clearly discerned under the same clothing.

I shall add that when the luminous object receives the brilliant coiffure which resides in the eye, its spreading is less limited by the size of the object than by the constitution of the (*p.* 216) eye itself. This we see by compressing the eyelid so that very long rays appear to surge from the luminous object, for those which come from the moon are not perceived as longer than those from Venus or from a torch or lantern. Fancy to yourself some definite size for the wig, and in the center of this imagine a very tiny luminous body. The shape of this will be lost, being crowned by excessively long hairs. But if you now imagine a larger and larger body, its real image will eventually be able to occupy so much of the eye that the wig will extend beyond it little or not at all. Thus the image of the moon, for instance, may occupy more space in the eye than does the usual irradiation. This being so, suppose the actual disc of Jupiter to occupy in your

vision a little circle whose diameter is one-twentieth the breadth of the radiant wig; in this great space, the tiny actual circle remains indistinguishable. Now I bring a telescope to bear upon it, and this magnifies for me the image of Jupiter by twenty diameters; but it does not magnify the irradiation, as that does not come through the lenses; hence I shall see Jupiter no longer as a very small star, but as a round moon quite large and well defined. If the star is much smaller than Jupiter but shines very vividly and brightly (as does the Dog Star, for example, whose diameter is not one-tenth of Jupiter's, but whose irradiation is little less), then the telescope, by enlarging the star but not the wig, makes the tiny disc which originally was imperceptible in the midst of so broad a light four hundred or more times as large, so that its shape may be well distinguished. If you will establish your reasoning upon such foundations, you may get rid of all difficulties by yourself.

Next I reply to your objections against our statement when Sig. Mario and I say that the telescope robs the stars of their resplendent crown. This statement was not put forth in the expectation of its being called to account by so punctilious a person as you, who, when you have nothing else to take hold of, (*p.* 217) declaim at length against people who employ the ordinary term 'infinite' in place of 'very great.' If we said that the telescope robs the stars of this irradiation, we meant that it operates upon them in such a way as to cause their bodies to appear bounded and shaped as if they were bare and shorn of that obstacle which conceals their figure from the naked eye. Is it not true, Sig. Sarsi, that Saturn, Jupiter, Venus and Mars reveal to the naked eye no difference in shape, and that none of them show much variation in size from one time to another? Yet with the telescope Saturn is seen as in [the first figure of] the adjoining diagram; Jupiter and Mars in this way always [as in the next two figures]; Venus is seen in all these shapes [at the bottom]—and what is more remarkable, with the indicated differences in size—so that when crescent-shaped she looks forty times as large as when round. Mars looks sixty times as large at perigee as at apogee, though the naked eye shows this as only four or five times. You cannot deny these things, because they are visible and permanent; nor can you

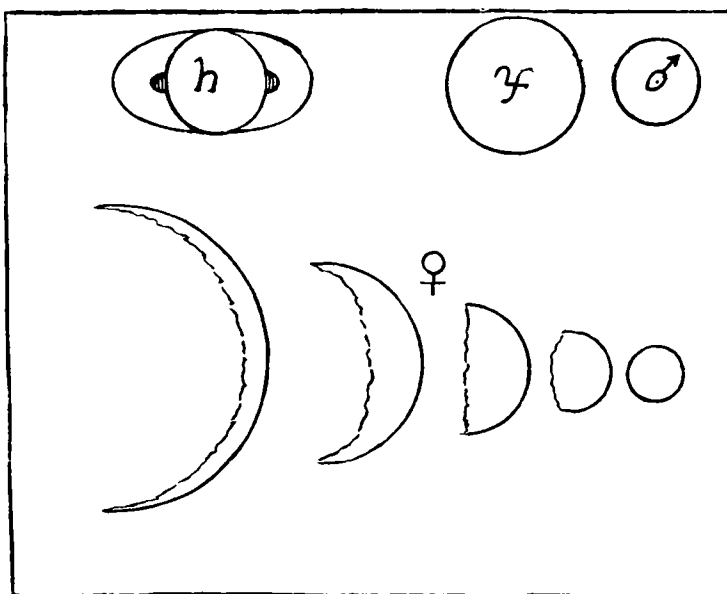


Fig. 22

hope by means of syllogisms to make (*p.* 218) people believe that things fall out otherwise. Now what we meant by saying the telescope ‘robs the stars of irradiation’ (for the mere words here are of no moment or consequence) is that it operates upon these stars in such a way [as to circumvent]⁴¹ the irradiation which disturbs the naked eye and impedes precise perception. This action is most momentous, and has remarkable and weighty consequences, but if our words offend you, who are still a pupil, you may alter them to suit yourself—just as you have previously changed our ‘enlargement’ into your ‘passage from nonexistence into existence.’

You think it reasonable to suppose that as the lucid image comes through the medium free of irradiation produced at the eye, so it should continue to do while passing through the lenses of the telescope. I reply by freely granting this to you, and I tell you that exactly the same thing happens to objects seen through the telescope as to those seen without it. The disc of Jupiter, for example, when

seen with the naked eye remains lost within the spread of its irradiation by reason of its small size; but that of the moon does not, for it occupies in our pupil so much larger a space than its radiant circle that it is seen as shorn, and not as hairy. In the same way, using the telescope to bring Jupiter's disc to my eye six hundred or a thousand times larger than its simple image, will, by reason of this amplitude, cause it to swallow up its coiffure of rays, and makes it look similar to the full moon. But the very tiny disc of the Dog Star, though enlarged a thousand times by the telescope, cannot equal its irradiated size so as to appear completely shorn. Yet since the rays near the outer edge are somewhat weak and separated, the disc is visible and the continuity of the little globe of the star is very easily seen amid the discontinuity of the rays. The more the telescope enlarges this, the more distinct and the less irradiated it appears. That is how matters stand, Sig. Sarsi, and we chose to call this 'robbing Jupiter of his crown'; if the words do not (*p.* 219) please you, you certainly have license to alter them as you will, and I give you my word that in the future I shall profit by your correction. But do not wear yourself out trying to alter the fact, because you will never succeed in doing that.

Since at the end you repeat that it is still necessary to grant that the surrounding air is illuminated and that the star appears larger for that reason, I reply to you once more that the circumfused vapors are indeed illuminated, but that it is not for this reason that the luminous body is enlarged. The light of the vapors being incomparably less than the primary light, the lucid body if large remains barren, and if small it is still sharply bounded, including the irradiation occurring at the eye, and is quite distinct from the very weak light of the vaporous air. And I repeat also to you, since you give me so many occasions for doing so, that you must completely abandon your false idea that the sun and moon when close to the horizon look larger because of a garland of illuminated air adjoining their discs; for this is very naive, as I have stated and proved above. And in order not to leave out anything that may lift you from error and enable you to understand this affair, I shall answer your closing words; there you say that these luminous rays around the star must

come through the telescope, so one cannot take away that small enlargement caused by them simply by saying that they are lost, for they are not lost. I reply that the enlargement is very great, as it is in all other objects, and that your error (as stated repeatedly) lies in comparing the star taken together with its irradiation when seen with the naked eye to the body of the star alone when seen with the telescope and distinguished from the irradiated region. Sometimes it appears larger than the latter, and sometimes equal, according to the size of the actual star and the power of the telescope; but even when it is seen smaller than this irradiation, the disc will still be perceived within the edges of the coiffure, as I have said. A very simple proof that great enlargement takes place as in all other objects is to pick out Jupiter with the telescope before dawn and then to follow it until sunrise and after. (*p.* 220) With the telescope, its disc will be seen as always of the same size, but as seen by the naked eye it will progressively shrink as the dawn brightens. Near sunrise, Jupiter—which in the dark surpassed all stars of the first magnitude—is reduced to a smaller appearance than a star of the fifth or sixth magnitude, and finally at sunrise it is reduced to an indivisible point and becomes completely lost. Yet after it has disappeared to the naked eye, it continues to be seen through the telescope large and round all day; I have an instrument which shows Jupiter to me when it is near the earth as large as the moon is when seen with the naked eye. So there is not here a small enlargement or none at all, but a great one, just as in all other objects.

If I cannot catch you on the run, Sig. Sarsi, I shall try to wear you out. Do you want another proof that objects at all distances increase in the same ratio? Listen, then, to this one. Let us place several visible objects at varying distances in such a way that all are seen in the same straight line, with the closest one hiding the others. Now keeping the eye in the same place and looking at these objects through the telescope, do you see them still placed in a straight line, or does the closest one no longer hide the rest from you, allowing you now to see them? I believe that you will reply that they appear to you in a straight line, since they are actually arranged that way. Now this being the case, imagine several straight rods placed parallel to one

another and at various distances from the eye; these rods are to be of unequal lengths, the most distant being the longest and the closer ones being progressively shorter in such a manner that their ends appear to be placed along two straight lines, one to your left and one to your right. Now take up the telescope and look at them through it. By your previous concession, the ends to the right and left will be seen along two straight lines as before, though the angle between the lines will open more widely. But how can this be, Sig. Sarsi? Among geometers it is required that all such lines shall increase (*p.* 221) in the same ratio; not that the closer ones shall grow more than the farther ones. Therefore yield, and be silent.

L

'FOURTH PROPOSITION . . . [*p.* 222] (*p.* 223) . . . cannot be a flame.' (*p.* 127, l. 21 to *p.* 130, l. 11).

Your Excellency, it is time that we bring this long discourse to a close, so let us pass on to this fourth and last proposition. Here, as you see, Sarsi says that he cannot contain his astonishment that I, having some reputation as a prudent observer and a man quite familiar with experiments, should be reduced to affirming things that can be very easily refuted by plain and simple experiments. He then adduces (*p.* 224) many of these, making himself appear to be as careful and honest an experimenter as I am an imprudent and mendacious one. Well, first, I shall mention briefly what it was that persuaded Sig. Mario to write and me to agree that if the comet were a flame it ought to conceal the stars from us. Then I shall consider Sarsi's examples and arguments, and finally I shall leave it to your Excellency to judge which of us is the more deficient and imprudent in his experimenting and reasoning.

We considered transparency of a body to be nothing but its permitting objects beyond it to be seen; hence we were convinced that the less visible a transparent body was, the better it might be seen through. Thus the very transparent air is completely invisible, and limpid water or well-polished glass is scarcely to be seen when placed between ourselves and visible objects, and it appeared to us

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that one might readily infer from this conversely that the more visible a thing was, the less transparent it should be.

Now since flames would not seem to be the least among things inherently visible, we judged that they should be but little transparent, and we were confirmed in this opinion when the authority of Aristotle and the Peripatetics was added to this reasoning. With respect to that authority, I note that Sarsi tries to give a different interpretation from that which the words will easily bear; he says that he knows perfectly well that it is true, but that its sense is that bodies may not be transparent in so far as they illuminate, and not that lucid bodies are not transparent. Of course, if Sarsi takes the proposition in this sense because the proposition thus appears true to him, he is abandoning the other because that appears to him to be false; yet as to the words themselves, they are better adapted to the latter sense than to the former. And he himself, a little farther on, not only affirms, but proves by experiments, that luminous bodies impede the view of things placed beyond them, when he writes: 'These also impede the appearance of things placed beyond themselves,' and what follows there.

But returning to the original argument, I say that beyond the authority of the Peripatetics we were still further confirmed in this by finally observing by experience that a heated glass (*p.* 225) very much impedes the view of objects which a cold glass allowed to be seen distinctly. The flame of a candle has the same effect, especially its upper part, for this is brighter than the lower part around the wick, where there are fumes that are not well ignited by the real flame. We observed, moreover, that the thickness of a body was quite significant even when the body was intrinsically not very opaque. Thus, for example, a cloud which did not prevent a tree trunk being seen at a thickness of twenty or thirty yards, when increased to two or three hundred cut off the visibility of the sun itself. So we thought it would be reasonable to believe that a flame could never be so little opaque as to be unable to impede our view of tiny stars, if this flame were increased in thickness to hundreds and hundreds of yards. From this we concluded that if the curl of the comet were a flame, its thickness (which may not be seventy

miles as Sarsi and his master say, but should at any rate be that many yards) ought to hide the stars from us; and since it did not do so, this appeared to us quite a conclusive argument against its being a fire.

Now Sarsi, caring little or naught for the main point of this very reasonable argument and seizing hold of Sig. Mario's single statement that a candle's flame is not transparent, became convinced that victory was his because he could prove that such a flame was somewhat transparent. He says that if one will hold some writing so close to it as almost to touch it, and will then take great pains to read the letters, he will be able to do so—to which I add, 'provided that he has perfect vision'; for I, not being near-sighted, can scarcely see things that close even if I make use of spectacles.

It is true that Sarsi adduces many other experiments besides the foregoing. Among these I should consider first, out of reverence and piety and because it is derived from the supreme authority, that one which Sarsi himself puts in first place and which he takes from the Holy Scriptures. Here Sig. Mario and I call attention to the scriptural words immediately preceding (p. 226) those selected by Sarsi; these appear to me to say that, before the King saw the angel and the three young men walking through the furnace, the flames had been removed from it. This seems to me to be the sense of the words of the Sacred Text, which are these: *Angelus autem Domini descendit cum Azaria et sociis eius, et excussit flammam ignis de fornace, et fecit medium fornacis quasi ventum roris flantem* ('But the angel of God went down with Azarius and his companions, and he drove the flame of the fires out of the furnace, and he made the midst of the furnace like the blowing of a wind bringing dew'). I note that the Scripture, by saying *flammam ignis*, seems to distinguish between 'flame' and 'fire'; and when one reads later that the King saw four persons walking, there is mention of fire but not of flame: *Ecce ego video quatuor viros solutos et ambulantes in medio ignis* ('Behold, I see four men unbound and walking in the midst of the fire'). But since I might be much mistaken in fathoming the true sense of matters which far transcend the powers of my mind, I leave such determinations to the care of doctors of divinity, and simply go on reasoning about these lesser doctrines, declaring that I am always prepared to

accept every decree of the superiors despite any demonstration or experiment which might appear contrary thereto.⁴²

Returning to Sarsi's experiments for making us see different objects through various flames, I say that I may freely concede all this to him as true. But it is of no help to his cause; it is insufficient for that purpose to interpose a flame one inch thick between objects as close as one inch to it, with the observer not much farther away, or to speak of objects inside the flame itself, and in its lower part, too, which is the least bright. Unless Sarsi is to be left behind, he must make it evident that a flame hundreds of yards thick, and very distant both from the observer and from the visible objects, does not impede their view—which is as much as to say that he must prove that the flame presents much less impediment than it would if it were so much cloud. Now the nature of a cloud is such that when interposed to the extent of several yards, let alone an inch, it introduces no impediment whatever, while at a thickness (*p.* 227) of one or two hundred yards it hides the very sun, to say nothing of the stars.

And finally, I cannot resist returning for a moment to Sarsi's amazement at my hopeless ineptitude in the use of experiments. You reproach me, Sig. Sarsi, as a poor experimentalist, and yet you yourself err as badly as a man can in that same employment. You must show us that counter to our assertion the interposed flame was insufficient to hide the stars, and in order to convince us by experiences you say that if we try to look at people, firebrands, coals, writings, and candles placed behind flames we shall see them quite plainly. Did it not enter your head to tell us that we might try looking at stars? Why, for heaven's sake, did you not tell us at the outset: 'Interpose a flame between the eye and some star, and the star will become neither more nor less visible'; surely stars are not lacking in the sky. Is this to be a skillful and prudent experimentalist? Let me ask you whether the comet's flame is like our flames or whether it is of a different nature. If of a different nature, experiments made with our kind of flame have no conclusive force; if like ours, then you might at once have made us look at the stars through our flames, leaving out firebrands, candle-snuffs, and such things, and, instead of saying that letters may be perceived through a

candle's flame, you might have asserted that a star is thus perceived. Anyone who wished to do business with you in the marketplace with an honest and sensitive balance, Sig. Sarsi, would say that you are under obligation to kindle a very distant flame as large as a comet and to make us see the stars through that, inasmuch as the size of the flame and its distance from the eye are very important in this matter and must be taken into account. But to give you every advantage and to put you at your ease, I will content myself with much less, and shall prepare for you a much more convenient means for your needs. First of all, since proximity to the eye is significant in seeing objects well, instead of placing the fire as far away as the comet I shall be content with a distance of only one hundred yards. Next, because the thickness (*p.* 228) and density of the medium is also very important, in place of the thickness of a comet—which as you know is many hundreds of yards—merely ten yards will satisfy me. Furthermore, since the object which is to be seen derives a great advantage from being bright, as you declare, I shall be content that this object shall be one of those stars which was seen through the curl of our comet—for those stars, according to your statement here, are far brighter than any flame. And now, with all these preparations which are so advantageous to your cause, if you can make the star visible through the transparency of this bonfire I want to confess myself vanquished, and to place you among the most prudent and subtle experimenters in the world.

But if you do not succeed, I desire no more from you than silence, by which you will put an end to this dispute, as indeed I hope you will. For if you should ever happen to see this essay of mine—and it rests with the gentleman to whom I write to show it to whomever he pleases—you will see what a man ought to do when he undertakes the task of attempting to examine the remarks of others; that is, he should pass over nothing without giving it consideration and should not go about, as you have done, like a blind chicken poking its beak into the ground now here and now there, hoping to find some grain of wheat to bite and peck at.

To conclude this section, you cannot deny that you yourself have understood and admitted that from the interposed flame some

impediment occurs even for your eye, for if absolutely no obscuration had taken place you would have said boldly (and without all these warnings and cautions about objects being farther from or nearer to the flame, or this flame being generated by sulfur or brandy rather than straw or wax): 'Let the flame and the object be what they will, no impediment occurs for me; I see freely as through pure air.' Besides, a little later, in speaking of things which are not self-lucid, like a flame, but are illuminated by something else, you say that these also impede the view of objects; now this word 'also' shows (*p.* 229) that you concede some impediment to the flame. Nor is that all. If a flame gave no hindrance, who would ever have thought of saying that it was not transparent? So even for you there is some small sensible beclouding (for you, I say, because for the rest of us the impediment is very great), and your experiments are made around so small a flame that surely the impediment from a cloud of equal size would be completely imperceptible. It follows that your flame is more of a hindrance than an equivalent cloud; but a cloud as thick as a comet would completely veil and remove from sight the sun itself; hence if the comet were a flame, it ought to be sufficient to hide the sun from us; and since it did not conceal the stars, let alone the sun, it was not a flame.

The resources for sustaining a falsehood are scarce indeed, but for the establishment of a truth supporting truths abound; I wish to hint to your Excellency a certain particular which seems to me to confirm the falsity of Aristotle's opinion. Inasmuch as nature directs upward all flames known to us while their source and origin remain toward the bottom, if the beard of the comet were a flame and its head were the material from which it took its origin, the curl would have to be pointed straight toward the sky. From this, two things would follow. Either the curl would be seen always in the guise of a garland around its head, which would be the case if the comet were situated very high, or else, if it were near the earth, at its birth the tip of its beard would appear first and the head last; then, as it rose toward our zenith, the beard would appear shorter, and at the zenith itself it would either disappear or would closely surround the head. And finally, as it went toward the west, the beard would necessarily

appear turned the opposite way so that the head would be seen to incline toward the west before the rest did. For if the beard should go first, as at its birth, (p. 230) it would be necessary for the flame to point down, contrary to its natural tendency and in contradiction of what it did when it was westerly. But such events are not observed in comets and their movements; hence comets are not flames.

L I

(Sixth argument) 'Also it ought not . . . has sufficient talent.' (p. 130, l. 12 to p. 131, l. 7). (p. 231)

Here, as you see, Sarsi tries to twist my own argument against me, but let us briefly examine how happily he has succeeded. First, I note that in order to carry out this intention of his, he runs into some self-contradictions; and what is more remarkable, does so without any need to. Previously, because it accorded with his case, he made every attempt to prove that flames are transparent so that stars might be seen through them. Here, to conquer me with my own weapons, he requires luminous bodies not to be transparent, so he sets about proving this by means of several experiments. From this it appears that he wants luminous bodies to be transparent and not transparent according to his necessities. He falls into this inconvenience without any need, since without incurring a shadow of contradiction from trying now to prove what he denied a little while ago, it was sufficient for him to say without attempting any proof that we ourselves have affirmed in general that luminous bodies are not transparent. Nor had he any occasion to fear that I would bring up distinctions between that which is luminous in itself and that which is lighted by something else, for I have always held that such devices are of service only to those who do not know how to express themselves clearly in the first place. If Sig. Mario had made any distinction between these, he would have declared so at the outset, and would not have waited for his adversary to call attention to his deficiency.

I say, then, that it is quite true that any illumination, whether intrinsic or external, would impede the transparency of a luminous body. But you need not understand by our saying this, Sig. Sarsi,

that we mean to imply that the body must become as opaque as a wall from receiving some tiny light. In proportion to the greater or less brightness, it would lose more or less of its transparency, and thus at the commencement of dawn, in proportion as the vaporous region begins to acquire a little light, the lesser stars become lost. Later, the splendor increasing, (*p.* 232) the larger stars are also lost, and finally at full illumination the moon itself is almost veiled. Furthermore, when through some rift in the clouds we see those long rays of the sun descend to earth, you will by taking pains perceive a noticeable difference in seeing the parts of a mountain you are facing; for those parts which lie behind the luminous rays are more obscured than those which lie off to the sides and which are not seen through these diagonal rays. Likewise, when a ray of sunlight descends through a small window into a darkened room, as it is sometimes seen to do where some small pane is broken in a church, all the objects beyond the path of the ray are seen less distinctly—at least when the observer is situated where he may plainly see the luminous ray, which is not true of every position.

All these things being true, I say (and have always said) that the material of the comet may be very much more subtle than the vaporous air and much less suited for illumination. I am persuaded of this by our seeing the comet disappear at dawn and twilight, when the sun lies well below the horizon. As to brightness, there is no more reason for the comet to hide the stars from us than for the vaporous region to do so. Then as to depth, the vaporous region is many miles thick, so we need not assume an immeasurable thickness for the beard of the comet, not having determined the diameter of its head, nor whether it is round, nor what its distance is. Despite all this, if anyone wished to assume it to be even eight or ten miles thick, this would not cause any difficulty; for even the vaporous air allows us to see the stars despite its being much thicker than the beard of the comet, and as much illuminated.

LII

‘In addition, Galileo objects . . . (*p.* 233) . . . many others?’ (*p.* 131, l. 8 to *p.* 132, l. 6)

The example in virtue of which Sarsi believes he can defend Aristotle and show Sig. Mario's objection to be invalid does not appear to me to agree very well with what it is supposed to exemplify. Seeing a great quantity of oats along the roadside and in the markets would very reasonably argue a greater abundance of oats than when these things are not seen, for it lies within the owners' power to expose it or conceal it; and besides, oats are not consumed or diminished in the least by being seen. These two circumstances have no counterparts in the matter of the comet. (*p.* 234)

Perhaps this is a more apt example. That the island of Cuba abounds in cinnamon is strongly argued by the knowledge that the islanders are continually burning it. The reasoning is conclusive, for it is within their control whether they burn it or not, and if they were short of cinnamon they would use it only as a condiment, as do we. But now suppose news to arrive that some months ago the cinnamon forest was attacked by a great fire through some accident, and that the islanders were far from the place and therefore could not extinguish the flames, so the forest was irreparably burned. Then if someone were to try to predict to our spice merchants that there would be an extraordinary abundance, and based this upon that unusual accident in which, where ordinarily they burn fagots of cinnamon, this time a whole forest was burned, I believe that he would be accounted a very foolish man. And anyone who should rejoice upon seeing flames devour the mature oats in his possession, and promise himself that because oats were burning by the bushel he would fill his granary much better than usually, would be considered completely crazy, I think.

The material of which the comet is made is either the same as that which produces the winds, or it is different. If it is different, one may not argue from the one material an abundance of the other—any more than a man might promise himself a great store of oil from seeing plenty of grapes. And if the material is the same, then the whole would burn when attacked by fire.

LIII

'I consider that . . . (p. 235) . . . which we have presented.' (p. 132, l. 7 to l. 36).

Here Sarsi makes two points, as you see; the first contains implicitly the judgment men should make of the fundamental weakness of our doctrine which he pretends to have proved to be supported upon mistaken experiments and defective reasoning. In the second place, he gives a catalogue and summary of the conclusions contained in Sig. Mario's *Discourse* which are impugned and refuted by him. In reply to the first, I imitate Sig. Sarsi and freely refer whatever judgment is to be made about the soundness of our doctrine to those who have attentively pondered the reasoning and the experiments on both sides. I hope that my cause will be not a little favored for my having examined point by point every reason and experiment adduced by Sarsi and replied to all, whereas he has skipped over the greater number of those given by Sig. Mario, and the most conclusive ones. (p. 236) I had thought of reviewing all these here in exchange for Sarsi's catalogue; but though I set myself to the task, I lacked the spirit and the strength, seeing that I should have to transcribe all over again little less than Sig. Mario's entire treatise. Hence, to save both your Excellency and myself this tedium, I have decided instead to refer your Excellency to a rereading of that *Discourse*.